Disclosure Policy in Contests with Sabotage and Group Size Uncertainty*

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September 9, 2025

Abstract

In many contests, players are not aware of how many competitors they face. While existing studies examine how disclosing this number affects their productive effort, this paper is the first to consider its impact on destructive behavior. To do so, I theoretically and experimentally study how revealing the number of contestants affects both effort and sabotage compared to concealing this information. Further, I evaluate contestants' productivity by comparing the resulting performances, which are shaped by the combination of the exerted effort and the received sabotage. I show that the overall performance can be higher under concealment, even though the disclosure policy does not affect average effort and sabotage levels. The experimental results largely confirm these theoretical predictions and demonstrate the significance of accounting for the effects of sabotage, as it induces performance differences between the group size disclosure policies. By concealing the number of contestants, a designer can mitigate the destructive effects of sabotage, without curbing the provision of productive effort.

Keywords: Sabotage, Contests, Group Size Uncertainty, Group Size Disclosure, Experiment

JEL: C72, C91, D62, D74, D82

*The published version is available at `https://doi.org/10.1016/j.euroecorev.2025.105120`. I thank Henrik Orzen, Wladislaw Mill, Aleksandra Khokhlova, Mark Spills, Anna Abate Bessomo, Thomas Eisfeld, Theodore Turocy, Bernhard Ganglmair, and Subashish Chowdhury for their helpful comments. I appreciate comments from participants from the ZEW/ University of Mannheim Experimental Seminar, the CDSE Seminar, the GESS research day, the Max Planck Summer School on the Political Economy of Conflict and Redistribution 2021, the 16th RGS Doctoral Conference in Economics, the MaCCI IO Day 2023, the Enter Jamboree 2023, the 9th Annual 'Contests: Theory and Evidence' Conference 2023, the 2023 ESA World Meeting, the VfS Annual Conference 2023, the ESA's job-market candidates seminar series 2023, the 19th HeiKaMax, the Mannheim Economics Alumni Symposium 2023, as well as from the Microeconomics Seminar at the University Carlos III of Madrid,the Experimental Seminar at the University of Innsbruck, and from the ECONtribute and C-SEB Design & Behavior Seminar at University of Cologne. This work was supported by the University of Mannheim's Graduate School of Economic and Social Sciences. Declarations of interest: none.

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1 Introduction

Contests exist in many settings, including job promotion tournaments, crowdsourcing contests, academic research grant applications, and patent races. In these competitive situations, agents spend non-refundable resources to outperform one or more competitors to enhance their chances of winning a valuable prize. However, in many cases, agents are not aware of how many other contestants they are facing, and whether there is another competitor at all (e.g., Boosey et al. 2017, Morgan et al. 2012, Lim & Matros 2009). In those cases, a contest designer, seeking to increase value-creating effort provisions, may decide to disclose the number of contestants or leave uncertainty about the group size. For instance, in a workplace setting, a manager may decide to reveal the number of short-listed candidates being considered for promotion. Similarly, when companies or the government offer inducement prizes for innovations, they may choose to disclose the number of participating competitors.[1]

For contest designers, disclosing the number of participants is an easy-to-implement tool. For contestants, this decision can have implications for their effort levels. That is because winning chances are determined by contestants' performances *relative* to the performances of their competitors. Relative performances are shaped by each contestant's effort level and, thus, deciding how much effort to exert depends on the number of competitors and beliefs about their effort levels. In line with theoretical equilibrium predictions, the experimental literature shows that if contestants know how many other contestants there are, effort usually decreases with an increasing group size (Dechenaux et al. 2015). If they do not know this number, it becomes more difficult to determine how much effort is needed to outperform others. In such cases, the theoretical equilibrium decisions are a weighted sum of the equilibrium choices conditional on the group sizes (Lim & Matros 2009), which lead to no difference in the average effort levels between disclosing and concealing the number of contestants (Fu et al. 2016, 2011). In these standard settings, even though effort choices become more difficult when players do not know the number of competitors, the experimental literature is consistent with theory as it does not find significant differences between the two disclosure policies (Jiao et al. 2022, Boosey et al. 2020, Aycinena & Rentschler 2019).

Yet, the choice of the disclosure policy may not only influence contestants' constructive efforts,

[1] See (Fu et al. 2016) for a more detailed discussion of these examples.

but also induce destructive behavior such as sabotage. Along with effort, sabotage is another strategy to increase one's relative performance – not through own productive effort but by negatively distorting one's competitors' performances, and a substantial literature has emerged on this topic (e.g. Chowdhury et al. 2023, Dato & Nieken 2020, Chowdhury & Gürtler 2015, Charness et al. 2014, Gürtler et al. 2013, Harbring & Irlenbusch 2011, Carpenter et al. 2010, Lazear 1989). Such sabotage can take various different forms. For instance, in workplace promotion tournaments, co-workers may withhold important information, skills, or experiences, share only partial information, or even provide wrong information to reduce the productivity of their colleagues (e.g., Serenko 2020, Pan et al. 2018, Kumar Jha & Varkkey 2018, Evans et al. 2015, Ford & Staples 2010).[2] I focus on sabotage that is used to decrease the productivity of competitors and thus destroys value. In particular, I concentrate on sabotage that can be easily scaled towards multiple potential competitors. For instance, employees could manipulate shared files, infrastructure, or resources while being prepared to continue working on their local computer, or simply omit crucial information to all of their colleagues.[3] If such destructive behavior happens, effort is spent less productively, which results in a decrease in the overall created value. For example, sabotaged co-workers may work with less efficient tools, focus on less important tasks, or spend their time recovering damaged files and resources. Although sabotage is of such importance for overall productivity, we still do not know how the choice of a group-size disclosure policy affects sabotage behavior. Yet, a policy that aims to increase productivity should take the adverse effects of sabotage into account. In this paper, I address this research gap by theoretically modeling and experimentally testing the differences between concealing and disclosing the number of contestants, taking into account not only contestants' effort choices but their sabotage decisions, as well. I first analyze the comparative statics of the realized group sizes under disclosure and the comparative statics of different enter probabilities and number of potential contestants under concealment. Then, I compare the resulting efforts and sabotage levels, expected payoffs, and performances between the disclosure policies. As a measure of contest-related productivity, I focus on the sum of individual performances (*group*

[2]Other common sabotage examples include the denigration of potential competitors' products or services (Nissen & Haugsted 2020), negative campaigning in political races (Lau & Rovner 2009), or fouls in sports (Deutscher et al. 2013)

[3]Sabotage can also occur between companies, for instance through cyberattacks on competitors' resources or infrastructure (Bitkom 2018). In those cases, easy-to-scale sabotage technologies include the targeting of widely used providers of Infrastructure-as-a-Service (IaaS), communication services, or malware phishing emails.

performance), as it shows the overall output in the presence of sabotage-induced value losses.

In my theoretical analysis, I model sabotage in a Tullock contest (Tullock 1980) and employ exogenous enter probabilities to model group size uncertainty, following Lim & Matros (2009). I introduce a designer, who commits to always conceal or disclose the number of contestants, but not their identities, following Fu et al. (2011). The number of potential contestants (and their enter probabilities) are common knowledge. As a consequence, players can sabotage all those that potentially compete with them independent of whether they know the number of actual competitors or whether there is uncertainty about it. For example, in a workplace context, players may have a sense of who potentially also applies for a position based on the position's requirements, allowing them to (preemptively) sabotage all of them. This sabotage could include not sharing crucial information or skills, or even providing wrong information and advice. Thus the knowledge of the set of potential competitors allows contestants to sabotage all of them, irrespective of the disclosure policy, as the designer merely discloses the number of active contestants, but not their identities. Similar dynamics can arise in patent races or other contests, where there are well-defined sets of 'usual suspects'.

A natural and important feature of group size uncertainty is the possibility of being the only actual contestant. For instance, there may be only a single applicant for a job promotion or, instead, a pre-determined favorite, which practically eliminates the contest and guarantees the prize to the unique player with certainty.

The theoretical results show that average effort and sabotage levels, as well as expected payoffs, are not different between the disclosure policies. However, the average group performance is higher under group size concealment compared to a disclosure of the realized group sizes. This is because group performance increases in own effort levels and decreases in the received sabotage. Contestants can adjust their effort and sabotage level to the specific realized group size under disclosure, while they have to choose one effort and one sabotage level, which will be used for any realized group size when it is concealed. Consequently, the distribution of effort and sabotage across group sizes differs, which induces differences in the group performances. The highest performance difference occurs in the case when there is only one contestant, who will win the prize with certainty. In this case, the one contestant does not exert any effort, when knowing to be the only contestant, compared to taking into account also other possible group size realizations, when not knowing the

realized group size. This leads to the exertion of a substantive amount of effort in anticipation of other group size realizations and receiving sabotage, while actually being the only contestant and not receiving any sabotage. Hence, the resulting performance is particularly high, which shapes the overall increase in group performance under group size concealment compared to disclosure. This is because for all other group size realizations, the performance differences between the disclosure policies are small, resulting in a higher average performance under concealment when there is at least a 1 percent chance of being the only contestant.

To evaluate the validity of these theoretical predictions, I conduct an experiment[4]. As sabotage is difficult to observe in the field,[5] a laboratory experiment is an optimal environment to test theories involving the possibility of sabotage. This holds especially true for this paper's setting because it involves a complex setting with several sources of uncertainties and best responses to competitors' effort *and* sabotage levels.[6] In the experiment, subjects play a Tullock contest with group size uncertainty, where each group member has the same exogenous enter probability. I vary the disclosure policy (concealment vs. disclosure) within subjects, and enter probabilities (0.25 vs. 0.75) and the size of the group (3 vs. 5) between subjects, leading to different probabilities of being the only contestant (0.4%, 6%, 32%, and 56%). Subjects receive an endowment that they can use to invest in 'Option A' (effort) to improve their own performance, or in 'Option B' (sabotage) to negatively affect everyone else's. Under group size disclosure, subjects make effort and sabotage decisions conditional on the realized group size via the strategy method, whereas under concealment they make one effort and one sabotage decision to fit all realized group sizes. To create the notion of productive effort and value-destroying sabotage in the experiment, money is donated to a non-profit charity, and the amount depends on the absolute performance of the group. Hence, by investing in effort, subjects increase both their own performance and donations, whereas by investing in sabotage, they increase their own relative performance by decreasing their opponents' performances but at the cost of decreasing donations. Importantly, the inclusion of

[4]The experiment was preregistered at aspredicted.org `https://aspredicted.org/blind.php?x=VB2_4DF` and received ethical approval from the Ethics Committee of the University of Mannheim.

[5]Observational studies usually rely on sports data to identify sabotage, which they typically define as the breaking of rules (e.g. Brown & Chowdhury 2017, Deutscher et al. 2013, Balafoutas et al. 2012, Del Corral et al. 2010).

[6]As a consequence, behavior may be influenced by other factors such as bounded rationality, probability distortions, risk aversion, and many others. Additionally, contests typically also induce non-monetary utilities such as joy of winning, which can lead to heterogeneous behavior (Dechenaux et al. 2015). With the existence of sabotage, other motives such as spitefulness may become relevant. Therefore, this experiment can be viewed as a robustness test for the theoretical predictions, which allows for these additional factors.

this externality does not change the theoretical predictions, even if subjects have a preference for donations. The experimental results are largely in line with theory and add to our understanding of contestants' behavior under the two disclosure policies. The first key finding is that group performance is significantly higher under concealment compared to disclosure but only when the probability of being the only contestant is not too low. As predicted, this difference is driven by the possibility of being the only contestant, where subjects do not receive any sabotage and exert much higher effort under concealment compared to disclosure. Consequently, there is no difference in group performance, when the probability of being alone is 0.4%, but in all other treatments where this probability is at least 6%, concealment leads to a higher group performance. The second key finding is that there is no evidence for a difference in average sabotage and effort levels, as well as in expected payoffs between the two disclosure policies. he only exception is when the number of potential contestants is 3 and enter probabilities 0.25. In this case, concealment leads to a slight increase in sabotage levels. Nonetheless, even in this case, the expected payoffs do not differ between the disclosure policies.

As additional results, I confirm the predicted comparative statics of disclosed group sizes, where a larger group size reduces sabotage and effort levels. At the same time, there is above-equilibrium sabotage in groups of sizes 3, 4, and 5. This behavior can be explained by joy of winning that increases in the number of competitors (constant winning aspiration) (Boosey et al. 2017), or by spiteful preferences (Morgan et al. 2003, Levine 1998). As to the comparative statics of group size uncertainty, I find that an increase in the number of potential contestants decreases sabotage levels for high enter probabilities, as theory suggests. For low enter probabilities, however, I do not find evidence for the hypothesized increase.

The contribution of this paper is as follows. I add to the discussion of group size disclosure policies, by examining contestants' behavior in a more nuanced setting, that allows not only for constructive behavior but also for destructive behavior. The literature shows that competition also induces cheating, fraud, and sabotage besides productive efforts (Piest & Schreck 2021, Chowdhury & Gürtler 2015, Carpenter et al. 2010, Faravelli et al. 2015), and thus a more realistic contest setting should account for such behavior. Moreover, the inclusion of sabotage is indispensable for policy evaluations, as sabotage destroys value and therefore has negative implications for the contest-induced productivity. In the most standard contest setting without sabotage, the disclosure

policy does not influence the average exerted effort and hence the created value (Lim & Matros 2009). I show that when sabotage in contests is accounted for, higher performances can be induced by concealing the number of competitors. This has substantial implications for contests' design. A designer can mitigate the value-destroying effects of sabotage by concealing the number of contestants.

I also contribute to the sabotage literature by suggesting a policy that mitigates the destructive effects of sabotage without curbing productive efforts. The theoretical and experimental literature shows ways of how to decrease sabotage altogether, including reducing the prize spread (Harbring & Irlenbusch 2011, 2005, Del Corral et al. 2010, Vandegrift & Yavas 2010, Lazear 1989), increasing the number of contestants (Konrad 2000), increasing the penalties for sabotage (Balafoutas et al. 2012), revealing the identity of the saboteur (Harbring et al. 2007), or not revealing intermediate relative performances or rank (Charness et al. 2014, Gürtler et al. 2013, Gürtler & Münster 2010) as sabotage is directed against the most able or best-performing contestant (Deutscher et al. 2013, Harbring et al. 2007, Münster 2007, Kräkel 2005, Chen 2003). For broader literature reviews on sabotage in contests see Piest & Schreck (2021), Amegashie et al. (2015), or Chowdhury & Gürtler (2015).

This paper also informs other theoretical contest settings without sabotage, where there are already differences in effort choices between the two group size disclosure policies. Accounting for the effects of sabotage may interact with their identified effects and possibly change the conclusions. These settings include different prize valuations together with different enter probabilities (Fu et al. 2016), different prize valuations with endogenous entry (Chen et al. 2023), the existence of bid caps (Wang & Liu 2023, Chen et al. 2020), either convex or concave cost structures (Jiao et al. 2022, Chen et al. 2017), and either strictly convex or concave characteristic functions of the Tullock contest (Feng & Lu 2016, Fu et al. 2011).

I further add to the experimental contest literature without sabotage (Jiao et al. 2022, Boosey et al. 2020, Aycinena & Rentschler 2019), which, in most settings, finds no difference in average effort levels between the two disclosure policies. In more specific settings, the experimental literature finds that disclosure can lead to higher effort levels, for instance when the outside option is high and entry endogenous (Boosey et al. 2020), or when effort costs are concave (Jiao et al. 2022). In this paper, I show that concealment leads to a higher performance, even though there are no

differences in the average effort and sabotage levels.

By also studying the comparative statics of group size, I provide evidence for the influence of known group sizes on sabotage, which so far lacks empirical evidence as pointed out by Piest & Schreck (2021) and Chowdhury & Gürtler (2015).[7] Additionally, I find substantial over-sabotage for larger group sizes. Therefore, increasing group size may not be an apt tool to decrease overall sabotage and should therefore be used with caution.

Moreover, my paper is the first to consider group size uncertainty in a contest with sabotage. For contests without sabotage, the literature shows that group size uncertainty matters for effort levels of contestants.[8] Yet, the existing sabotage literature assumes that the number of contestants is common knowledge.[9] I experimentally confirm that effort and sabotage decisions under uncertainty can be described by a weighted sum of the level choices for the known group sizes.[10]

Finally, I also contribute to the literature on optimal information disclosure more broadly (e.g., Du & Shahriar 2024, Antsygina & Teteryatnikova 2023, Mariotti et al. 2023, Serena 2022, Fu & Wu 2022, Mathevet et al. 2020, Bergemann & Morris 2019, Schweizer & Szech 2018, 2017, Benndorf et al. 2015), as well as to unmoral behavior and market institutions (e.g., Serra-Garcia & Szech 2022, Huck et al. 2021, Bartling et al. 2015). In my setting, information disclosure informs not only players' constructive effort choices but, crucially, also their sabotage choices, which leads to a higher destructive effect of sabotage and an associated productivity loss. I therefore caution that discussions about information disclosure and transparency should also consider their effects on destructive behaviors, as neglecting this may lead to erroneous conclusions.

The structure of this paper is as follows: In Section 2, I set up a theoretical model in order to derive equilibrium predictions. Section 3 describes the experimental design. In Section 4, I present the results before I provide a discussion and conclusion in Section 5.

[7] So far, there is only one experimental study that investigates a known number of competitors but in a rank-order tournament, which predicts no differences in sabotage levels across group sizes. Thus, the authors do not find any differences in their experiment (Harbring & Irlenbusch 2008).

[8] Gu et al. (2019), Boosey et al. (2017), Chen et al. (2017), Ryvkin & Drugov (2020), Kahana & Klunover (2016, 2015), Morgan et al. (2012), Fu et al. (2011), Lim & Matros (2009), Münster (2006), Myerson & Wärneryd (2006), Higgins et al. (1988)

[9] Chowdhury et al. (2023, 2022), Dato & Nieken (2020, 2014), Benistant & Villeval (2019), Brown & Chowdhury (2017), Leibbrandt et al. (2017), Charness et al. (2014), Deutscher et al. (2013), Gürtler et al. (2013), Amegashie (2012), Balafoutas et al. (2012), Harbring & Irlenbusch (2011), Carpenter et al. (2010), Vandegrift & Yavas (2010), Gürtler & Münster (2010), Harbring & Irlenbusch (2008), Münster (2007), Harbring et al. (2007), Kräkel (2005), Chen (2003), Konrad (2000), Lazear (1989)

[10] I also contribute to many of Nora Szech's works, where she has modeled uncertainty about the number of agents in various settings (e.g. Rothenhäusler et al. 2018, Szech 2011, 2010).

2 Theoretical Model and Predictions

2.1 Setup

I partially follow Konrad (2000) and model sabotage in a Tullock contest (Tullock 1980) and employ exogenous enter probabilities to model group size uncertainty, following Lim & Matros (2009).[11]

Let N be the set of all homogeneous and risk-neutral potential contestants, and n the number of potential contestants indexed by $i \in N, N = \{1, ..., n\}$. Every potential contestant has the same enter probability of $q \in (0, 1]$. The set of potential contestants N and their enter probabilities q are common knowledge. Let N_i be the set of possible opponents of player i. Conditional on player i participating, let $M_i \subseteq N_i$ be the set of other active players except for player i in the contest. M_i is not known to the players. Let m be the number of active contestants including player i with $M = \{1, ..., m\}$ being the set of all active contestants including player i.

There is a contest designer, who ex-ante commits to always conceal or reveal the number of active contestants m.[12] She does not reveal the identities of the active players. Because of this, players can only choose to sabotage all other potential contestants N_i, because they know who potentially enters, but they do not know who actually entered.

Active players compete to win a single prize W. They choose to spend effort $e_i \geq 0$ with linear costs $C(e_i) = e_i$ and sabotage $s_i \geq 0$ with linear sabotage costs $C(s_i) = s_i$. Unlike Konrad (2000), I explicitly model sabotage costs as independent of n and discuss this model assumption in Section 2.5. I also assume that they can only sabotage all other potential competitors by the same amount.[13]

Contestant i is subjected to total sabotage of $\sum_{j \in M_i} s_j$. Only active players are affected by the

[11] Exogenous enter probabilities may arise when a contest is exposed to specific regulations and entry barriers, that include certain quality and safety standards of a product in a patent race, or specific requirements concerning skills and characteristics of employees for a promotion (Boosey et al. 2017). Likewise, they may arise as mixed-strategy equilibrium enter choices (Fu et al. 2015) determined by the value of the prize, entry fees, and the outside option.

[12] If the designer decides to partially disclose the number of contestants, contestants can anticipate the specific realized group sizes, where the designer would prefer to disclose. Lim & Matros (2009) show in a contest without sabotage, that if a designer can not credibly commit to always either conceal or disclose, she would always disclose the number of contestants. Similar dynamics would arise in this more specific setting but is beyond the scope of this paper.

[13] As it does not cost anything extra to include an additional competitor, players would choose to sabotage all other homogeneous competitors by the same amount even if they had the option to allocate sabotage selectively – except when a competitor exerts zero effort (or receives infinite sabotage), in which case players would be indifferent to including them. Additionally, some sabotage technologies are naturally targeted against all others: e.g., not sharing information or manipulating shared resources or files. An important benefit of this assumption is that it substantially simplifies the complex experimental choice situation that participants face.

exerted sabotage as only they exert contest-induced additional efforts.[14] The effort and sabotage levels translate into individual performance y_i as follows:

$$y_i = \frac{e_i}{1 + \sum_{j \in M_i} s_j}$$

Individual performances are increasing in contestants' own effort levels and decreasing in the total amount of received sabotage (i.e., their opponents' sabotage levels).[15] The impact on the performance function shows the fundamental difference between constructive effort and destructive sabotage: while effort e_i enhances one's own performance, sabotage s_i destroys value by lowering the performance of others. Consequently, the performance function reflects a player's overall productivity in the presence of sabotage-induced productivity losses. Player i's probability of winning is determined by the following contest success function:[16]

$$p_i(y_i, y_{-i}, M_i) := \begin{cases} \frac{y_i}{y_i + \sum_{j \in M_i} y_j} & \text{if } \max\{y_1, ..., y_m\} > 0 \\ \frac{1}{m} & \text{otherwise,} \end{cases}$$

With this contest success function, relative performances determine individual winning probabilities. Therefore, players have two options to increase their winning chances. They can either increase their own performance by providing additional effort or decrease their opponents' performances by sabotaging more. An essential feature of group size uncertainty is the possibility of being the only contestant. In this case, the one only active player i wins the contest with certainty independent of her effort and sabotage choices.

The timing of the game is as follows. Before the contest, the designer ex-ante commits to always conceal or disclose the number of contestants. Then, nature determines who becomes active and enters the contest. Conditional on participating, active contestants simultaneously make their effort and sabotage choices. Afterwards, the contest is resolved according to the winning probabilities.

[14] This assumption isolates the effect of the disclosure policy on the contest-induced performances. Additionally, if the sabotage is specifically directed towards only contest-related efforts, such as withholding information about promotion-relevant work activities, there is no effect on non-active players. Even if there is an effect on the base productivity of non-active players, and this base productivity is small enough or the effectiveness of sabotage on this base productivity is small, the results remain the same. See Section 5 for a more detailed discussion.

[15] The results extend to performance functions with less pronounced marginal returns in the received sabotage: $y_i = \frac{e_i}{(1+\sum_{j \in M_i} s_j)^t}$ with $t < 1$. For $\lim_{t \to 0}$, however, sabotage has no effect anymore and the performance differences between the disclosure policies disappear. See Appendix A.6 for a more detailed analysis.

[16] For an axiomatization see Skaperdas (1996).

2.2 Group Size Disclosure

Under group size disclosure, the designer ex-ante commits to reveal the number of contestants but not their identities. Therefore, players only know the identities of all *potential* contestants N_i, allowing them to sabotage them all, which include their actual competitors. Conditional on being active, player i chooses e_i and s_i to maximize her expected payoff:

$$\arg\max_{e_i, s_i \in \mathbb{R}_+} p_i(y_i, y_{-i}, m)W - e_i - s_i \tag{1}$$

The associated first-order and second-order conditions can be found in Appendix A.1. Conditional on being active, all contestants simultaneously choose effort and sabotage. The following proposition characterizes the static symmetric equilibrium:

Proposition 1. *Consider a contest as described above. The static symmetric equilibrium is characterized as follows:*

$$e^* = \frac{(m-1)}{m^2}W \tag{2}$$

$$s^* = \begin{cases} \frac{1}{m^2}W - \frac{1}{m-1} & \text{if } W \geq \frac{m^2}{(m-1)} \text{ and } m \geq 2 \\ 0 & \text{else} \end{cases} \tag{3}$$

Proof. See Appendix A.2. □

The proposition shows that effort and sabotage levels depend on the realized group size m. When there is only one active competitor ($m = 1$), this one active player wins with certainty, making it optimal to not exert any effort or sabotage. When there is at least one other contestant ($m > 1$), equilibrium effort and sabotage levels decrease with increasing group size.

2.3 Group Size Uncertainty

Under group size uncertainty, the contest designer ex-ante commits to conceal the number of contestants. Hence, contestants only know the set of all other potential contestants N_i and their enter probabilities q. With these, they can compute the expected number of contestants. Because they know the identities of the potential contestants, as under group size disclosure, players can exert sabotage against all of them. Conditional on being active, player i chooses e_i and s_i as follows:

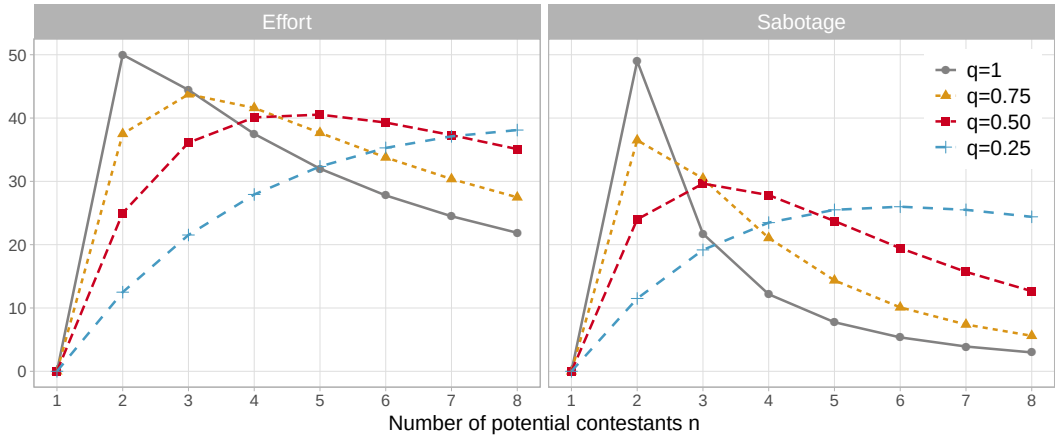


Figure 1: Equilibrium effort and sabotage levels conditional on the number of potential contestants and concealment for different enter probabilities. The prize is set to $W = 200$.

$$\underset{e_i, s_i \in \mathbb{R}_+}{\arg\max} \sum_{M_i \in \mathcal{P}^{N_i}} q^{|M_i|}(1-q)^{|N_i/M_i|} p_i(y_i, y_{-i}, M_i)W - e_i - s_i \tag{4}$$

where $\mathcal{P}^{N_i}$ is the powerset of N_i. Conditional on participating, players simultaneously maximize their expected profit function by choosing e_i and s_i. The following proposition characterizes the static symmetric equilibrium:

Proposition 2. *Consider a contest with group size uncertainty as described above. Conditional on being active, the optimal effort in the static symmetric equilibrium is described by:*

$$e^* = \sum_{(m-1)=0}^{n-1} \underbrace{\frac{(n-1)!}{(m-1)!(n-m)!} q^{m-1}(1-q)^{n-m}}_{\text{probability of } m-1 \text{ others}} \times \underbrace{\frac{m-1}{m^2} W}_{\text{effort choice for } m-1 \text{ others}} \tag{5}$$

A numerical solution to the following equation describes the optimal sabotage level s^*:

$$\sum_{(m-1)=0}^{n-1} \underbrace{\frac{(n-1)!}{(m-1)!(n-m)!} q^{m-1}(1-q)^{n-m}}_{\text{probability of } m-1 \text{ others}} \times \frac{m-1}{m^2} \frac{1}{1+(m-1)s} W = 1 \tag{6}$$

Proof. See Appendix A.3. □

Proposition 2 shows that effort decisions under group size uncertainty are a weighted sum of the

equilibrium choices for known realized group sizes.[17] Numerical results show that sabotage follows a similar weighted-sum pattern.[18] Figure 1 illustrates the comparative statics of Proposition 2 and reveals single-peaked functions for all enter probabilities. These peaks stem from optimal zero effort and sabotage when being the only contestant ($m = 1$), along with the maximum choices for $m = 2$, and their subsequent decline with larger m when group sizes are known ($q = 1$). For lower enter probabilities, the influence of being the only contestant dominates longer with an increasing number of potential competitors n compared to higher enter probabilities. As a result, equilibrium levels initially increase with n, with this increase lasting longer for lower q. For higher q, the peak occurs earlier, and the subsequent decline happens more quickly as n increases.

2.4 Comparing Disclosure Policies

In the following, I analyze how the disclosure policy affects expected effort, sabotage, and individual payoffs. Additionally, I compare the contest-induced productivity between the two disclosure policies by analyzing the expected sum of individual performances, which reflects both the productive effects of effort and the destructive effects of sabotage.[19]

2.4.1 Expected Effort, Sabotage, and Payoffs

There is no difference in the expected effort between the two policies, as under concealment, contestants take the weighted sum of their equilibrium effort levels for the known group sizes. A numerical analysis shows that there are also no substantial differences in sabotage levels, which leads to no substantial differences in the expected individual payoffs: because of symmetry, everyone exerts the same effort and sabotage levels, leading to the same winning probabilities and thus payoffs.

Corollary 1. *There are no substantial differences in expected effort, in expected sabotage, and in expected individual payoffs between disclosing and concealing the number of competitors.*

Proof. See Appendix A.4 and A.5. □

[17] This result relies on the assumptions of linear effort costs (see Chen et al. (2017)) and the linearity of the characteristic function (i.e., $H(\cdot) = y(\cdot)/y'(\cdot)$) of the Tullock contest success function (see Fu et al. (2011)).

[18] There is no closed form solution because in the performance function, 1 is added to the received sabotage to ensure a solution in the special case of not receiving any sabotage $y_i = \frac{e_i}{1+\sum_{j\neq i} s_j}$.

[19] Rather than assessing absolute value creation and welfare, I focus on productivity differences between the two disclosure policies. A precise quantitative evaluation would require a more refined performance function that translates effort into performance beyond a simple one-to-one mapping.

2.4.2 Expected Group Performance

Next, I study how the disclosure policy affects the expected group performance. As a first step, I compute the equilibrium group performance P^* conditional on the realized group size m as follows:

$$P^*(m) = \underbrace{\sum_{i=1}^{m} y_i(m)}_{\text{sum of individual performances}} = \underbrace{\frac{e^*}{1+(m-1)s^*}}_{\text{individual performance}} \times \underbrace{m}_{\text{realized number of contestants}} \tag{7}$$

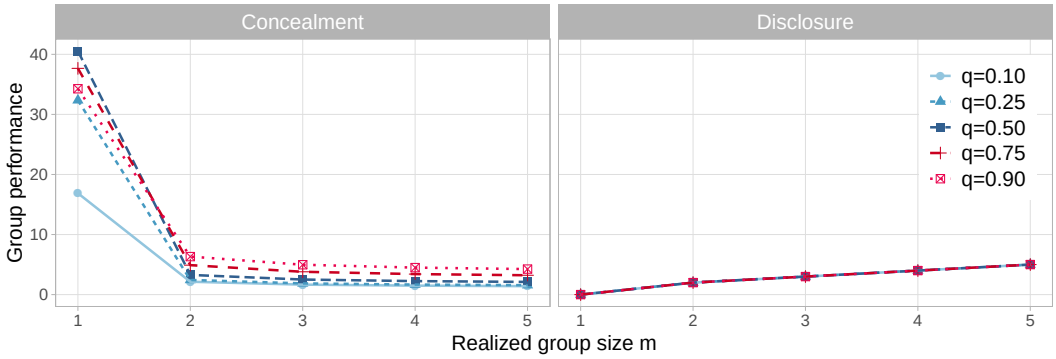


Figure 2: Equilibrium group performance conditional on the realized number of contestants under concealment (left graph) and disclosure (right graph). The different colors indicate different enter probabilities. Under disclosure, all lines are exactly the same. The prize is $W = 200$ and $n = 5$.

When players know the number of competitors, they can adjust their effort and sabotage levels according to the realized group size m ($e^*(m)$, $s^*(m)$), while they have to choose one effort and one sabotage level for all realized group sizes ($e^*(n,q)$, $s^*(n,q)$) under group size concealment. Consequently, group performance differs between the disclosure policies (see Figure 2). While there are only small group performance differences for $m > 2$, they are substantial for $m = 1$. This is because under disclosure, the sole active player knows to win by default and thus does not exert any effort, resulting in a performance of 0. Under concealment, she is not aware of this fact and chooses her effort level in anticipation of all possible realized group sizes. Consequently, if $m = 1$, she exerts a substantive amount of effort without receiving any sabotage, leading to a particularly high performance.[20] The overall expected impact on the productivity depends on the probability of a single active contestant ($m = 1$). Figure 3 shows the expected group performance[21] conditional on the number of potential contestants. The following corollary summarizes the effect

[20] This difference is also pronounced for performance functions that have a less pronounced decrease in the marginal returns of the received sabotage (see Appendix A.6).

[21] The expected group performance is computed as follows: $\mathbb{E}[P^*] = \sum_{m=1}^{n} \frac{n!}{m!(n-m)!} q^m (1-q)^{n-m} \times P^*(m)$

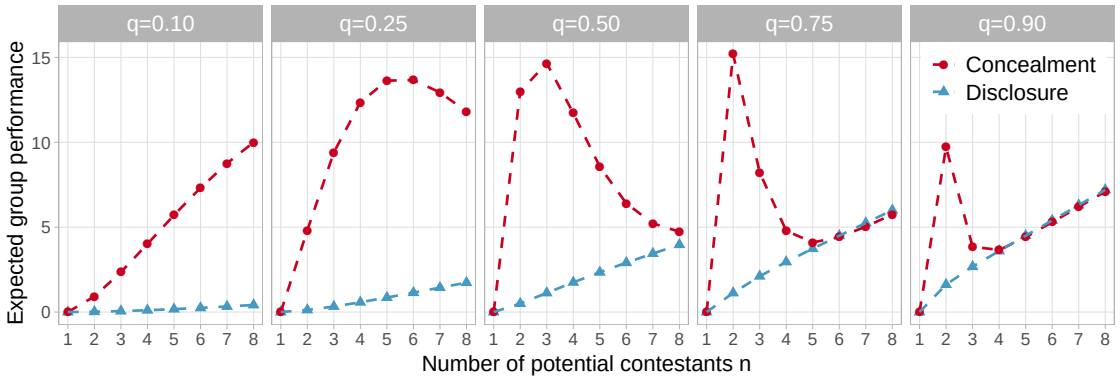


Figure 3: Expected equilibrium group performance conditional on the disclosure policy for different enter probabilities and number of potential contestants. The prize is $W = 200$.

of the disclosure policy on the expected group performance:

Corollary 2. *When the probability of being along is high (low q, low n, but < 1), expected group performance is higher under concealment. When this probability gets lower (larger q, larger n), expected group performance converges across the two disclosure policies.*

Proof. Numerical analysis illustrated in Figure 3. □

2.5 On Sabotage Costs

In this section, I discuss how sabotage costs may or may not depend on the number of targeted competitors. Exerting destructive sabotage is costly, as it requires time and resources for planning and execution. The extent to which these costs depend on the number of targeted opponents varies based on the specific sabotage technology, contextual factors, and the relative importance of planning compared to the actual execution.

In some cases, sabotage costs may increase linearly in the number of targeted opponents (i.e. $C(s) = (n-1)s$). This occurs when all sabotage efforts need to be individually tailored and repeated for every single opponent or when the execution costs far exceed the initial investment in planning and discovering the specific sabotage technology. One example could be individually-tailored cyberattacks against different competitors, which require an individual adaption to each competitor's potentially unique operating and security systems. Another example includes deliberately misdirecting or overloading colleagues with unnecessary tasks, in case that each act of sabotage must be individually customized to the specific role and responsibilities of the targeted colleague.

In other cases, sabotage costs may not linearly increase in the number of opponents if the technology incurs (almost) no additional costs per additional target or if the execution of sabotage activities is relatively small compared to its planning. In both cases the marginal costs of including an additional opponent may be negligible. For instance, withholding important information, critical updates, or productivity-enhancing skills in the workplace affects everyone else independent of the number of colleagues. But also more active sabotage, such as sharing wrong advice or overloading everyone with the same unnecessary tasks, can be easily scaled to many when colleagues work on similar topics or with similar tools. Another example is the disruption of shared resources – such as servers or shared work files – while being prepared to work offline, which can decrease the productivity of many others at the same time.[22] Lastly, anonymously sending a malware phishing email (e.g., by hiring an independent hacker) can cause serious damage to competitors' work resources while additional opponents can easily be including simply by expanding the recipient list.

Efforts to discover and employ such easy-to-scale sabotage technologies may be especially beneficial when the identities of the actual competitors are unknown because contestants have to (preemptively) target all potential competitors.

For this reason and for the simplicity of the experiment, I employ sabotage costs that are independent of the number of competitors, i.e., $C(s) = s$. This is in contrast to Konrad (2000), who employs $C(s) = (n-1)s$. To examine the implications of this modeling difference, I compare the theoretical results for these two sabotage cost regimes (details can be found in Appendix A.7).

The main takeaway is that equilibrium sabotage levels are generally lower under both disclosure policies when sabotage becomes relatively more costly with increasing targets. While it does not influence the comparative statics of disclosed group sizes and the comparative statics under group size concealment at high enter probabilities, it changes the comparative statics of the number of potential contestants under group size concealment at low enter probabilities. Specifically, whereas $C(s) = s$ leads to an initial increase in sabotage as n rises at low enter probabilities for $n \geq 2$, $C(s) = (n-1)s$ results in an overall decrease at all enter probabilities for $n \geq 2$ (see Figure 18 in

[22]Similarly, companies can engage in cyberattacks targeting shared infrastructure – such as distributed denial-of-service (DDoS) attacks – to disrupt widely used cloud services, collaboration platforms, or communication services. The global rise in spendings on Infrastructure-as-a-Service (IaaS) Statista (2024) amplifies these risks, as a single breach can affect numerous firms (e.g., the 2023 Microsoft Azure DDoS attack).

Appendix A.7).

Crucially, this difference does not influence the comparison of the two disclosure policies, as equilibrium effort and sabotage choices under uncertainty remain (roughly) the weighted average of the equilibrium known group size choices. Similarly, the mechanism that induces the expected group performance difference remains the same under both cost regimes.

3 Experimental Design

In this section, I present the experimental conditions, hypotheses, the experimental design and experimental procedures. Following the model in Section 2, the main part of the experiment consists of a Tullock contest with exogenous enter probabilities.[23] Subjects are part of a fixed group of potential contestants with size n and each of them becomes active with the same enter probability q. The value of the prize is worth EUR 18, so the contest is highly incentivized.

3.1 Experimental Conditions and Hypotheses

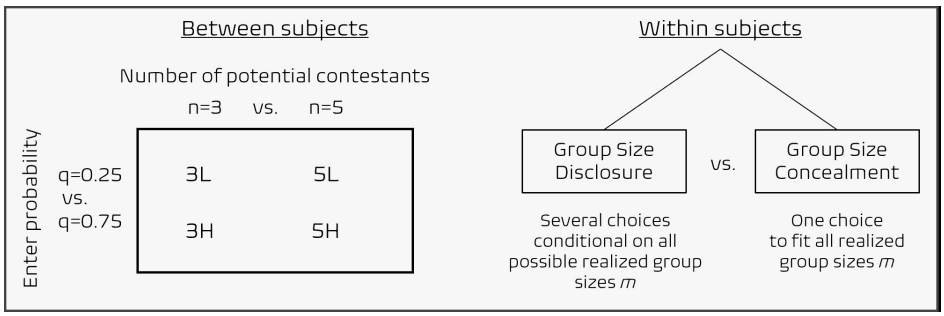


Figure 4: Experimental conditions

Figure 4 shows an overview of the experimental conditions. I set $W = 200$ to ensure non-zero equilibrium sabotage, and exogenously vary the disclosure policy within subjects (full disclosure of the number of contestants m vs. full concealment), meaning that every subject makes decisions under both disclosure policies. At the same time, I vary the enter probability (low $q = 0.25$ vs. high $q = 0.75$) and number of potential contestants (small $n = 3$ vs. large $n = 5$) between subjects to study both disclosure rules under different scenarios, resulting in the treatments $3L$, $5L$, $3H$, and $5H$ with different probabilities of being the only contestant (56% in *3L*, 32% in *5L*, 6% in *3H*,

[23]The experiment received ethical approval from the Ethics Committee of the University of Mannheim.

and 0.4% in *5H*). Lastly, under group size disclosure, subjects make several decisions conditional on all possible realized group sizes m, which allows me to study the comparative statics of different known group sizes m.

Based on Section 2, I formulate pre-registered hypotheses for the experimental conditions.[24] Table 1 provides an overview of these hypotheses and the place of their derivation.[25]

	Hypothesis	Derivation
Group Size Disclosure		
Hypothesis 1.1.	*A larger disclosed group size decreases effort and sabotage levels for* $m > 1$.	Proposition 1
Group Size Uncertainty		
Hypothesis 1.2.	*For high enter probabilities (*$q = 0.75$*), effort and sabotage levels decrease when the number of potential contestants increases from* $n = 3$ *to* $n = 5$.	Proposition 2, Figure 1
Hypothesis 1.3.	*For low enter probabilities (*$q = 0.25$*), effort and sabotage levels increase when the number of potential contestants increases from* $n = 3$ *to* $n = 5$.	Proposition 2, Figure 1
Comparison Expected Effort, Sabotage, and Payoff		
Hypothesis 2.1.	*There are no substantial differences in expected effort and expected sabotage levels between concealing and disclosing the number of contestants.*	Corollary 1
Hypothesis 2.2.	*There is no substantial difference in expected payoffs between disclosure and concealment.*	Corollary 1
Comparison Expected Group Performance		
Hypothesis 3.1.	*Concealing the number of contestants increases the expected group performance compared to disclosure when the probability of being the only contestant is not too low (at least 6%, treatments 3L, 5L, 3H).*	Corollary 2, Figure 3
Hypothesis 3.2.	*For a low enough probability of being the only contestant (0.4%, treatment 5H), there is no substantial difference in the expected group performance between disclosure and concealment.*	Corollary 2, Figure 3

Table 1: Overview hypotheses

[24] The hypotheses are pre-registered on `https://aspredicted.org/VB2_4DF`

[25] Hypothesis 2.1. was not preregistered and added later. However, it follows directly from the model that remained unchanged.

3.2 Experimental Design and Procedures

The main part of the experiment consists of 35 rounds of the contest. To ensure incentive-compatibility of each single round, I pay the average of 3 randomly determined rounds only.[26] These randomly chosen payments are displayed on the last page of the experiment only. Depending on the treatment, participants are assigned to a corresponding fixed group of 3 or 5. They stay in that group until the end of the main part and only interact with other participants of this group. Therefore, I can treat each group as a statistically independent observation. Additionally, the provided feedback of the other group members is not tied to their identities but is presented anonymously in a randomized order to reduce dynamic effects such as retaliation, reputation building, and tacit collusion across rounds.

To reduce experimenter demand and priming effects, the instructions are held on an abstract level, without using the words 'effort', 'sabotage', 'contest', or 'opponents'. Instead, I call effort 'Option A' and sabotage 'Option B'. Without this framing, both choices are simply tools to increase own winning probabilities with different marginal returns. Therefore, to capture the value-creating effects of effort and the value-destroying effects of sabotage, I incentivize the resulting sum of individual performances (group performance), which is positively affected by effort and negatively by sabotage. Specifically, to incorporate these value-creating and value-destroying externalities, I include donations to a charity that depend on the group performance.[27] In this way, when players exert effort, they increase their winning probabilities and the donations, and when they exert sabotage, they increase their winning probabilities but at the additional cost of decreasing the donations.[28] Note that the individually rational equilibrium predictions are not influenced by the inclusion of donations, as they do not influence the individual payoffs. Additionally, even if contestants have a preference for donations, effort and sabotage levels are only marginally different, and the comparative statics remain unchanged (see Appendix A.8).[29]

[26] See Azrieli et al. (2018) for a theoretical discussion on incentive compatibility. I decided to pay the average of three rounds instead of one single round, to contribute to the maintenance of a more reliable and satisfied subject pool. Empirically, I do not observe any last-round effects.

[27] Former experimental literature on sabotage in contests includes a principal in their experiment whose payoff is determined by the performance of the contestants (Harbring & Irlenbusch 2011, 2008). While this procedure requires an additional participant per group, the same goal can be achieved by including donations to a charity.

[28] The donations are calculated as follows: $donations = \sum_{i=1}^{m} y_i + 10$, where m is the number of all active players and y_i the individual performance of player $i \in M$.

[29] To eliminate heterogeneous preferences for specific charities across participants, I include five charities from various sectors (Amnesty International, Doctors Without Borders, German Red Cross, Greenpeace, and UNICEF).

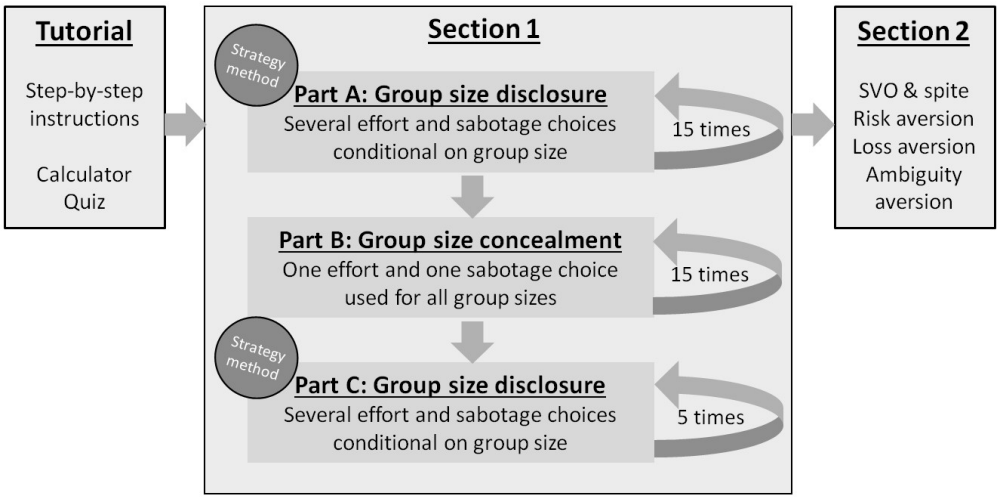


Figure 5: Experimental design

Figure 5 depicts an overview of the experimental design. The experiment starts with an extensive Tutorial and is followed by section 1. Section 1 contains the main part of the experiment, where Part A is designed to study decisions under group size disclosure and the comparative statics about the influence of a known realized group size m. Part B is designed to study decisions under group size concealment and the comparative statics of the influence of the number of potential contestants n and their enter probabilities q. Part C is identical to Part A. By comparing the choices of Part B to the choices of Part A and C, I compare the effects of the disclosure policies. Part A is repeated 15 times, followed by 15 repetitions of Part B, followed by 5 rounds of part C. The reason why I repeat another 5 rounds of group size disclosure in part C is to control for potential order effects.[30] In section 2, I elicit social value orientation (SVO), spiteful preferences, risk, loss, and ambiguity aversion, and standard demographics.

I now describe the experimental procedure in detail (see Appendix D for the experimental instructions). To make sure that participants understood the experiment, they started with an extensive tutorial. In this tutorial, the rules were explained carefully and subjects could make practice choices with the computer making random choices for their opponents. The tutorial started with a simple contest scenario and successively added layers to facilitate understanding. At the end

After all sessions were conducted, one of the charities was randomly selected for all groups. Subjects were instructed about the random selection of one charity.

[30] Appendix C.1.2 shows a small negative time trend over all rounds. The results, however, are not impacted by the time trend. Specifically, the impact of the disclosure policy is very similar between the change from disclosure to concealment in round 16 and from concealment to disclosure in round 31. Additionally, the comparative statics of disclosure and concealment are not impacted by the slight time trend (see Appendix C.2.2 and Appendix C.3.2).

of the tutorial, participants had to answer comprehension questions to ensure understanding and could only proceed until they answered all of them correctly. During the tutorial and throughout section 1, participants had access to a probability calculator, where they could try out different effort and sabotage levels (see Figure 25 in Appendix B).[31] As the contest's prize was EUR 18, participants had high incentives to work through the tutorial thoroughly and were given many tools to understand the game properly.

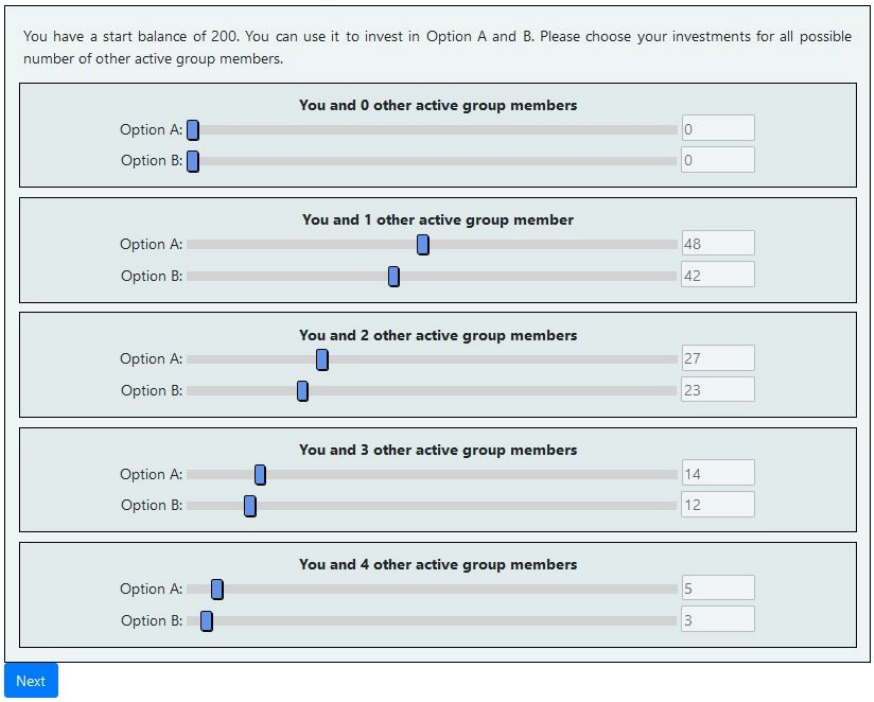


Figure 6: Effort and sabotage elicitation under group size disclosure (Part A & Part C)

After the tutorial, participants started with Part A. Figure 6 depicts the elicitation procedure of Part A. In each round of Part A, subjects received an endowment of 200 points[32] and could use this to invest in effort ('Option A') and sabotage ('Option B').[33] They were asked for their choices for all possible realized group sizes prior to their realization.

[31] Participants could enter their own levels of effort and sabotage and do the same for all other active participants. In the simplified version, the calculator assumed all others to make the same decision. Subjects could switch to the advanced version, where they could indicate different choices for every other active participant. The probability calculator then dynamically showed them their winning probabilities for all possible group size realizations with dynamic pie charts. Additionally, the donations for the specific group sizes were shown, as well as their payoffs conditional on winning or losing.

[32] I used an experimental currency called 'points' with an exchange rate of 100 points = EUR 9.

[33] Using a chosen effort and sabotage design goes in line with (e.g. Harbring & Irlenbusch 2011, 2008) and allows me to more cleanly test the theoretical predictions. For instance, effort provision in real-effort tasks seems to be insensitive to monetary incentives (Erkal et al. 2018).

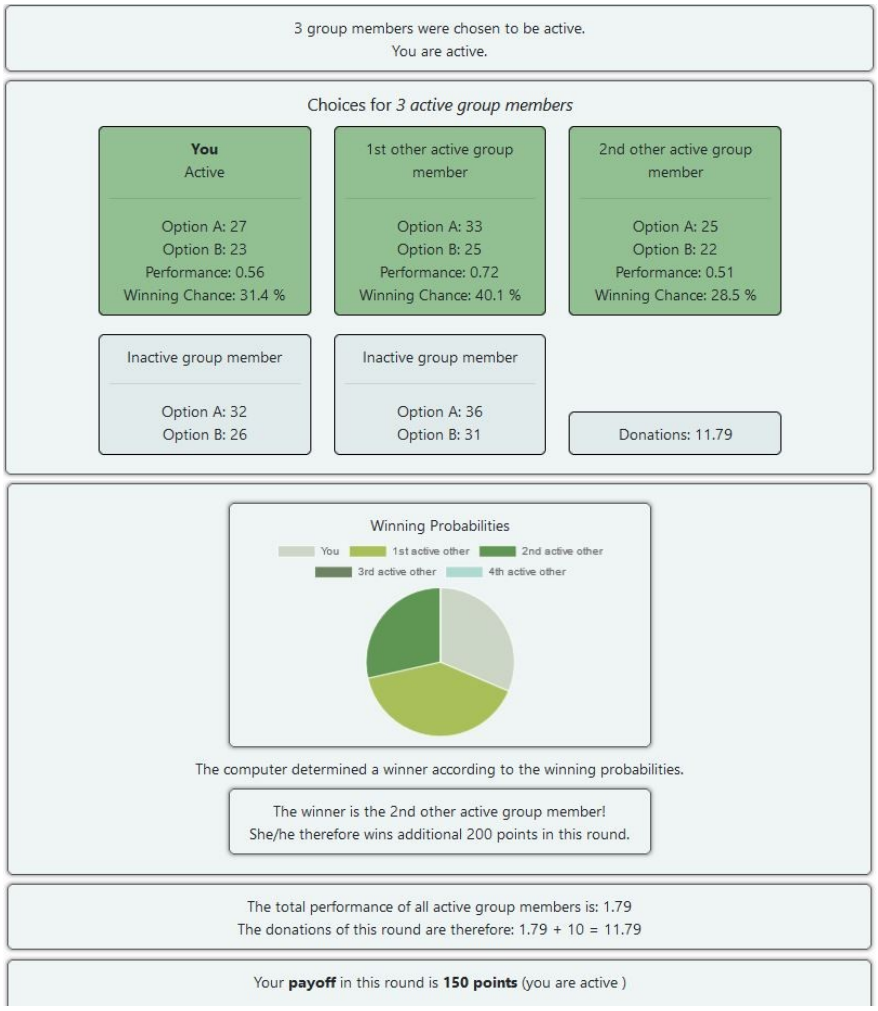


Figure 7: Contest realization and feedback after effort and sabotage elicitation

After all group members made their choices, the contest was realized as follows (see Figure 7): First, the computer decided who became active according to the enter probabilities.[34] After that, the computer calculated their performances and winning probabilities with the choices for the specific realized group size. It then randomly determined a winner according to the winning probabilities and calculated the donations. Then, participants received feedback about all other group members' effort and sabotage levels (including from the inactive group members) as well as the performances, winning probabilities, the winner, and the group's donations. The identities of their other group members were not disclosed in the feedback, as they were called either 'other active player' or 'other non-active player' in a randomized order.[35] Additionally, the computer

[34]If none of the participants were chosen to become active, the computer decided for everyone anew. This procedure does not influence the relevant group size probabilities conditional on being active.

[35]Including all (active and inactive) group members' effort and sabotage levels in the feedback minimizes learning

calculated and showed the individual payment of the round.[36] Then the next round began.

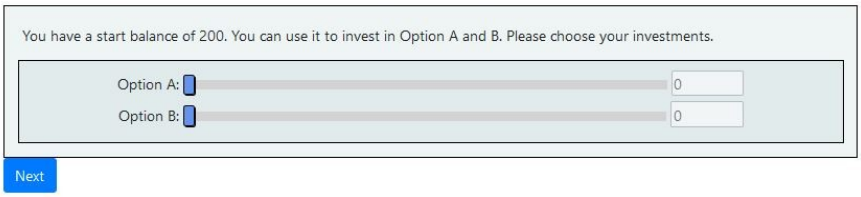


Figure 8: Effort and sabotage elicitation under group size concealment (Part B).

After finishing all 15 rounds of Part A, participants received short instructions for Part B, and went through 15 rounds of Part B. In the instructions of Part B, I communicated the group size probabilities conditional on participation instead of enter probabilities for better understanding, following Boosey et al. (2017). Participants could access these probabilities throughout the whole Part B (see Figure 24 in Appendix B). In Part B, participants had to indicate one effort and one sabotage decision prior to the group size realization (see Figure 8). This one decision each was then taken for any group size realization. The contest realization and the feedback were the same as in Part A, with the only difference that the one effort and sabotage levels were taken for any number of active contestants. After finishing Part B, participants completed 5 additional rounds of group size disclosure in Part C.

In section 2, I used the 6-item primary scale of the SVO Slider Task (Murphy & Ackermann 2014, Murphy et al. 2011) to elicit prosocial preferences (see Table 14). The choices result in a continuous measure, the SVO-angle, which ranges from $-16.26°$ to $61.39°$. It represents a participant's prosociality, where a higher angle represents a higher prosociality. I additionally included the 3 items of the spite task to elicit spiteful preferences also used by Mill & Stäbler (2023), Mill & Morgan (2022*b*,*a*), Kirchkamp & Mill (2021). The spite score is calculated by dividing the destroyed points relative to the maximally possible points and hence ranges between 0 and 1. One of the 9 items was randomly determined for payment. Afterwards, I elicited risk aversion, loss aversion, and ambiguity aversion using a lottery list similar to the methods used by Holt & Laury (2002)

effect differences between the treatments. Otherwise, as enter probabilities are different across the treatments, there would be more feedback in the *5H*, and *3H* treatments compared to the *5L*, and *3L* treatments. Additionally, the display of the order of the group members was randomized such that it was more difficult to identify another participant's dynamic decisions.

[36] If a player was chosen to be active, all costs for sabotage and effort were deducted from the endowment. If this active player won, the prize was added to the payment. Inactive players received the endowment and costs for the stated investments were not deducted.

and Sutter et al. (2013), following Boosey et al. (2017) (see tables 15, 16, and 17).[37] The risk and loss aversion lists were presented in a random order, ambiguity aversion was always in third place because its elicitation builds on the risk aversion list. One row of one of the lists was chosen randomly for payment. At the very end, participants answered a questionnaire to elicit standard demographics that included age, gender, highest degree, the field of study, and a self-report of how concentrated they were and how well they understood the experiment.

The experiment was conducted online using the subject pool from the Mannheim Laboratory for Experimental Economics. Sessions were organized through Zoom meetings, where the experimenter welcomed the participants and distributed individual participation links to the software. Subjects could not turn on their microphones or videos and also could not chat with each other in the Zoom meetings. Additionally, the experimenter ensured anonymity by removing the subjects' names when admitting them from the waiting room.

This online setting has several advantages. First, it ensures anonymity, and thus decreases reputational concerns, which may be especially important for sabotage decisions with negative externalities on donations. Second, it excludes social ties and peer effects, as subjects do not know, who else participates in the session. Lastly, relying on the university's subject pool may increase motivation, concentration, and accuracy in the decision-making process compared to other online samples.

4 Results

In this section, I present the results of the experiment. I conducted the experiment online with the subject pool of the Mannheim Laboratory for Experimental Economics (mLab). Subjects were recruited via ORSEE (Greiner 2015), the experiment was programmed in oTree (Chen et al. 2016), and the online sessions were implemented with Heroku servers. Overall, 196 subjects participated in the experiment.[38] The average duration was about 80 minutes and the average payoff was EUR

[37] In each list, participants chose between a gambling lottery and a certain amount of money. Risk aversion and loss aversion are constructed with the row number, where participants switched between the gamble and the certain amount. Ambiguity aversion is constructed by taking the difference from the row number where participants switched in the risk list and the ambiguity list.

[38] I excluded one participant who dropped out due to internet problems, in accordance with the preregistration, which indicated the exclusion of subjects, who leave early or have continuous technical problems. Hence, I analyze the behavior of 195 subjects.

21.50 (min = EUR 10.56, max = EUR 32.25). The average donations per group amounted to EUR 2.83. The mean age was 23.6 years and 50% of the subjects were female.

Throughout the results section, I rely on non-parametric Wilcoxon signed-rank tests for within-subjects comparisons and on non-parametric Mann–Whitney U tests for between-subjects comparisons. The unit of analysis is the fixed groups. As everyone makes their effort and sabotage decisions conditional on being active, but prior to knowing whether they become active or not, I analyze all effort and sabotage decisions of all the participants in each round, including those who were not chosen to become active in a specific round.

I start with the main results about the differences between the disclosure policies in Section 4.1, and subsequently also show the comparative statics with respect to realized group sizes m under disclosure, and with respect to the number of potential contestants n and their enter probabilities q under group size concealment in Section 4.2.

4.1 Comparing Disclosure Policies

In this section, I compare the effects of disclosing the number of contestants compared to concealment. In Section 4.1.1, I find no differences in average effort, sabotage, and expected payoffs between the disclosure policies. Subsequently, in Section 4.1.2, I find that the sum of individual performances (group performance) is higher under concealment, provided that the probability of being alone is at least 6%. Given that the sum of individual performances reflects the amount of value that is induced by the contest, the designer prefers concealing the number of contestants in this case.

To compare the choices of the two policies, I compute the average expected values based on the elicited values. For this, I take the weighted sum of all elicited values over all possible group size realizations (and combinations of opponents) weighted by their probabilities. In this way, I use all the elicited choices of every player in each round. As in the theory part, I do this conditional on at least one player being active. The results thus show the average expected effort, sabotage, received sabotage, payoffs, and the resulting expected group performance from a player's view conditional on being active. All results can be replicated by focusing on the actually implemented choices (see Appendix C.1.5).[39]

[39]The realized values do not rely on all elicited values, but only on the random draw of active contestants in each

Furthermore, there are slight time trends in the expected effort and sabotage levels, as well as in the expected group performance (see Appendix C.1.2). Therefore, as robustness checks, first, I analyze only 5 rounds each around the changes of the disclosure policy (i.e., rounds 10-20 and 25-30) to focus on the induced differences.[40] Second, I run regressions that include the pre-registered controls.[41]

4.1.1 Effort, Sabotage, and Expected Payoff

Figure 9 shows the differences in the average expected effort, sabotage, and average expected payoff between the disclosure policies pooled over all treatments. As theory predicts (see Hypothesis 2.1.), I do not find any significant difference in the average expected effort and sabotage levels across the two disclosure policies. Even though there is a marginally significant ($p < 0.1$) increase in effort under disclosure, this difference is not robust to either focusing on the subset of rounds around the change or the regression analysis, which among other variables, controls for the time trend (see Appendix C.1.3).[42] Moreover, I also do not find any significant difference in the expected individual payoffs, as predicted (see Hypothesis 2.2).[43] All robustness checks do not find any significant difference (see Appendix C.1.3).

The insensitivity of the exerted effort, sabotage, and the resulting expected payoffs towards the disclosure policy does not depend on the specific setting, as I do not find differences in effort, sabotage, or expected payoffs between the disclosure rules in any of the treatments individually (see Appendix C.1.1). The only exceptions are sabotage levels in treatment *3L*, which are slightly higher under concealment ($p < 0.05$ in the robustness checks, otherwise $p < 0.1$). Summarizing, I find the following:

Result 1.1. *Concealing the number of contestants does not significantly change average expected*

round and their elicited values for the randomly realized group size and therefore add noise in each round.

[40] I did not specify this robustness check in the pre-analysis. However, it is consistent with analyzing this subset of rounds around the policy changes and controlling for time effects.

[41] The controls are: Being active in the round before, having won in the round before, average sabotage and effort levels of other participants in the rounds before, round, the treatments, realized group size in the round before, how often won in the rounds before, SVO, spite, risk, loss and ambiguity aversion, age, gender, highest degree, the field of study, the degree of concentration and understanding.

[42] Instead, the regression analysis shows a significant ($p < 0.05$) positive increase in sabotage under concealment. However, a Cohen's D of -0.09 for sabotage shows that even if there are significant differences between the disclosure policies, this difference can not be considered to be very substantive. Additionally, in no other robustness check do I find this significant increase.

[43] The expected payoffs exclude the 200-point endowment in each row and thus represent the expected payoff from the contest.

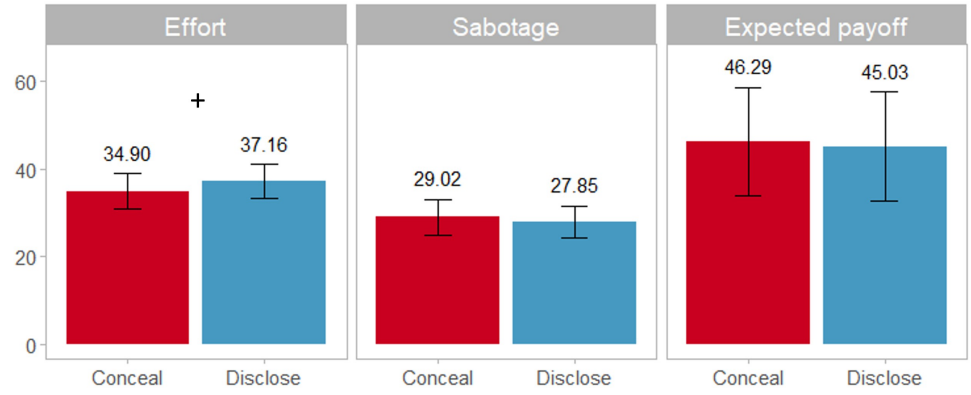


Figure 9: The figure shows average effort, sabotage, and expected payoffs conditional on the disclosure policy, pooled over all treatments. Error bars show 95% confidence intervals. Significance levels: $+$ $p < 0.10$

effort and sabotage levels, except when the probability of being alone is high (52%, treatment 3L), concealment leads to higher sabotage levels.

Result 1.2. *The average expected payoff does not significantly differ between concealing and disclosing the number of contestants.*

4.1.2 Group Performance

So far, I showed that subjects do not significantly change their average effort and sabotage levels between the disclosure policies and, as a consequence, their expected payoffs do not differ. Hence, they are ex-ante indifferent towards the chosen disclosure policy. From a productivity point of view, theory predicts that concealment leads to a higher sum of individual performances and hence to a relatively higher productivity compared to disclosure. In this section, I study whether the experiment shows that group performances are indeed higher under concealment.

Figure 10 overall illustrates how the group performance is shaped, pooled over all treatments.[44] The group performance is defined as: $\sum_{i \in M} y_i = \sum_{i \in M} \frac{e_i}{1+\sum_{j \in M_i} s_j}$, and rows 1 and 2 show the average effort and sabotage levels (e_i and s_i), row 3 the average received sabotage ($\sum_{j \in M_i} s_j$), and row 4 the average group performance. The panels on the LHS show these values depending on the realized group size, whereas the panels on the RHS show the weighted averages.

[44]The data for the realized group size of $m = 4$ and $m = 5$ come from treatments *5L* and *5H* only. Comparative statics look relatively similar across treatments.

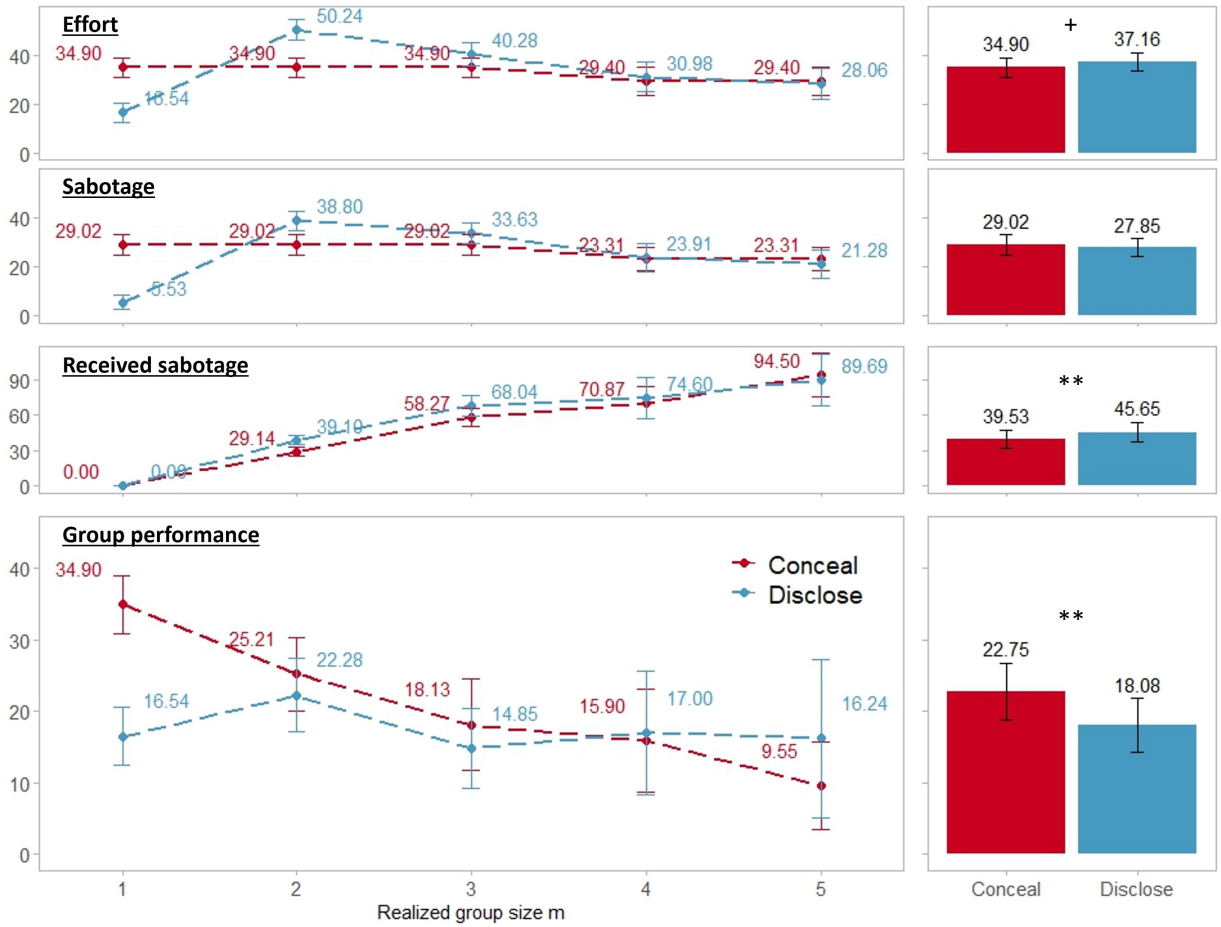


Figure 10: The right panels show differences between the two disclosure policies in the expected effort and sabotage levels, the received sabotage, and the resulting group performance. The left panels show them conditional on the realized group size. Error bars show 95% confidence intervals. Significance levels: $+$ $p < 0.1$, $*$ $p < 0.05$, $**$ $p < 0.01$, $***$ $p < 0.001$

The bar chart on the RHS of row 4 shows that concealing the number of contestants significantly increases group performance ($p < 0.01$). This result can be replicated in both robustness checks ($p < 0.05$ and $p < 0.1$) (see Appendix C.1.4). The panel on the LHS of row 4 depicts those differences depending on the realized group size m. It shows that the group performance difference between the disclosure policies is primarily driven by the case when contestants do not face any competitors ($m = 1$). This goes in line with theory because, under concealment, subjects have to choose one effort level without knowing the group size and hence exert a large amount of effort even if they end up being the only contestant and win the contest with certainty. If they know that they are the only contestant, they exert much less effort.[45] The important factor that induces

[45]They still provide non-zero effort because effort creates value and increases the donations in the experiment.

performance differences is the combination of the exerted effort and the received sabotage depending on the realized group size. Specifically, when a contestant does not face any competitor, she is not subjected to any sabotage, simply because there are no others who sabotage her. Because of this, the substantive effort difference between the disclosure rules when alone ($m = 1$), translates directly into a large performance difference. For all other realized group sizes, contestants do receive sabotage and hence, even if there are effort differences, the resulting group performances are not significantly different because of the sabotage that they receive from each other. Specifically, for realized group sizes of $m = 2$ and $m = 3$, subjects exert significantly ($p < 0.01$) higher effort under disclosure (row 1), yet, also receive significantly ($p < 0.01$) higher levels of sabotage (row 3), leading to not significantly different group performances. For realized group sizes of $m = 4$ and $m = 5$, there are no significant differences in the exerted effort, received sabotage, and resulting group performances between the disclosure policies.[46]

Even though I find a significant difference between the disclosure policies, the difference is not as pronounced as predicted. There are two reasons for this. First, subjects on average provide a substantial amount of effort, even when they know that they are the only contestant. This is because effort is constructive and increases the donations. Second, for all other realized group sizes ($m > 2$), group performances are overall much higher than predicted under both policies (see Figure 2), reducing the effect of the difference for a realized group size of one. This is because of substantial heterogeneity in the exerted effort and sabotage between group members. The group member, who exerts the most effort, on average receives the least sabotage and thus maintains a higher performance (see Appendix C.1.6). Despite these heterogeneities, I still find the predicted increase in group performance under concealment.

Next, I test my theoretical prediction (see Hypothesis 3.1. and Hypothesis 3.2.) that concealment leads to higher group performances only when the probability of being the only contestant is roughly larger than 1% (56% in treatment *3L*, 32% in *5L*, and 6% in *3H* opposed to 0.4% in treatment *5H*). Figure 11 depicts the average group performance per treatment conditional on the disclosure policy in comparison to the Nash equilibrium predictions (dashed black lines). It shows a significant increase in the group performance under concealment for *3L* ($p < 0.05$) and *3H* ($p < 0.01$) and a non-significant increase for *5L*. For *5H*, there is no significant difference between

[46] Only the group performance difference for a realized group size of 5 is marginally significant ($p = 0.09$).

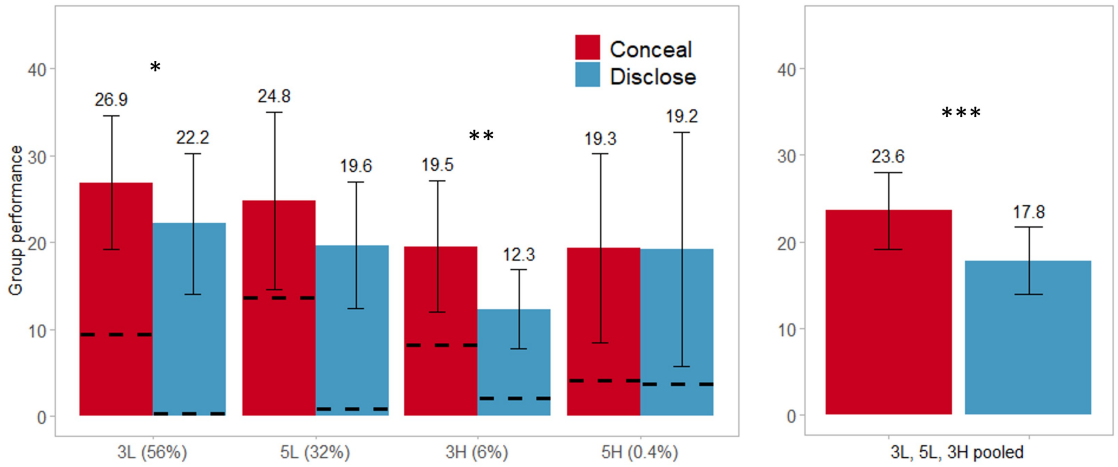


Figure 11: The bar charts show the average group performance conditional on the disclosure policy and on treatments. Percentages in parentheses show the probability of being the only contestant in each of the treatments. Black dashed lines show the Nash equilibrium predictions. The error bars show 95% confidence intervals. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

the disclosure policies, as predicted. Moreover, because theory predicts an increase for treatments *3L*, *5L*, and *3H*, I pool them and find a significant increase in group performances under concealment ($p < 0.001$). This difference is quite substantial with an increase under concealment compared to disclosure of around 30%. The robustness checks (see Appendix C.1.4) confirm the significant differences in all cases but for treatment *3H*, where I do not find any significant differences, yet the increase is qualitatively replicated.

Result 2.1. *When the probability of being alone in the contest is at least 6% (3H, 3L, 5L), concealment leads to higher group performance.*

Result 2.2. *When the probability of being alone in the contest is 0.4% (5H), I do not find any differences between the disclosure policies.*

4.2 Comparative Statics under Disclosure and Concealment

In this section, I study how different known group sizes m influence effort and sabotage levels in Section 4.2.1 and how the number of potential contestants n and their enter probabilities influence effort and sabotage levels under group size uncertainty in Section 4.2.2.

4.2.1 Known Group Sizes (Group Size Disclosure)

Figure 12 depicts mean effort and sabotage levels under group size disclosure conditional on the realized group size compared to the Nash equilibrium predictions. Averages are computed over all rounds (of Part A and Part C) and pooled over all treatments. The figure suggests that effort levels follow very closely the equilibrium predictions. Specifically, I find a significant decrease in effort from a realized group size of 2 to 5 ($p < 0.001$). This decrease in effort is in line with the experimental contest literature without sabotage (Anderson & Stafford 2003, Sheremeta 2011, Morgan et al. 2012, Aycinena & Rentschler 2019).

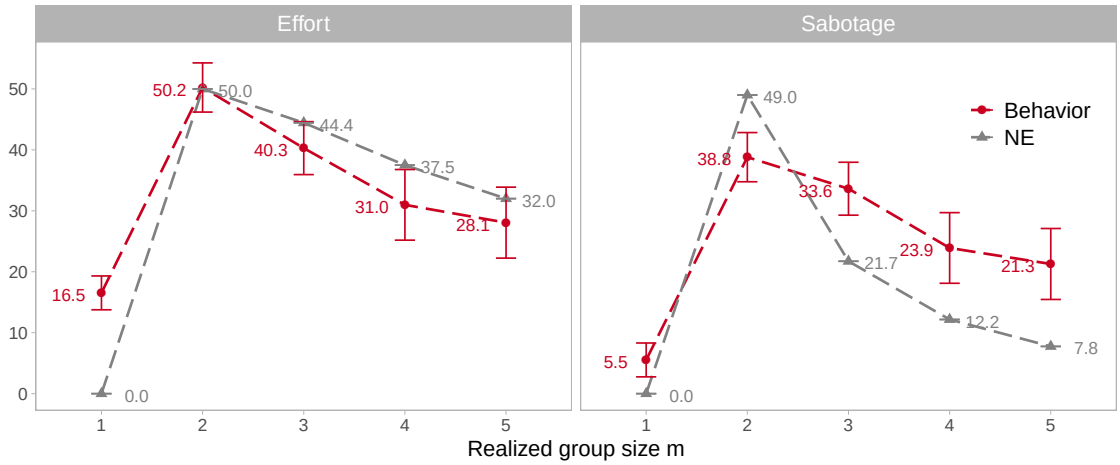


Figure 12: Mean effort and sabotage levels under group size disclosure as a function of the realized group size. Yellow lines show the equilibrium predictions. Red lines show the elicited behavior of the experiment. The error bars show 95% confidence intervals.

Furthermore, I find that sabotage also decreases significantly from $m = 2$ to $m = 5$ ($p < 0.001$), as predicted by theory. A regression analysis reveals a significant ($p < 0.001$) negative effect of a realized group size on effort and sabotage for group sizes $m > 1$ (see Appendix C.2.3).

The decrease in sabotage is slightly less steep than predicted, leading to over-sabotage for larger group sizes. In particular, sabotage levels are significantly ($p < 0.001$) below the Nash equilibrium for a group size of 2, and significantly ($p < 0.001$) above for the group sizes of 3, 4, and 5. In the experimental contest literature, it is common that subjects over-invest in effort compared to the Nash equilibrium (Sheremeta 2018, Dechenaux et al. 2015, Sheremeta 2013).[47] As the total

[47]There is no overbidding in effort and no joint overbidding when aggregating both effort and sabotage levels. The sum of effort and sabotage levels is not significantly higher than the sum of the Nash equilibrium predictions. $m = 3$: $p = 0.078$, $m = 4 : p = 0.430$, $m = 5 : p = 0.114$

amount of exerted sabotage is added up over the number of active contestants, this over-sabotage is particularly harmful, as it leads to more destroyed value for larger realized group sizes.

All results remain robust when analyzing different pre-registered sets of subrounds (see Appendix C.2.1) and when running a regression analysis (see Appendix C.2.3).[48] To summarize, I find the following:

Result 3.1. *An increase in the group size (for $m > 1$) decreases effort and sabotage levels.*

Result 3.2. *There are above-equilibrium sabotage levels for realized group sizes larger than* 2.

I propose two concepts that can explain the increases in over-sabotage in the group size. First, a modified version of joy of winning – constant winning aspiration – postulates that joy of winning increases linearly with group size Boosey et al. (2017). Hence, subjects experience greater joy, when they win against more competitors, which makes them over-invest for larger group sizes. This cannot explain, however, why there exists over-sabotage but not an overexertion of effort.

Second, if subjects have spiteful preferences (e.g. Levine 1998, Morgan et al. 2003), they receive additional utility for harming others. As sabotage's harm increases in the number of competitors, spite's utility gains also increase in the number of competitors and thus can explain the above-equilibrium sabotage levels for larger group sizes. This explanation is supported by the significant positive correlation between spiteful preferences and sabotage (see Table 10 in Appendix C.2.3).

4.2.2 Group Size Uncertainty

I now turn to the case of group size uncertainty. Figure 13 depicts average effort and sabotage levels conditional on the enter probability (high vs. low) and the number of potential contestants. For high entering probabilities ($q = 0.75$), I find a significant ($p < 0.01$) decrease in effort and sabotage levels when the number of potential contestants increases from 3 to 5, as predicted. This part of the comparative statics is therefore supported by the evidence. In the case of low enter probabilities ($q = 0.25$), however, I do not find the predicted increase in effort and sabotage levels. Instead, I find a slight (non-significant) decrease in effort and sabotage. This deviation from theory does not come from distortions related to the uncertainty of the group size, as subjects take the weighted average of their known group size choices (see Section 4.1.1). This result complements

[48] Additionally, the comparative statics remain stable over time (see Appendix C.2.2).

the results of Boosey et al. (2017) who find a significant increase in effort levels for an increasing group size for low entering probabilities but no significant increase for high entering probabilities.

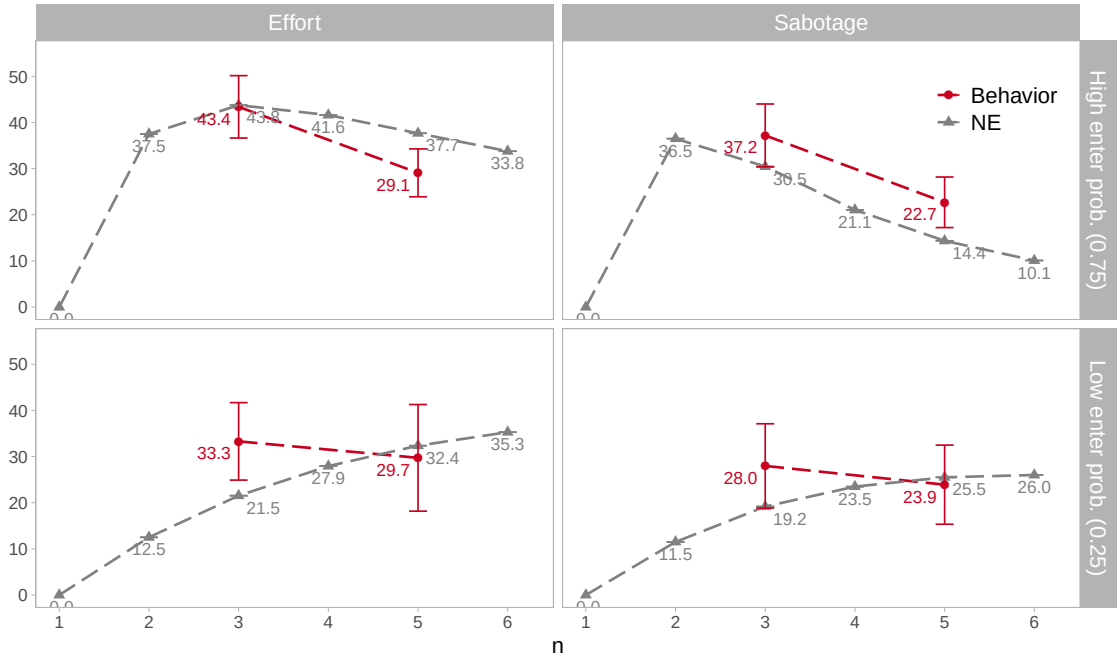


Figure 13: Mean effort (left panels) and sabotage (right panels) decisions under group size uncertainty (Part B) for high (upper panels) and low (lower panels) enter probabilities. The x-axes show the number of potential contestants n. Yellow lines show the equilibrium predictions and red lines the elicited choices. Error bars show 95% confidence intervals.

The results remain robust to analyzing only the pre-registered subrounds and running a regression analysis with the pre-registered controls (see Appendix C.3.1 and Appendix C.3.3).[49] Summarizing, I find:

Result 4.1. *For high enter probabilities ($q = 0.75$), an increase in the group size (from $n = 3$ to $n = 5$) decreases effort and sabotage levels.*

Result 4.2. *For low enter probabilities ($q = 0.25$), an increase in the group size (from $n = 3$ to $n = 5$) does not significantly change effort and sabotage levels.*

[49]Specifically, I find a significant decrease (at least $p < 0.05$) in effort and sabotage levels from $3H$ to $5H$ for the subset of rounds 1-7 and rounds 8-15 and in the regression analysis. For the single round 1, I only qualitatively replicate the decrease ($p = 0.1994$). I do not find any significant differences between the treatments 3L and 5L in any of the robustness checks. Moreover, I find a slight decrease in effort and sabotage over time, however, the comparative statics remain stable (see Appendix C.3.2).

5 Discussion and Conclusion

In this paper, I provide a theoretical and experimental analysis of how disclosing the number of contestants affects effort and sabotage levels compared to concealing this information. Since contests are often used to increase the productivity of workers or companies, I compare the resulting differences in the contest-induced productivity. To do so, I compare the resulting sum of individual performances (group performance) because they incorporate the productive effects of effort on their own performance and the destructive effects of sabotage on others' performances.

I model sabotage in a Tullock contest with exogenous enter probabilities, where the designer commits to either always conceal or disclose the number of contestants. According to the theoretical analysis, this decision should not affect average effort, sabotage, or expected payoffs. The choice of the disclosure policy, however, leads to a higher resulting group performance under concealment compared to disclosure – provided that the probability of being the only contestant is not too low.

I run an experiment to test these theoretical predictions. In line with theory, the first key result is that a designer can increase the sum of individual performances, by concealing the number of contestants. However, this only works if the probability of being the only contestant is not too low. This is because, as predicted, the difference in group performance is driven by substantially higher performances under concealment when contestants do not face any competitors. For all other realized group sizes, I find no significant difference in the resulting group performance. Therefore, I do not find any difference in group performances, when the probability of being alone is negligible (0.4%) but otherwise (at least 6%), I find that concealment leads to higher group performances.

The second key result is that contestants are ex-ante indifferent between the two disclosure policies.[50] This is because I do not find any significant difference in average effort, sabotage, or expected payoffs between the disclosure policies, as predicted. The only exception is when the probability of being the only contestant is high (56%), where I find that concealment leads to a slight increase in sabotage. In all other cases, I find that under concealment, choices are the weighted sum of subjects' choices under disclosure.

These theoretical and experimental results demonstrate that it is important to consider sabotage when comparing group size disclosure policies, as omitting the possibility of sabotage can lead to

[50] If they do not have preferences over the produced value. Otherwise, they would prefer concealment when the probability of being the only contestant is not too low.

wrong conclusions. By incorporating the destructive effects of sabotage, I show that the choice of the disclosure policy matters. This is because the resulting group performance – and thus contestants' productivity – is different between concealing and disclosing the number of active players.

The practical implication for a designer is that she can induce higher performances by concealing the number of contestants without curbing their productive efforts or expected payoffs. This is a notable result because the resulting productivity can be increased without requiring contestants to exert additional effort. Rather, it enhances the productivity of the exerted effort by mitigating the destructive effects of potential sabotage. Furthermore, concealing the number of contestants is an easy-to-implement tool because not disclosing the number of contestants simply requires deliberately omitting information about the group size. Whether concealment should be implemented, however, depends on the specific setting. This is because concealment is effective only when the probability of a player being the only contestant is not too low (larger than 6%). At the same time, if this probability is too high (56%), concealment can lead to higher sabotage levels (while still leading to overall higher performances). A designer should therefore carefully counterbalance the effects of a specific setting.

As additional results, I find evidence for the comparative statics of known group sizes under group size disclosure. When contestants know the number of competitors, a higher group size decreases effort and sabotage levels. I also observe significant above-equilibrium sabotage levels for realized group sizes of 3, 4, and 5. This behavior can be explained by a modified version of joy of winning (constant winning aspirations), where the experienced joy increases in the number of outperformed competitors (Boosey et al. 2017), or by spiteful preferences (Morgan et al. 2003), where agents receive utility by harming others, and hence this utility increases in the number of harmed competitors. [51]

Observing higher-than-equilibrium sabotage and at the same time, not higher effort levels is bad news for productivity, especially for larger groups. The individually exerted over-sabotage leads to a drastic increase in the received sabotage when the number of sabotage-exerting contestants increases. Hence, more value is destroyed and individual performances diminished. As a conse-

[51] Empirically, I find an overall positive correlation of spite with sabotage, which suggests that the observed over-sabotage is, at least, partially driven by spiteful preferences. The importance of spiteful preferences adds to other literature which shows that spiteful preferences matter in competitive settings (Mill & Stäbler 2023, Mill & Morgan 2022*a*, Mill 2017).

quence, increasing the number of contestants to decrease sabotage may not be an apt tool. Instead, if a designer can set and reveal the number of competitors, my findings suggest that she should rather determine a smaller number of competitors, as less value is destroyed overall.

Another additional result is that when contestants do not know the realized number of contestants, I find that an increase in the number of potential contestants decreases effort and sabotage levels when enter probabilities are high ($q = 0.75$), as predicted. When enter probabilities are low ($q = 0.25$), however, I do not find evidence for the theoretical decrease in effort and sabotage.

The study comes with certain limitations. For instance, I abstract from any spillovers from the exerted sabotage on a baseline productivity of all potential contestants, including non-active players. If the sabotage activities also harm all sabotaged players' baseline productivities, concealment increases this harm, as subjects exert sabotage even when they are alone in the contest. How much harm is done, and which of the counteracting forces between the disclosure policies prevails, depends on the specific parametrization of the baseline productivity and the effectiveness of sabotage on the baseline productivity. For a small enough baseline productivity or small enough effectiveness of sabotage, the results of this paper still hold. Furthermore, I assume exogenous enter probabilities, whereas many times entry into a contest may be an endogenous choice. In this paper, exogenous entry probabilities can be thought of as fixing enter beliefs and significantly reducing complexity for subjects. Endogenizing enter probabilities provides an interesting avenue for future research. This is because the possibility of exerting sabotage and getting sabotaged may attract in particular spiteful and tough players, which may potentially lead to additional differences between the disclosure policies and to even more over-sabotage for larger group sizes.

Future work should study group sizes larger than five to explore whether the behavioral pattern of over-sabotage further increases. Additionally, it would be interesting to expand the disclosure policy to not only disclosing the number of contestants but also to revealing contestants' identities. While this is not always feasible due to data protection concerns, such a policy could result in different sabotage levels. Finally, as sabotage can destroy value, future work that assesses the value-inducing effects of any kind of policies should account for the possibility of such destructive behaviors. Otherwise, these assessments might lead to wrong conclusions.

Declaration of Generative AI and AI-assisted Technologies

During the preparation of this work the author used ChatGPT 3.5 in order to get assistance with a small number of language reformulations to improve readability. After using this tool/service, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

References

Amegashie, J. A. (2012), 'Productive versus destructive efforts in contests', *European Journal of Political Economy* **28**(4), 461–468.

Amegashie, J. A. et al. (2015), 'Sabotage in contests', *A companion to rent seeking and political economy* pp. 138–149.

Anderson, L. R. & Stafford, S. L. (2003), 'An experimental analysis of rent seeking under varying competitive conditions', *Public Choice* **115**(1-2), 199–216.

Antsygina, A. & Teteryatnikova, M. (2023), 'Optimal information disclosure in contests with stochastic prize valuations', *Economic Theory* **75**(3), 743–780.

Aycinena, D. & Rentschler, L. (2019), 'Entry in contests with incomplete information: Theory and experiments', *European Journal of Political Economy* **60**, 101803.

Azrieli, Y., Chambers, C. P. & Healy, P. J. (2018), 'Incentives in experiments: A theoretical analysis', *Journal of Political Economy* **126**(4), 1472–1503.

Balafoutas, L., Lindner, F. & Sutter, M. (2012), 'Sabotage in tournaments: Evidence from a natural experiment', *Kyklos* **65**(4), 425–441.

Bartling, B., Weber, R. A. & Yao, L. (2015), 'Do markets erode social responsibility?', *The Quarterly Journal of Economics* **130**(1), 219–266.

Benistant, J. & Villeval, M. C. (2019), 'Unethical behavior and group identity in contests', *Journal of Economic Psychology* **72**, 128–155.

Benndorf, V., Kübler, D. & Normann, H.-T. (2015), 'Privacy concerns, voluntary disclosure of information, and unraveling: An experiment', *European Economic Review* **75**, 43–59.

Bergemann, D. & Morris, S. (2019), 'Information design: A unified perspective', *Journal of Economic Literature* **57**(1), 44–95.

Bitkom (2018), 'Spionage, sabotage und datendiebstahl – wirtschaftsschutz in der industrie'.
URL: *September 28, 2023,* `https://www.bitkom.org/sites/default/files/file/import/181008-Bitkom-Studie-Wirtschaftsschutz-2018-NEU.pdf`

Boosey, L., Brookins, P. & Ryvkin, D. (2017), 'Contests with group size uncertainty: Experimental evidence', *Games and Economic Behavior* **105**, 212–229.

Boosey, L., Brookins, P. & Ryvkin, D. (2020), 'Information disclosure in contests with endogenous entry: An experiment', *Management Science* **66**(11), 5128–5150.

Brown, A. & Chowdhury, S. M. (2017), 'The hidden perils of affirmative action: Sabotage in handicap contests', *Journal of Economic Behavior & Organization* **133**, 273–284.

Carpenter, J., Matthews, P. H. & Schirm, J. (2010), 'Tournaments and office politics: Evidence from a real effort experiment', *American Economic Review* **100**(1), 504–17.

Charness, G., Masclet, D. & Villeval, M. C. (2014), 'The dark side of competition for status', *Management Science* **60**(1), 38–55.

Chen, B., Galariotis, E., Ma, L., Wang, Z. & Zhu, Z. (2023), 'On disclosure of participation in innovation contests: a dominance result', *Annals of Operations Research* **328**, 1615–1629.

Chen, B., Jiang, X. & Knyazev, D. (2017), 'On disclosure policies in all-pay auctions with stochastic entry', *Journal of Mathematical Economics* **70**, 66–73.

Chen, B., Ma, L., Zhu, Z. & Zhou, Y. (2020), 'Disclosure policies in all-pay auctions with bid caps and stochastic entry', *Economics Letters* **186**, 108805.

Chen, D. L., Schonger, M. & Wickens, C. (2016), 'otree—an open-source platform for laboratory, online, and field experiments', *Journal of Behavioral and Experimental Finance* **9**, 88–97.

Chen, K.-P. (2003), 'Sabotage in promotion tournaments', *Journal of Law, Economics, and Organization* **19**(1), 119–140.

Chowdhury, S., Danilov, A. & Kocher, M. (2022), Affirmative action policies and sabotage behavior in promotional tournaments: An experiment, Technical report, Working Paper.

Chowdhury, S. M., Esteve-González, P. & Mukherjee, A. (2023), 'Heterogeneity, leveling the playing field, and affirmative action in contests', *Southern Economic Journal* **89**(3), 924–974.

Chowdhury, S. M. & Gürtler, O. (2015), 'Sabotage in contests: a survey', *Public Choice* **164**(1-2), 135–155.

Dato, S. & Nieken, P. (2014), 'Gender differences in competition and sabotage', *Journal of Economic Behavior & Organization* **100**, 64–80.

Dato, S. & Nieken, P. (2020), 'Gender differences in sabotage: the role of uncertainty and beliefs', *Experimental Economics* **23**(2), 353–391.

Dechenaux, E., Kovenock, D. & Sheremeta, R. M. (2015), 'A survey of experimental research on contests, all-pay auctions and tournaments', *Experimental Economics* **18**(4), 609–669.

Del Corral, J., Prieto-Rodriguez, J. & Simmons, R. (2010), 'The effect of incentives on sabotage: The case of spanish football', *Journal of Sports Economics* **11**(3), 243–260.

Deutscher, C., Frick, B., Gürtler, O. & Prinz, J. (2013), 'Sabotage in tournaments with heterogeneous contestants: Empirical evidence from the soccer pitch', *The Scandinavian Journal of Economics* **115**(4), 1138–1157.

Du, N. & Shahriar, Q. (2024), 'Information disclosure in mitigating moral hazard: An experimental investigation', *Games and Economic Behavior* **144**, 284–299.

Erkal, N., Gangadharan, L. & Koh, B. H. (2018), 'Monetary and non-monetary incentives in real-effort tournaments', *European Economic Review* **101**, 528–545.

Evans, J. M., Hendron, M. G. & Oldroyd, J. B. (2015), 'Withholding the ace: The individual-and unit-level performance effects of self-reported and perceived knowledge hoarding', *Organization Science* **26**(2), 494–510.

Faravelli, M., Friesen, L. & Gangadharan, L. (2015), 'Selection, tournaments, and dishonesty', *Journal of Economic Behavior & Organization* **110**, 160–175.

Feng, X. & Lu, J. (2016), 'The optimal disclosure policy in contests with stochastic entry: A bayesian persuasion perspective', *Economics Letters* **147**, 103–107.

Ford, D. P. & Staples, S. (2010), 'Are full and partial knowledge sharing the same?', *Journal of knowledge management* **14**(3), 394–409.

Fu, Q., Jiao, Q. & Lu, J. (2011), 'On disclosure policy in contests with stochastic entry', *Public Choice* **148**(3-4), 419–434.

Fu, Q., Jiao, Q. & Lu, J. (2015), 'Contests with endogenous entry', *International Journal of Game Theory* **44**(2), 387–424.

Fu, Q., Lu, J. & Zhang, J. (2016), 'Disclosure policy in tullock contests with asymmetric stochastic entry', *Canadian Journal of Economics/Revue canadienne d'économique* **49**(1), 52–75.

Fu, Q. & Wu, Z. (2022), 'Disclosure and favoritism in sequential elimination contests', *American Economic Journal: Microeconomics* **14**(4), 78–121.

Greiner, B. (2015), 'Subject pool recruitment procedures: organizing experiments with orsee', *Journal of the Economic Science Association* **1**(1), 114–125.

Gu, Y., Hehenkamp, B. & Leininger, W. (2019), 'Evolutionary equilibrium in contests with stochastic participation: Entry, effort and overdissipation', *Journal of Economic Behavior & Organization* **164**, 469–485.

Gürtler, O. & Münster, J. (2010), 'Sabotage in dynamic tournaments', *Journal of Mathematical Economics* **46**(2), 179–190.

Gürtler, O., Münster, J. & Nieken, P. (2013), 'Information policy in tournaments with sabotage', *The Scandinavian Journal of Economics* **115**(3), 932–966.

Harbring, C. & Irlenbusch, B. (2005), 'Incentives in tournaments with endogenous prize selection', *Journal of Institutional and Theoretical Economics (JITE)/Zeitschrift für die gesamte Staatswissenschaft* pp. 636–663.

Harbring, C. & Irlenbusch, B. (2008), 'How many winners are good to have?: On tournaments with sabotage', *Journal of Economic Behavior & Organization* **65**(3-4), 682–702.

Harbring, C. & Irlenbusch, B. (2011), 'Sabotage in tournaments: Evidence from a laboratory experiment', *Management Science* **57**(4), 611–627.

Harbring, C., Irlenbusch, B., Kräkel, M. & Selten, R. (2007), 'Sabotage in corporate contests–an experimental analysis', *International Journal of the Economics of Business* **14**(3), 367–392.

Higgins, R. S., Shughart, W. F. & Tollison, R. D. (1988), *Free Entry and Efficient Rent-Seeking*, Springer US, Boston, MA, pp. 127–139.

Holt, C. A. & Laury, S. K. (2002), 'Risk aversion and incentive effects', *American economic review* **92**(5), 1644–1655.

Huck, S., Kajackaite, A. & Szech, N. (2021), 'Honesty and moral behavior in economic games', *Frontiers in Psychology* **12**, 769856.

Jiao, Q., Ke, C. & Liu, Y. (2022), 'When to disclose the number of contestants: Theory and experimental evidence', *Journal of Economic Behavior & Organization* **193**, 146–160.

Kahana, N. & Klunover, D. (2015), 'A note on poisson contests', *Public Choice* **165**(1-2), 97–102.

Kahana, N. & Klunover, D. (2016), 'Complete rent dissipation when the number of rent seekers is uncertain', *Economics Letters* **141**, 8–10.

Kirchkamp, O. & Mill, W. (2021), 'Spite vs. risk: Explaining overbidding in the second-price all-pay auction: A theoretical and experimental investigation', *Games and Economic Behavior* **130**, 616–635.

Konrad, K. A. (2000), 'Sabotage in rent-seeking contests', *Journal of Law, Economics, and Organization* **16**(1), 155–165.

Kräkel, M. (2005), 'Helping and sabotaging in tournaments', *International Game Theory Review* **7**(02), 211–228.

Kumar Jha, J. & Varkkey, B. (2018), 'Are you a cistern or a channel? exploring factors triggering knowledge-hiding behavior at the workplace: evidence from the indian r&d professionals', *Journal of Knowledge Management* **22**(4), 824–849.

Lau, R. R. & Rovner, I. B. (2009), 'Negative campaigning', *Annual review of political science* **12**, 285–306.

Lazear, E. P. (1989), 'Pay equality and industrial politics', *Journal of Political Economy* **97**(3), 561–580.

Leibbrandt, A., Wang, L. C. & Foo, C. (2017), 'Gender quotas, competitions, and peer review: Experimental evidence on the backlash against women', *Management Science* **64**(8), 3501–3516.

Levine, D. K. (1998), 'Modeling altruism and spitefulness in experiments', *Review of economic dynamics* **1**(3), 593–622.

Lim, W. & Matros, A. (2009), 'Contests with a stochastic number of players', *Games and Economic Behavior* **67**(2), 584–597.

Mariotti, T., Schweizer, N., Szech, N. & von Wangenheim, J. (2023), 'Information nudges and self-control', *Management Science* **69**(4), 2182–2197.

Mathevet, L., Perego, J. & Taneva, I. (2020), 'On information design in games', *Journal of Political Economy* **128**(4), 1370–1404.

Mill, W. (2017), 'The spite motive in third price auctions', *Economics Letters* **161**, 71–73.

Mill, W. & Morgan, J. (2022*a*), 'Competition between friends and foes', *European Economic Review* **147**, 104171.

Mill, W. & Morgan, J. (2022*b*), 'The cost of a divided america: an experimental study into destructive behavior', *Experimental Economics* **25**, 974–1001.

Mill, W. & Stäbler, J. (2023), 'Spite in litigation', *CESifo Working Paper No. 10290,* .

Morgan, J., Orzen, H. & Sefton, M. (2012), 'Endogenous entry in contests', *Economic Theory* **51**(2), 435–463.

Morgan, J., Steiglitz, K. & Reis, G. (2003), 'The spite motive and equilibrium behavior in auctions', *Contributions in Economic Analysis & Policy* **2**(1), 1–25.

Münster, J. (2006), 'Contests with an unknown number of contestants', *Public Choice* **129**(3-4), 353–368.

Münster, J. (2007), 'Selection tournaments, sabotage, and participation', *Journal of Economics & Management Strategy* **16**(4), 943–970.

Murphy, R. O. & Ackermann, K. A. (2014), 'Social value orientation: Theoretical and measurement issues in the study of social preferences', *Personality and Social Psychology Review* **18**(1), 13–41.

Murphy, R. O., Ackermann, K. A. & Handgraaf, M. (2011), 'Measuring social value orientation', *Judgment and Decision making* **6**(8), 771–781.

Myerson, R. B. & Wärneryd, K. (2006), 'Population uncertainty in contests', *Economic Theory* **27**(2), 469–474.

Nissen, M. & Haugsted, F. (2020), 'Badmouthing your competitor's products: When does denigration become an antitrust issue?'.
URL: *September 28, 2023,* `https://awards.concurrences.com/IMG/pdf/feb20_nissen_2_13f.pdf?67369`

Pan, W., Zhang, Q., Teo, T. S. & Lim, V. K. (2018), 'The dark triad and knowledge hiding', *International Journal of Information Management* **42**, 36–48.

Piest, S. & Schreck, P. (2021), 'Contests and unethical behavior in organizations: a review and synthesis of the empirical literature', *Management Review Quarterly* **71**(4), 679–721.

Rothenhäusler, D., Schweizer, N. & Szech, N. (2018), 'Guilt in voting and public good games', *European Economic Review* **101**, 664–681.

Ryvkin, D. & Drugov, M. (2020), 'The shape of luck and competition in winner-take-all tournaments', *Theoretical Economics* **15**(4), 1587–1626.

Schweizer, N. & Szech, N. (2017), 'Revenues and welfare in auctions with information release', *Journal of Economic Theory* **170**, 86–111.

Schweizer, N. & Szech, N. (2018), 'Optimal revelation of life-changing information', *Management Science* **64**(11), 5250–5262.

Serena, M. (2022), 'Harnessing beliefs to optimally disclose contestants' types', *Economic Theory* **74**(3), 763–792.

Serenko, A. (2020), 'Knowledge sabotage as an extreme form of counterproductive knowledge behavior: the perspective of the target', *Journal of Knowledge Management* **24**(4), 737–773.

Serra-Garcia, M. & Szech, N. (2022), 'The (in) elasticity of moral ignorance', *Management Science* **68**(7), 4815–4834.

Sheremeta, R. M. (2011), 'Contest design: An experimental investigation', *Economic Inquiry* **49**(2), 573–590.

Sheremeta, R. M. (2013), 'Overbidding and heterogeneous behavior in contest experiments', *Journal of Economic Surveys* **27**(3), 491–514.

Sheremeta, R. M. (2018), 'Impulsive behavior in competition: Testing theories of overbidding in rent-seeking contests', *Available at SSRN 2676419* .

Skaperdas, S. (1996), 'Contest success functions', *Economic theory* **7**(2), 283–290.

Statista (2024), 'Public cloud infrastructure as a service (iaas) end-user spending worldwide from 2015 to 2025'.
URL: *February 04, 2025* *https://www.statista.com/statistics/505251/worldwide-infrastructure-as-a-service-revenue/*

Sutter, M., Kocher, M. G., Glätzle-Rützler, D. & Trautmann, S. T. (2013), 'Impatience and uncertainty: Experimental decisions predict adolescents' field behavior', *American Economic Review* **103**(1), 510–31.

Szech, N. (2010), Five Essays in Economic Theory, PhD thesis, Universitäts-und Landesbibliothek Bonn.

Szech, N. (2011), 'Becoming a bad doctor', *Journal of Economic Behavior & Organization* **80**(1), 244–257.

Tullock, G. (1980), Efficient rent seeking, *in* . G. T. James M. Buchanan, Robert D. Tollison, ed., 'Toward a theory of rent-seeking society', College Station: Texas A&M University Press, pp. 97–112.

Vandegrift, D. & Yavas, A. (2010), 'An experimental test of sabotage in tournaments', *Journal of Institutional and Theoretical Economics JITE* **166**(2), 259–285.

Wang, X. & Liu, S. (2023), 'Disclosure policies in all-pay auctions with bid caps and stochastic entry: The impact of risk aversion', *Bulletin of Economic Research* .

A Theory Appendix

A.1 First and Second Order Conditions for Maximization Problem

In order to derive the equilibrium effort and sabotage levels, I maximize the individual payoff function with respect to effort and with respect to sabotage for contestant i without loss of generality. Equation 8 is the first order condition with respect to effort and Equation 9 the first order conditions with respect to sabotage.

$$\frac{\partial \pi_i}{\partial e_i} = \frac{(\frac{1}{1+\sum_{j\in M_i} s_j})\sum_{l=1}^{m}(\frac{e_l}{1+\sum_{k\in M_l} s_k}) - (\frac{e_i}{1+\sum_{j\in M_i} s_j})(\frac{1}{1+\sum_{j\in M_i} s_j})}{(\sum_{l=1}^{m}\frac{e_l}{1+\sum_{k\in M_l} s_k})^2} W - C'(e_i) = 0 \quad (8)$$

Next, I assume without loss of generality that player i be the m-th player.

$$\frac{\partial \pi_i}{\partial s_i} = \frac{(\frac{e_i}{(1+\sum_{j\in M_i} s_j)})(\frac{e_1}{(1+\sum_{k\in M_1} s_k)^2} + ... + \frac{e_{m-1}}{(1+\sum_{k\in M_{m-1}} s_k)^2})}{(\sum_{l=1}^{m}\frac{e_l}{1+\sum_{k\in M_l} s_k})^2} W - C'(s_i) = 0 \quad (9)$$

The following two second order conditions hold true $\forall e_i > 0, \forall s_i, W \geq 0, \forall i \in M$, where M is the set of all active players. This indicates that the solutions to the first order conditions are maxima:

$$\frac{\partial^2 \pi_i}{\partial e_i^2} = -2\frac{(\frac{1}{(1+\sum_{j\in M_i} s_j)^2})\sum_{j\in M_i}(\frac{e_j}{1+\sum_{k\in M_j} s_k})}{(\sum_{l=1}^{m}\frac{e_l}{1+\sum_{k\in M_l} s_k})^3} W - C''(e_i) < 0 \quad (10)$$

$$\begin{aligned}\frac{\partial^2 \pi_i}{\partial s_i^2} &= -2\frac{(\frac{e_i}{(1+\sum_{j\in M_i} s_j)})(\frac{e_1}{(1+\sum_{k\in M_1} s_k)^3} + ... + \frac{e_{m-1}}{(1+\sum_{k\in M_{m-1}} s_k)^3})}{(\sum_{l=1}^{m}\frac{e_l}{1+\sum_{k\in M_l} s_k})^2} W \\ &+2\frac{(\frac{e_i}{(1+\sum_{j\in M_i} s_j)})(\frac{e_1}{(1+\sum_{k\in M_1} s_k)^2} + ... + \frac{e_{m-1}}{(1+\sum_{k\in M_{m-1}} s_k)^2})^2}{(\sum_{l=1}^{m}\frac{e_l}{1+\sum_{k\in M_l} s_k})^3} W - C''(e_i) < 0\end{aligned} \quad (11)$$

A.2 Proof Proposition 1

Proof. In a symmetric equilibrium, homogeneous contestants choose the same strategies. Hence, the chosen individual effort and sabotage levels are the same for everyone: $e_i = e_{-i} = e$ and $s_i = s_{-i} = s$. As there are m active contestants, everyone receives the sabotage of $m-1$ other contestants. Therefore, the received sabotage is $(m-1)s$. The first order condition with respect to effort (8) becomes:

$$\begin{aligned}
&\frac{(\frac{1}{1+(m-1)s})(m\frac{e}{1+(m-1)s}) - (\frac{e}{1+(m-1)s})(\frac{1}{1+(m-1)s})}{m^2(\frac{e}{1+(m-1)s})^2}W = C'(e_i) \\
\iff &\frac{(\frac{1}{1+(m-1)s})(m-1)(\frac{e}{1+(m-1)s})}{m^2(\frac{e}{1+(m-1)s})^2}W = C'(e_i) \\
\iff &\frac{(m-1)}{m^2}\frac{W}{e} = C'(e_i)
\end{aligned}$$

Assuming $C(e_i) = e_i$ and thus $C'(e_i) = 1$, the equation becomes:

$$\iff e^* = \frac{(m-1)}{m^2}W \tag{12}$$

Likewise, the first order condition with respect to sabotage (9) becomes:

$$\begin{aligned}
&\frac{(\frac{e}{1+(m-1)s})(m-1)(\frac{e}{(1+(m-1)s)^2})}{m^2(\frac{e}{1+(m-1)s})^2}W = C'(s_i) \\
\iff &\frac{(m-1)}{m^2}\frac{W}{(1+(m-1)s)} = C'(s_i)
\end{aligned} \tag{13}$$

Assuming $C(s_i) = s_i$ and thus $C'(s_i) = 1$, the equation becomes:

$$\iff s^* = \frac{1}{m^2}W - \frac{1}{(m-1)} \tag{14}$$

□

A.3 Proof Proposition 2

Proof. Under group size concealment, the expected profit function is as follows:

$$\mathbb{E}[\pi_i] = \sum_{M_i \in \mathcal{P}^{N_i}} q^{|M_i|}(1-q)^{|N_i/M_i|} p_i(y_i, y_{-i}, M_i)W - C_i(e_i) - C_i(s_i)$$

First, I take the first order condition of the expected profit function with respect to e_i:

$$\frac{\partial \mathbb{E}[\pi_i]}{\partial e_i} = \sum_{M_i \in \mathbb{P}^{N_i}} q^{|M_i|}(1-q)^{|N_i/M_i|} \frac{\frac{1}{1+\sum_{j\in M_i} s_j} \sum_{j\in M_i} \frac{e_j}{1+\sum_{k\in M_j} s_k}}{(\frac{e_i}{1+\sum_{j\in M_i} s_j} + \sum_{j\in M_i} \frac{e_j}{1+\sum_{k\in M_j} s_k})^2} W - C_i'(e_i) = 0$$

Next, I employ symmetry and by assumption $C'(e) = 1$. As every contestant is the same, the sum over all possible combinations of other active competitors relaxes to the binomial distribution. It describes the probabilities for each realized number of other contestants $(m-1)$ out of $n-1$ potential contestants. Hence, the equation becomes:

$$\sum_{(m-1)=0}^{n-1} \frac{(n-1)!}{(m-1)!(n-1-(m-1))!} q^{(m-1)}(1-q)^{n-1-(m-1)} \frac{\frac{(m-1)e}{(1+(m-1)s)^2}}{(m-1+1)^2 \frac{e^2}{(1+(m-1)s)^2}} W = 1$$

$$\iff \sum_{(m-1)=0}^{n-1} \frac{(n-1)!}{(m-1)!(n-1-(m-1))!} q^{(m-1)}(1-q)^{n-m} \frac{m-1}{m^2} \frac{W}{e} = 1$$

$$\iff e^* = \sum_{(m-1)=0}^{n-1} \frac{(n-1)!}{(m-1)!(n-m)!} q^{(m-1)}(1-q)^{n-m} \frac{m-1}{m^2} W \tag{15}$$

Subsequently, I take the first order condition with respect to s_i:

$$\frac{\partial \mathbb{E}[\pi_i]}{\partial s_i} = \sum_{M_i \in \mathbb{P}^{N_i}} q^{|M_i|}(1-q)^{|N_i/M_i|} \frac{\frac{e_i}{1+\sum_{j\in M_i} s_j} \sum_{j\in M_i} \frac{e_j}{(1+\sum_{k\in M_j} s_k)^2}}{(\frac{e_i}{1+\sum_{j\in M_i} s_j} + \sum_{j\in M_i} \frac{e_j}{1+\sum_{k\in M_j} s_k})^2} W - C_i'(s_i) = 0$$

After employing symmetry and the binomial coefficient, the equation becomes the following:

$$\sum_{(m-1)=0}^{n-1} \frac{(n-1)!}{(m-1)!(n-1-(m-1))!} q^{(m-1)}(1-q)^{n-1-(m-1)} \frac{\frac{(m-1)e^2}{(1+(m-1)s)^3}}{(m-1+1)^2 \frac{e^2}{(1+(m-1)s)^2}} W = C'(s) \tag{16}$$

After some algebra and $C'(s) = 1$ it becomes the following:

$$\sum_{(m-1)=0}^{n-1} \frac{(n-1)!}{(m-1)!(n-m)!} q^{(m-1)}(1-q)^{n-m} \frac{m-1}{m^2} \frac{1}{1+(m-1)s} W = 1$$

Which does not yield a closed form solution and hence I solve it numerically. Further, the second order conditions follow immediately from equation (10) and (11) and hold such that the FOC describe the maxima. □

A.4 Expected Effort and Sabotage

In this section, I compare the expected effort and sabotage levels between disclosure and concealment. When the group size realization is zero, i.e., when there are no contestants at all, there is no effort under both disclosure policies. Therefore, it suffices to show that the implemented effort is the same conditional on at least one player participating. Hence, conditional on at least one player participating, the expected efforts are as follows:

$$\textit{Concealment}: \mathbb{E}[e] = \sum_{(m-1)=0}^{n-1} \underbrace{\frac{(n-1)!}{(m-1)!(n-m)!} q^{m-1}(1-q)^{n-m}}_{\text{Probability of group size } m} \underbrace{e^*}_{\text{Effort concealment}} = \underbrace{e^*}_{\text{Effort concealment}}$$

$$\textit{Disclosure}: \mathbb{E}[e] = \sum_{(m-1)=0}^{n-1} \underbrace{\frac{(n-1)!}{(m-1)!(n-m)!} q^{m-1}(1-q)^{n-m}}_{\text{Probability of group size } m} \underbrace{\frac{(m-1)}{m^2} W}_{\text{Effort disclosure}} = \underbrace{e^*}_{\text{Effort concealment}}$$

Under concealment, conditional on one player being active, the expected effort is simply the equilibrium effort under concealment, because contestants exert the same effort for each realized group size. Under concealment, the weighted sum over the equilibrium choice for the specific realized group size is taken. This is exactly, how the equilibrium effort decision under concealment is computed (see Equation 15). Hence, the two expected efforts are equivalent.

As the equilibrium sabotage under concealment cannot be solved analytically, I compare the numerical solutions under both policies. Figure 14 depicts the expected sabotage, conditional on at least one player being active, under both policies and shows that there are only very small and

negligible differences, if any. Therefore, I show that there are no substantial differences between average sabotage levels under concealment and disclosure.

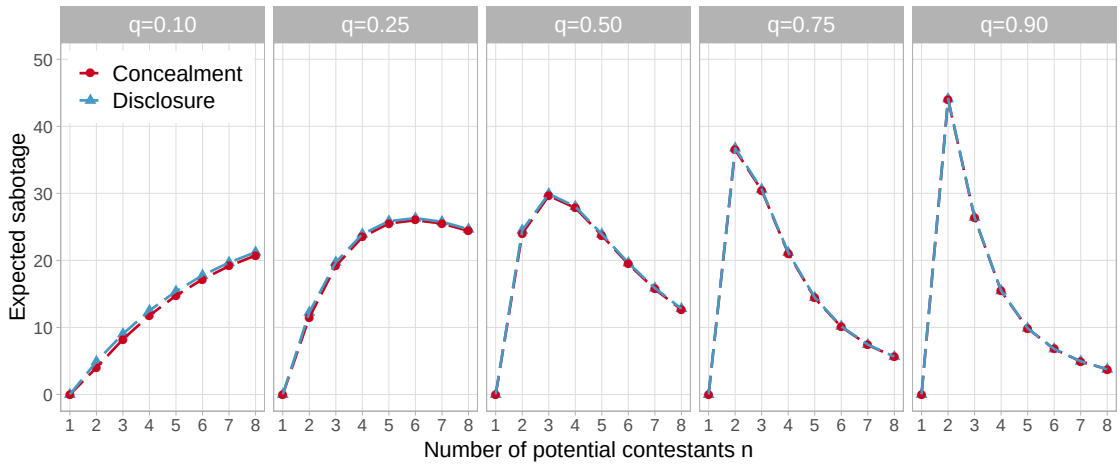


Figure 14: Expected sabotage under concealment compared to disclosure for low and high enter probabilities. The x-axes show the group size of all potential contestants (active and non-active). The y-axes show the average sabotage levels. Red lines indicate concealment and blue lines disclosure.

A.5 Expected Individual Payoff Simulation

In this section, I show that the individual expected payoff is not substantially different between disclosure policies. For this, I calculate the expected payoff conditional on participation. Because of symmetry, all players employ the same effort and sabotage. Therefore, performances are identical and all active players' winning probabilities are reduced to: $p_{win} = \frac{1}{m}$, with m being the realized number of active players. Under disclosure, conditional on participating, the expected payoff for player i is as follows:

$$E[\pi_i]_{Disclosure} = \sum_{(m-1)=0}^{n-1} \Bigg(\underbrace{\frac{(n-1)!}{(m-1)!(n-m)!} q^{m-1}(1-q)^{n-m}}_{\text{probability for } m-1 \text{ other active players}} \times \underbrace{\frac{1}{m}}_{\text{probability to win}} \times W - \underbrace{e^*(m) - s^*(m)}_{\text{costs}} \Bigg) \tag{17}$$

Conditional on concealment, the expected payoff for a participant, conditional on participating is as follows:

$$E[\pi_i]_{Concealment} = \sum_{(m-1)=0}^{n-1} \Bigg(\underbrace{\frac{(n-1)!}{(m-1)!(n-m)!} q^{m-1}(1-q)^{n-m}}_{\text{probability for } m-1 \text{ other active players}} \times \underbrace{\frac{1}{m}}_{\text{probability to win}} \times W \Bigg) - \underbrace{e^*(q,n) - s^*(q,n)}_{\text{costs}} \tag{18}$$

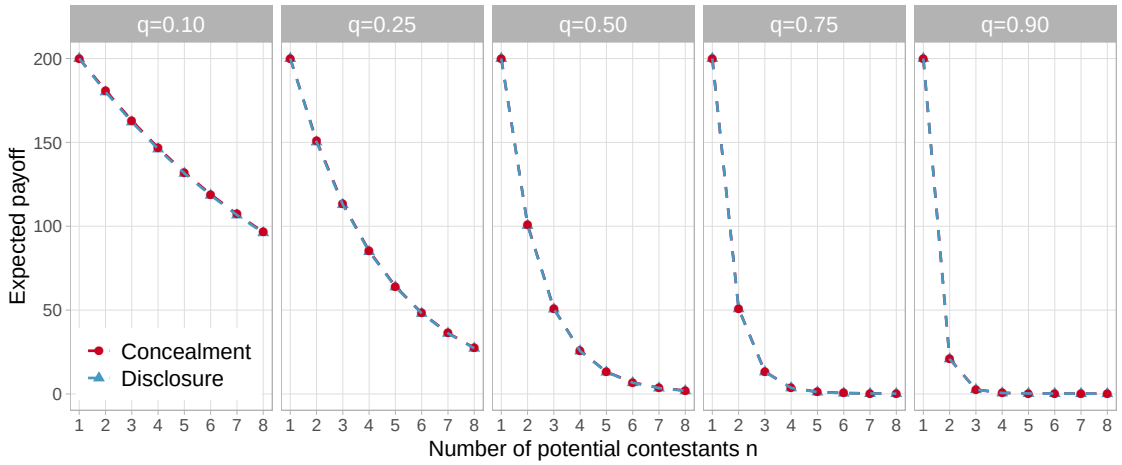


Figure 15: Expected individual utility conditional on low enter probability $q = 0.25$ (left graph) or high enter probability $q = 0.75$ (right graph). The x-axes show the group size of all potential contestants (active and non-active). The y-axes show the expected individual utility. Red lines indicate concealment and yellow lines disclosure of the number of active contestants.

Figure 15 shows the numerical solution and indicates that there is no substantial difference between the individual ex-ante expected payoffs between disclosing and concealing the number of participants. It further shows that expected payoffs decrease both in the number of potential contestants and in their enter probabilities. This is because players win with certainty if they are the only contestants, and the probability of being alone in the contest decreases with an increasing number of potential contestants and enter probabilities.

A.6 Robustness Effectiveness of Sabotage

This section provides a robustness check for the theoretical results, with a slightly different performance function that allows for a different effectiveness in the received sabotage. Specifically, I use the following performance function:

$$y_i = \frac{e_i}{(1 + (m-1)s)^t}$$

I show that the difference in the expected group performance still holds for different parameters of t. As $\lim_{t\to 0}$, however, the difference disappears. Yet, this is not surprising, because $\lim_{t\to 0}$ means that sabotage has no influence on the performance overall and thus cannot induce differences between the disclosure policies. When contestants know the number of active contestants m, The effort and sabotage levels in the symmetric equilibrium look as follows:

$$e^* = \frac{(m-1)}{m^2}W$$

$$s^* = \begin{cases} t\frac{1}{m^2}W - \frac{1}{m-1} & \text{if } W \geq \frac{m^2}{t(m-1)} \text{ and } m \geq 2 \\ 0 & \text{else} \end{cases}$$

Note that only the equilibrium sabotage levels are affected by the effectiveness of sabotage parameter t. Specifically, the less effective (higher t) sabotage, the lower are equilibrium sabotage levels. As sabotage levels are (almost) the weighted sum of the group size disclosure levels, a lower t also leads to lower sabotage levels under concealment. Figure 16 shows the numerical solutions of the expected group performances for different levels of t, i.e., $t \in \{0.25, 0.5, 1, 2\}$. It shows that even for a small effectiveness of sabotage, which also induces lower sabotage levels overall, the difference between concealment and disclosure is still pronounced. However, this difference becomes smaller, the smaller t, with no differences as $\lim_{t\to 0}$. Nonetheless, for not too low values of t, the differences are pronounced between the disclosure policies. Therefore, this analysis shows that the differences in the group performance between the disclosure policies remain robust, even if the performance function does not carry as pronounced decreasing marginal return in the received sabotage.

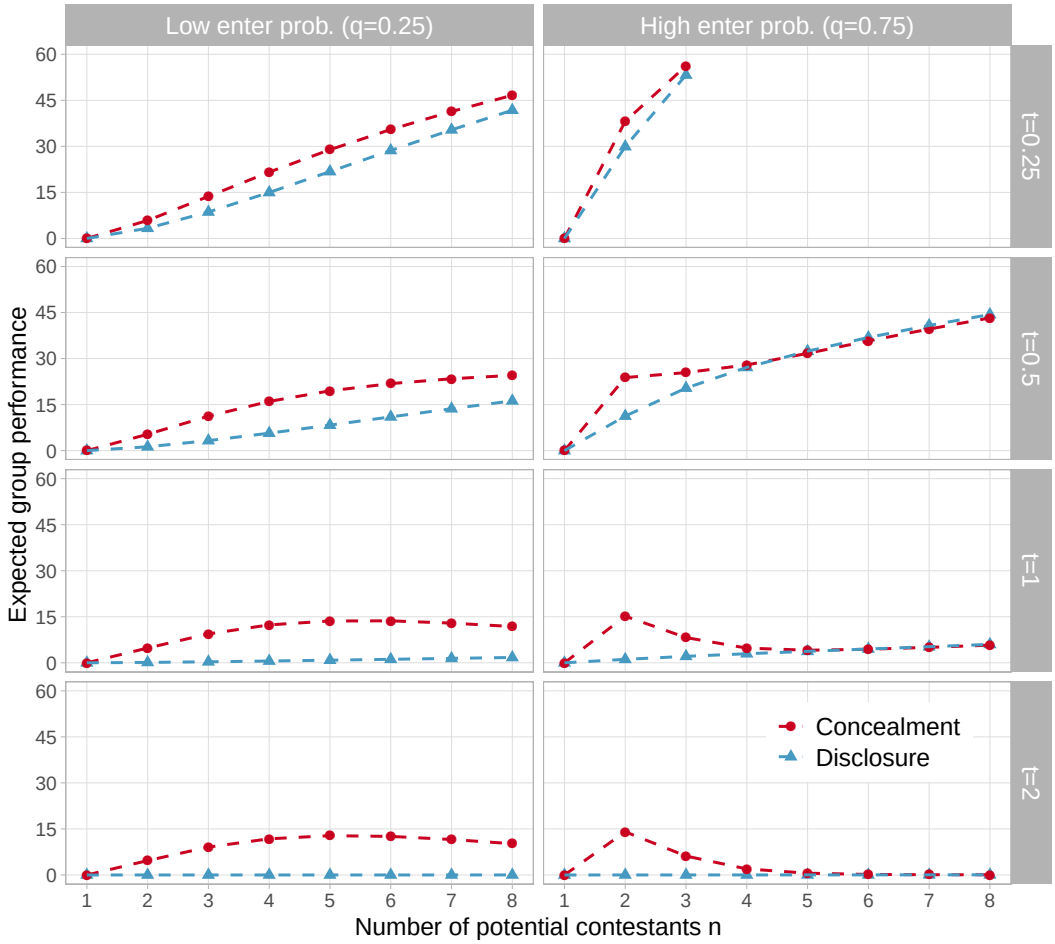


Figure 16: Expected group performance (sum of individual performances) conditional on the disclosure policy for low (left panel) and high (right panel) enter probabilities. The four rows vary the effectiveness of sabotage parameter $t \in \{0.25, 0.5, 1, 2\}$. The prize W is set to 200.

A.7 On Sabotage Costs

In this appendix section, I formally compare the equilibrium sabotage levels for the cost function $C(s) = (n-1)s$ compared to $C(s) = s$. Following equation 13, equilibrium sabotage levels under disclosure are described by $s^* = \frac{1}{m^2(n-1)}W - \frac{1}{m-1}$ (and $s^* = 0$ for $W \leq \frac{m^2}{m-1}(n-1)$ or $m = 1$) compared to $s^* = \frac{1}{m^2}W - \frac{1}{m-1}$ for $C(s) = s$. Figure 17 depicts this comparison and shows that for $n > 2$ (and $m > 1$), sabotage levels are strictly lower for $C(s) = (n-1)s$. Importantly, independent of the level of n, an increase in m leads to a decrease in sabotage (for $s^* \neq 0$).

Figure 18 depicts the numerical solutions for the equilibrium sabotage levels under concealment for both cost regimes (following equation 16). It shows that equilibrium sabotage levels are strictly lower for $n > 2$ for all enter probabilities q. As a result, the change in the comparative statics

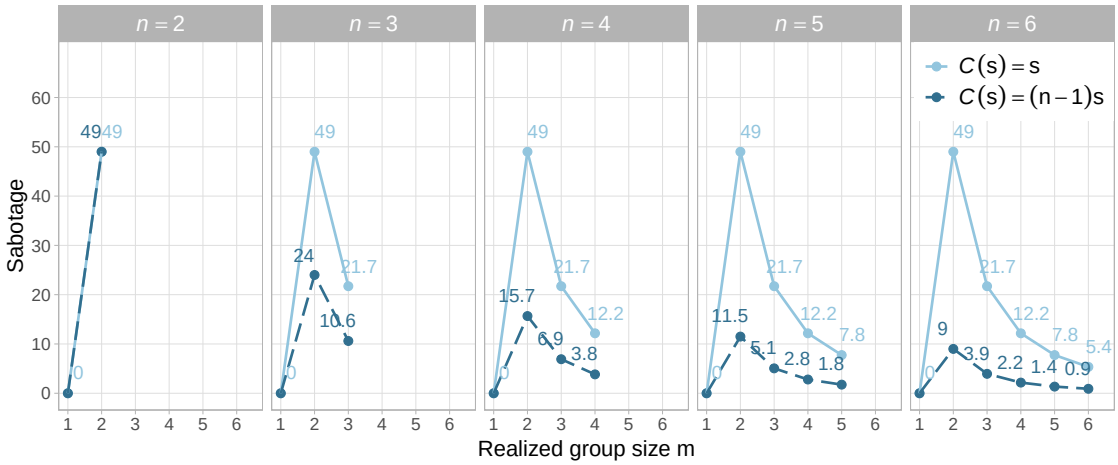


Figure 17: Equilibrium sabotage levels under group size disclosure conditional on the cost function. The prize is $W = 200$.

with respect to the number of potential contestants n for lower versus higher enter probabilities disappears. Instead, sabotage decreases for all enter probabilities. This is because the increase in sabotage costs with increasing n is more pronounced than the influence of higher sabotage levels for small realized group sizes.

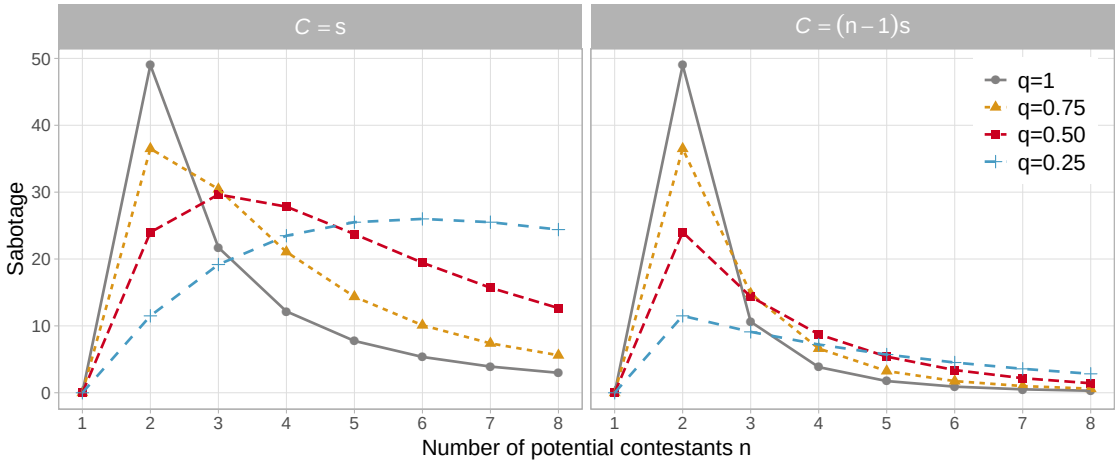


Figure 18: Equilibrium sabotage levels under group size concealment conditional on the cost function and enter probabilities. The prize is $W = 200$.

Crucially, sabotage levels under group size concealment remain (almost) the weighted sum of the choices for the realized group sizes m, and thus there are no substantial differences between the expected sabotage under the two disclosure policies (see Figure 19).

Similarly, the mechanism that induces the difference in the expected group performance remains

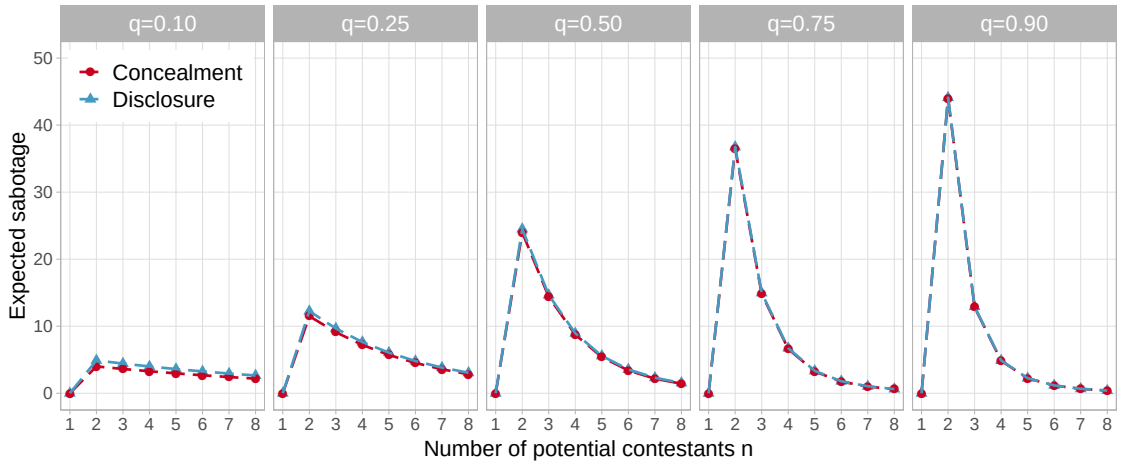


Figure 19: Expected sabotage for $C(s) = (n-1)s$ and $W = 200$ conditional on the disclosure policy and enter probabilities.

the same: the case of being the only contestant induces differences between the disclosure policies. If this probability becomes smaller, group performance converges between the two polices (see Figure 20). The only difference is that group performance is generally higher, especially as n grows because sabotage levels decrease and less performance is destroyed under $C(s) = (n-1)s$ compared to $C(s) = s$.

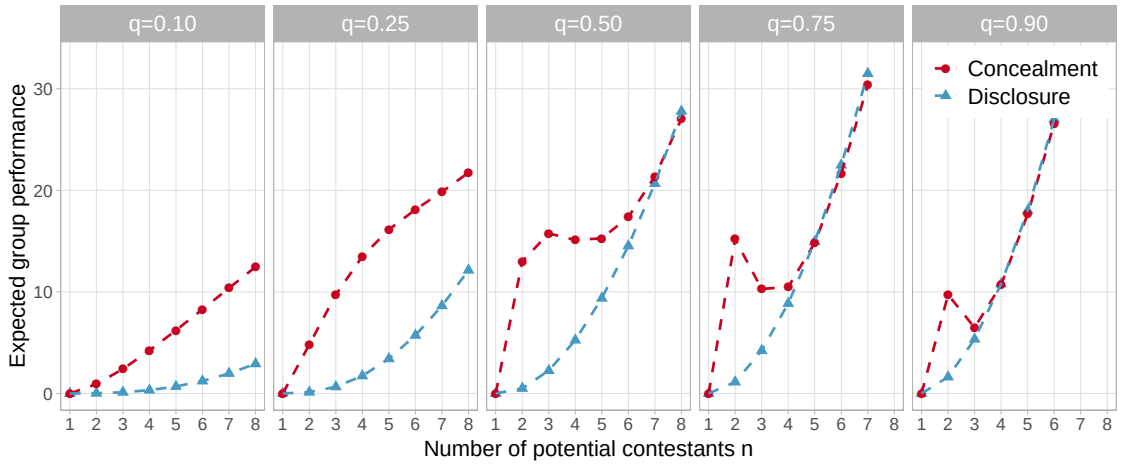


Figure 20: Expected group performance for $C(s) = (n-1)s$ and $W = 200$ conditional on the disclosure policy and enter probabilities.

A.8 Preference for Donations

Suppose that agents have a preference for donations D. Specifically, suppose that agents' utility from donations is described by $U_{donations} = \alpha D$ and for simplicity that the utility gains from own payoff and the donations are additive, such that $U_i = \pi_i + \alpha D$ with $\alpha \in [0, 1)$. The equilibrium levels under group size disclosure and group size uncertainty are only marginally influenced by the preference for donations parameter α. As a consequence, the comparative statics remain the same. Furthermore, the expected group performances and thus the difference between the expected performances between the disclosure policies are also only marginally affected by this preference for donations parameter. In conclusion, the main theoretical comparative statics are robust to the inclusion of preferences for donation.

A.8.1 Group Size Disclosure

Conditional on the realized group size m with the set of active player M, the overall expected utility under group size disclosure is then described by:

$$\mathbb{E}[U_i] = \frac{y_i}{\sum_{j \in M} y_j} W + \alpha(\sum_{j \in M} y_j + 10) - e_i - s_i \tag{19}$$

with $y_i = \frac{e_i}{1+\sum_{j \neq i} s_j}$ and $\frac{y_i}{\sum_{j \in M} y_j} = \frac{1}{n}$, if $y_i = 0 \; \forall i \in M$. Suppose, agents simultaneously maximize their expected utility by choosing e_i and s_i. Then the equilibrium effort and sabotage levels can be described by the following two equations:

$$e^* = \frac{-\sqrt{(m-1)^2 W^2 - 4(m-1)\alpha m^2 \left(m - \frac{1}{2}\right) W + \alpha^2 m^4} + (2W - \alpha) m^2 - 3Wm + W}{2m^2 (m-1)} \tag{20}$$

$$s^* = \frac{Wm - 2m^2 + \sqrt{2\sqrt{(m-1)^2 W^2 - 4a(m-1)\left(m - \frac{1}{2}\right) m^2 W + a^2 m^4}\, a m^2 + (m-1)^2 W^2 - 4a(m-1)\left(m - \frac{1}{2}\right) m^2 W + 2a^2 m^4} - W}{2m^2 (m-1)} \tag{21}$$

Proof. Suppose that agents simultaneously maximize their expected payoff:

$$\mathbb{E}[\pi_i] = \frac{y_i}{\sum_{j \in M} y_j} W + \alpha(\sum_{j \in M} y_j + 10) - e_i - s_i$$

with $y_i = \frac{e_i}{1+\sum_{j\neq i} s_j}$ and $\frac{y_i}{\sum_{j\in M} y_j} = \frac{1}{m}$, if $y_i = 0\ \forall i \in M$. First, I take the first order condition of the expected profit function with respect to e_i:

$$\frac{\partial \pi_i}{\partial e_i} = \frac{(\frac{1}{1+\sum_{j\neq i} s_j})\sum_{j=1}^{m}(\frac{e_j}{1+\sum_{l\neq j} s_l}) - (\frac{e_i}{1+\sum_{j\neq i} s_j})(\frac{1}{1+\sum_{j\neq i} s_j})}{(\sum_{j=1}^{m}\frac{e_j}{1+\sum_{l\neq j} s_l})^2} W + \alpha\frac{1}{1+\sum_{j\neq i} s_j} - 1 = 0$$

Applying symmetry yields:

$$\frac{(m-1)}{m^2}\frac{W}{e} + \alpha\frac{1}{1+(m-1)s} = 1 \tag{22}$$

Next, suppose without loss of generality that player i is the m-th player. I then take the first order condition with respect to s_i:

$$\frac{\partial \pi_i}{\partial s_i} = \frac{(\frac{e_i}{(1+\sum_{j\neq i} s_j)})(\frac{e_1}{(1+\sum_{l\neq 1} s_l)^2} + ... + \frac{e_{m-1}}{(1+\sum_{l\neq m-1} s_l)^2})}{(\sum_{j=1}^{m}\frac{e_j}{1+\sum_{l\neq j} s_l})^2} W$$
$$+\alpha(-\frac{e_1}{(1+\sum_{l\neq 1} s_l)^2} - ... - \frac{e_{m-1}}{(1+\sum_{l\neq m-1} s_l)^2}) - 1 = 0$$

Symmetry yields:

$$\frac{(m-1)}{m^2}\frac{W}{(1+(m-1)s)} - \alpha(m-1)\frac{e}{(1+(m-1)s)^2} = 1$$
$$\implies s_{1,2} = \frac{(m-1)W - 2m^2 \pm \sqrt{(m-1)^2W^2 - 4\alpha m^4(m-1)e}}{2m^2(m-1)}$$

As this yields two solutions, I check which of the two is admissible. For this, I plug in $\alpha = 0$ to see whether the expression collapses to the solution without preferences for donations. This is only true for $s_1 = \frac{(m-1)W-2m^2+\sqrt{(m-1)^2W^2-4\alpha m^4(m-1)e}}{2m^2(m-1)}$. I now take s_1 and plug it into Equation 22:

$$\frac{(m-1)}{m^2}\frac{W}{e}+\alpha\frac{1}{1+(m-1)\frac{(m-1)W-2m^2+\sqrt{(m-1)^2W^2-4\alpha m^4(m-1)e}}{2m^2(m-1)}}=1$$

$$\Longrightarrow e_{1,2}=\frac{\pm\sqrt{(m-1)^2W^2-4(m-1)\alpha m^2\left(m-\frac{1}{2}\right)W+\alpha^2m^4}+(2W-\alpha)m^2-3Wm+W}{2m^2(m-1)}$$

Similarly, to determine, which of the two solutions for e is admissible, I plug in $\alpha=0$ and see, if the solution relaxes to $e=\frac{(m-1)}{m^2}W$, which is the case without any preferences for donations. This is only the case for:

$$e^*=\frac{-\sqrt{(m-1)^2W^2-4(m-1)\alpha m^2\left(m-\frac{1}{2}\right)W+\alpha^2m^4}+(2W-\alpha)m^2-3Wm+W}{2m^2(m-1)} \tag{23}$$

Plugging this into s_1 yields:

$$s^*=\frac{Wm-2m^2+\sqrt{2\sqrt{(m-1)^2W^2-4a(m-1)\left(m-\frac{1}{2}\right)m^2W+a^2m^4}\,am^2+(m-1)^2W^2-4a(m-1)\left(m-\frac{1}{2}\right)m^2W+2a^2m^4}-W}{2m^2(m-1)} \tag{24}$$

which relaxes to $s=\frac{1}{m^2}W-\frac{1}{m-1}$ for $\alpha=0$ and hence is admissible. □

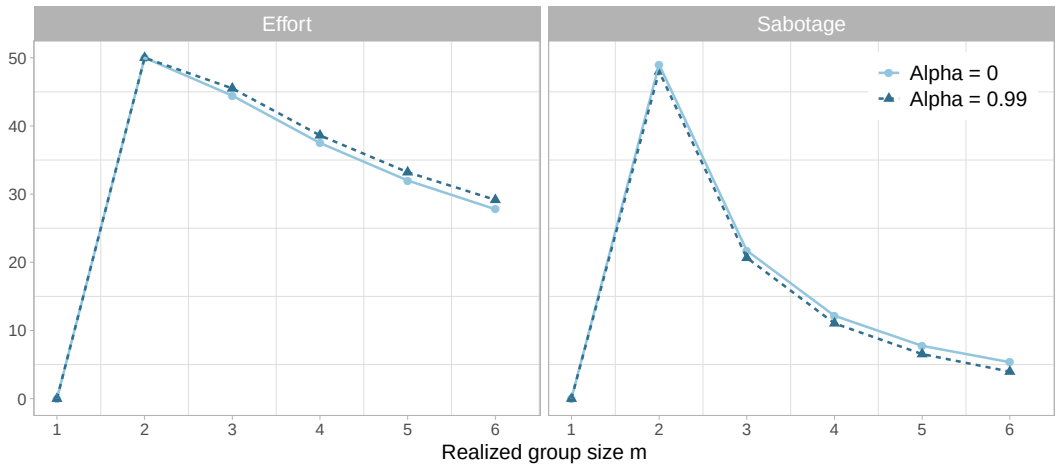


Figure 21: Equilibrium effort and sabotage levels conditional on the known realized group size m and the preference for donation parameter α. Dashed dark blue lines indicate a high preference for donations and light blue lines no preference for donations.

Figure 21 illustrates the equilibrium effort and sabotage levels for a preference for donations

parameters of $\alpha = 0$ and $\alpha = 0.99$. It shows that a preference for donations only marginally changes the equilibrium choices. Specifically, effort levels are slightly higher and sabotage levels slightly lower. More importantly, a preference for donations does not change the comparative statics of the realized group size. This is because the marginal benefit of increasing the winning probability of the prize W through higher effort is much greater than the marginal benefit of more donations through lower sabotage.

A.8.2 Group Size Concealment

The expected utility with a preference for donations under group size concealment is as follows:

$$\mathbb{E}[\pi_i] = \sum_{M_i \in \mathcal{P}^{N_i}} q^{|M_i|}(1-q)^{|N_i/M_i|} \left[\frac{y_i}{\sum_{j \in M} y_j} W + \alpha(\sum_{j \in M} y_j + 10) \right] - e_i - s_i$$

with $y_i = \frac{e_i}{1+\sum_{j \neq i} s_j}$ and $\frac{y_i}{\sum_{j \in M} y_j} = \frac{1}{m}$, if $y_i = 0 \; \forall i \in M$. The first order condition of the expected profit function with respect to e_i is:

$$\frac{\partial \pi_i}{\partial e_i} = \sum_{M_i \in \mathcal{P}^{N_i}} q^{|M_i|}(1-q)^{|N_i/M_i|} [\frac{(\frac{1}{1+\sum_{j \neq i} s_j}) \sum_{j=1}^{m} (\frac{e_j}{1+\sum_{l \neq j} s_l}) - (\frac{e_i}{1+\sum_{j \neq i} s_j})(\frac{1}{1+\sum_{j \neq i} s_j})}{(\sum_{j=1}^{m} \frac{e_j}{1+\sum_{l \neq j} s_l})^2} W$$
$$+ \alpha \frac{1}{1+\sum_{j \neq i} s_j}] - 1 = 0$$

I now apply symmetry. With homogeneous contestants, the sum over all possible sets of all other active contestants. relaxes to all possible number of others. For readability, I define $B_{m-1}^{n-1} = \sum_{(m-1)=0}^{n-1} \frac{(n-1)!}{(m-1)!(n-1-(m-1))!} q^{(m-1)}(1-q)^{n-1-(m-1)}$:

$$B_{m-1}^{n-1} \left[\frac{(m-1)}{m^2} \frac{W}{e} + \alpha \frac{1}{1+(m-1)s} \right] = 1$$

$$\iff e = \frac{B_{m-1}^{n-1} \frac{m-1}{m^2} W}{1 - B_{m-1}^{n-1} \alpha \frac{1}{1+(m-1)s}} \tag{25}$$

Next, suppose without loss of generality that player i is the m-th player. The first order condition

with respect to s_i is:

$$\frac{\partial \pi_i}{\partial s_i} = \sum_{M_i \in \mathcal{P}^{N_i}} q^{|M_i|}(1-q)^{|N_i/M_i|}[\frac{(\frac{e_i}{(1+\sum_{j\neq i} s_j)})(\frac{e_1}{(1+\sum_{l\neq 1} s_l)^2} + ... + \frac{e_{m-1}}{(1+\sum_{l\neq m-1} s_l)^2})}{(\sum_{j=1}^{m} \frac{e_j}{1+\sum_{l\neq j} s_l})^2} W$$
$$+\alpha(-\frac{e_1}{(1+\sum_{l\neq 1} s_l)^2} - ... - \frac{e_{m-1}}{(1+\sum_{l\neq m-1} s_l)^2})] - 1 = 0$$

Applying symmetry, and again with $B^{n-1}_{m-1} = \sum_{(m-1)=0}^{n-1} \frac{(n-1)!}{(m-1)!(n-1-(m-1))!} q^{(m-1)}(1-q)^{n-1-(m-1)}$, it becomes:

$$B^{n-1}_{m-1}\left[\frac{(m-1)}{m^2}\frac{W}{(1+(m-1)s)} - \alpha(m-1)\frac{e}{(1+(m-1)s)^2}\right] = 1 \tag{26}$$

Which does not yield a closed-form solution for s. Hence, I solve equations 25 and 26 numerically. Figure 22 shows the numerical solution for the parameters of interest. It shows the equilibrium effort and sabotage levels for a preference for donations parameter of $\alpha = 0$ and $\alpha = 0.99$. It reveals only marginal differences in the effort and sabotage levels between these parameters.

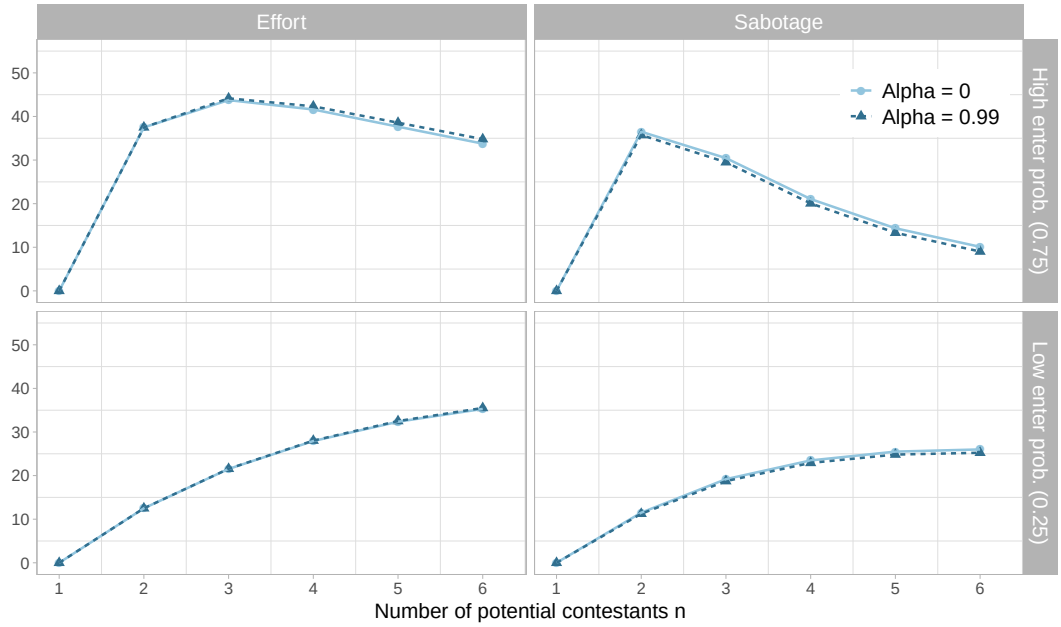


Figure 22: Equilibrium effort and sabotage levels under group size uncertainty conditional on the preference for donation parameter α. Dashed dark blue lines indicate a high preference for donations and light blue lines no preference for donations.

A.8.3 Comparison Disclosure Policies

Figure 23 depicts the difference in expected group performances between concealment and disclosure for a preference for donation parameter of $\alpha = 0$ (upper row) and $\alpha = 0.99$. The difference between the disclosure policies is almost the same between the two parameters. Consequently, a preference for donation parameter does also not change this comparative static.

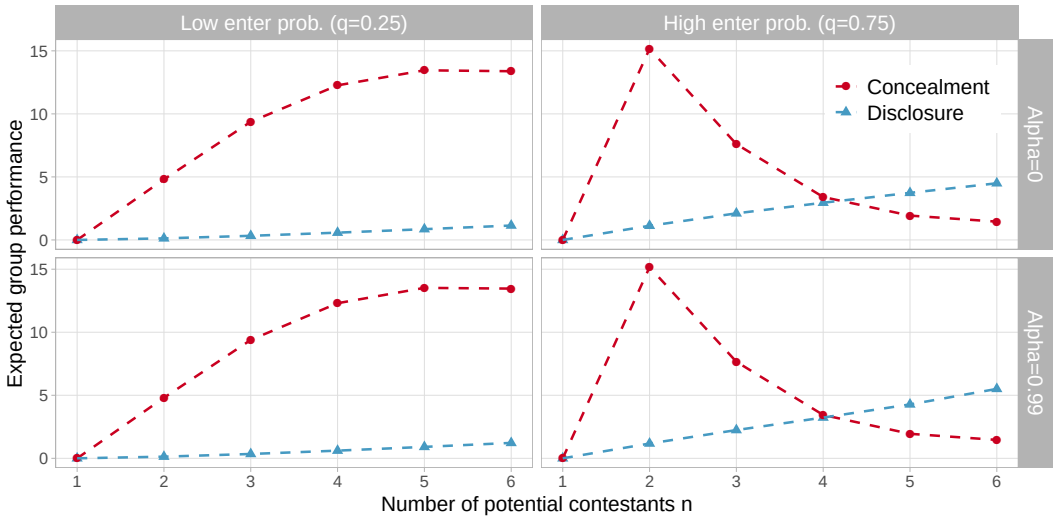


Figure 23: Expected group performance under concealment (red) and disclosure (blue) for low enter probabilities (left) and high enter probabilities (right) for either $\alpha = 0$ (upper row) or $\alpha = 0.99$ (lower row).

B Experimental Design Appendix

Figure 24 shows the communicated group size probabilities in Part B of the experiment for treatment *5H*. For all other treatments, this looked the same but only with the respective probabilities and possible number of active group members (only 0, 1, 2 for treatments *3L* and *3H*).

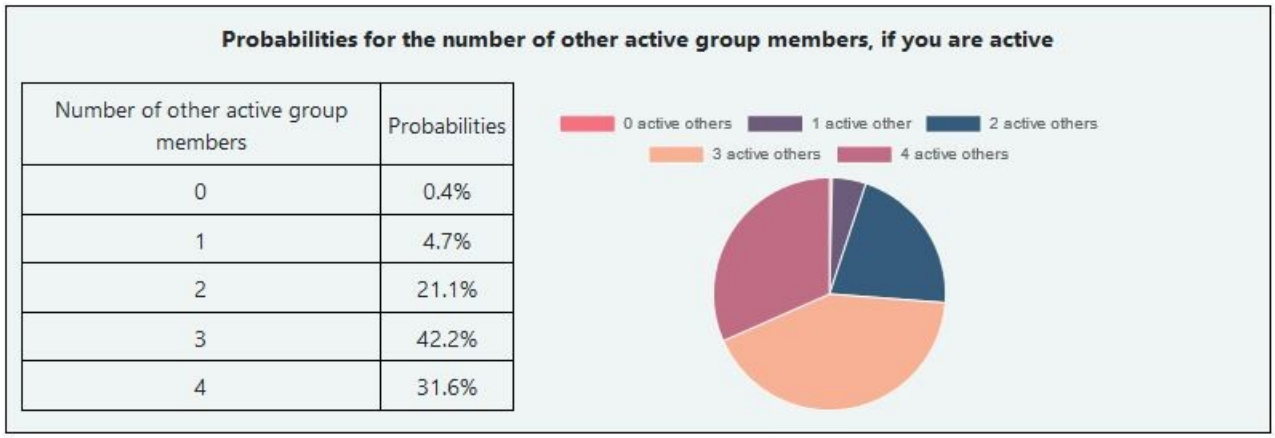


Figure 24: Communicated group size probabilities in Part B (for treatment *5H*)

Figure 25 shows the probability calculator that subjects had access to at any time. By clicking on the button 'advanced calculator', they could enter Option-A and Option-B choices for each of their potential competitors individually.



Figure 25: Probability calculator

C Results Appendix

C.1 Disclosure Policy

C.1.1 Effort, Sabotage, and Expected Payoff per Treatment

Figure 26 shows effort and sabotage differences across the policies conditional on the treatments. It shows that there are no significant differences in effort and sabotage levels between the disclosure policies for any of the treatments, except in treatment *3L*, where concealment increases sabotage levels ($p < 0.1$). The robustness checks confirm these results (see Appendix C.1.3), and find a significant ($p < 0.05$) increase in sabotage under concealment for treatment *3L* treatment.[52]

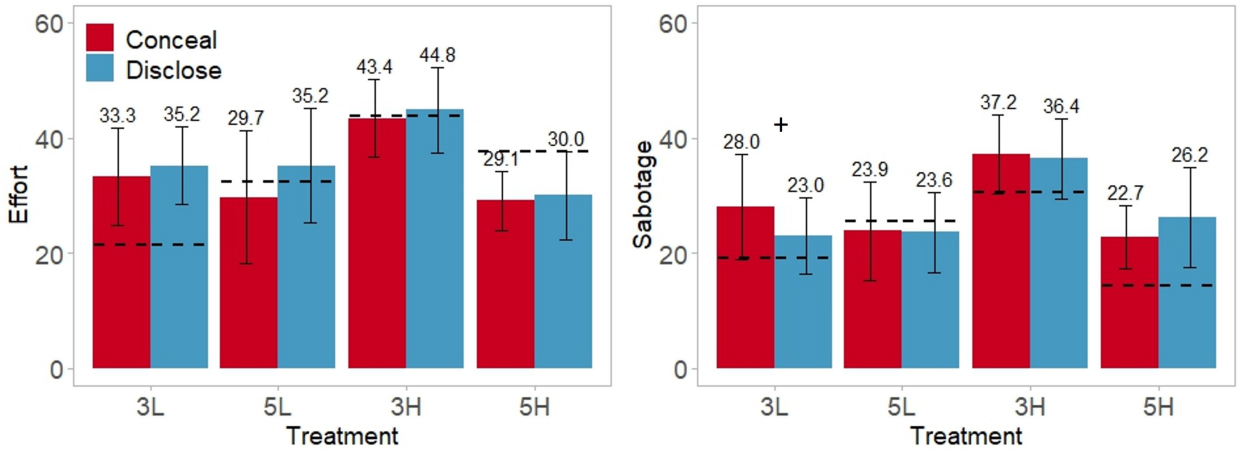


Figure 26: The bar charts show the average effort (left panel) and sabotage (right panel) conditional on the disclosure policy and the treatments. Black dashed lines show the Nash equilibrium predictions. The error bars show 95% confidence intervals. Significance levels: + $p < 0.10$

Figure 27 compares the expected payoff between the disclosure policies for each treatment (left panel) and pooled over all treatments (right panel). It shows that there are no significant differences between the disclosure policies for all treatments. The robustness checks also do not find any significant differences (see Appendix C.1.3).

[52]Specifically, I find a significant ($p < 0.05$) increase under concealment for treatment *3L*, when studying only the rounds around the policy changes and in the regression analysis. Apart from that, there is no significant difference in effort and sabotage levels between the disclosure policies in the robustness checks.

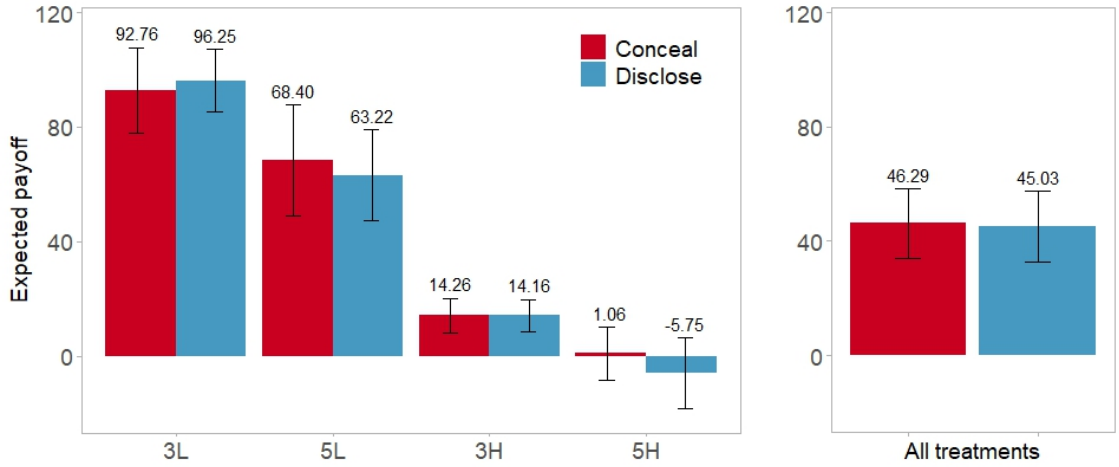


Figure 27: The figure depicts the individual expected payoffs based on the subjects' choices conditional on the treatments (left panel) or pooled over all treatments (right panel). Error bars show 95% confidence intervals.

C.1.2 Time Trends

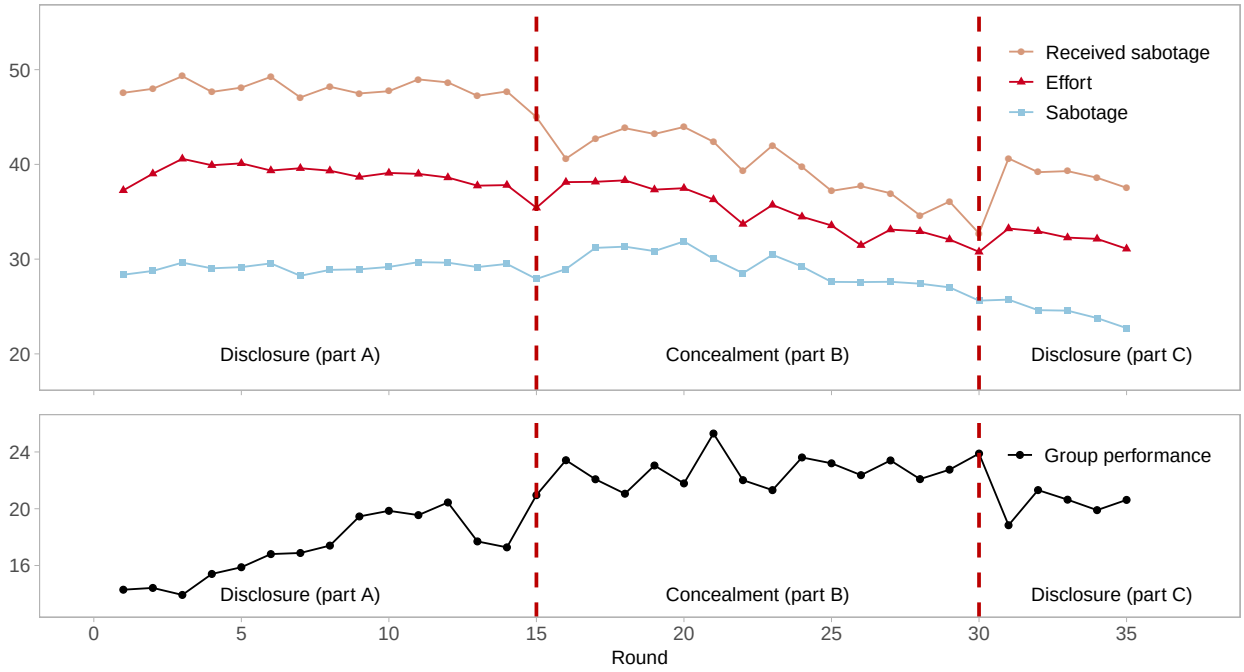


Figure 28: The two panels show the average expected received sabotage, the average expected effort and sabotage levels, and the average expected group performance, based on the subjects' choices over all rounds.

Figure 28 depicts the time trends of the expected received sabotage, the expected effort and sabotage levels, as well as the expected group performance over all rounds and pooled over all

treatments. The upper panel shows that there is a slight decrease in the expected choices over time, whereas the lower panel shows that there is a slight increase in the expected group performance over time. Therefore, I conduct the two robustness checks, where I first focus only on the rounds around the disclosure policy changes and on regression analyses that control for the time trend. The robustness checks support the results of the main section.

C.1.3 Robustness Check Effort, Sabotage, and Expected Payoff

Subset of Rounds around Policy Change

In this section, I focus on the rounds that are in the neighborhood of the disclosure policies, i.e. rounds 11-20 and 21-30. The following figure shows the pooled averages only for those rounds. Figure 29 shows the pooled averages over all treatments. It shows no significant differences in average expected effort, expected sabotage levels, and expected payoffs between the disclosure policies. Figure 30 shows average expected effort and sabotage levels conditional on the disclosure policy and each treatment. It shows no significant differences in levels between the policies, except for treatment *3L*, where concealment leads to significantly ($p < 0.05$) higher sabotage levels. Similarly, Figure 31 shows the expected payoff conditional on the disclosure policy for each treatment individually. It, too, shows no significant difference between the disclosure policies.

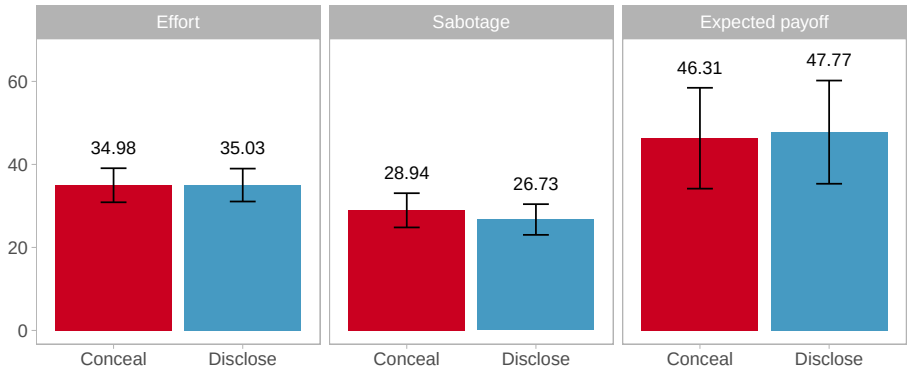


Figure 29: The figure shows average effort, sabotage, and expected payoffs conditional on the disclosure policy, pooled over all treatments. The figure is based solely on rounds 11-20 and 21-30 (around policy changes). Error bars show 95% confidence intervals. Significance levels: + $p < 0.10$

Regression Analyses

As an additional robustness check, I run linear regression models, which control for the time trend.

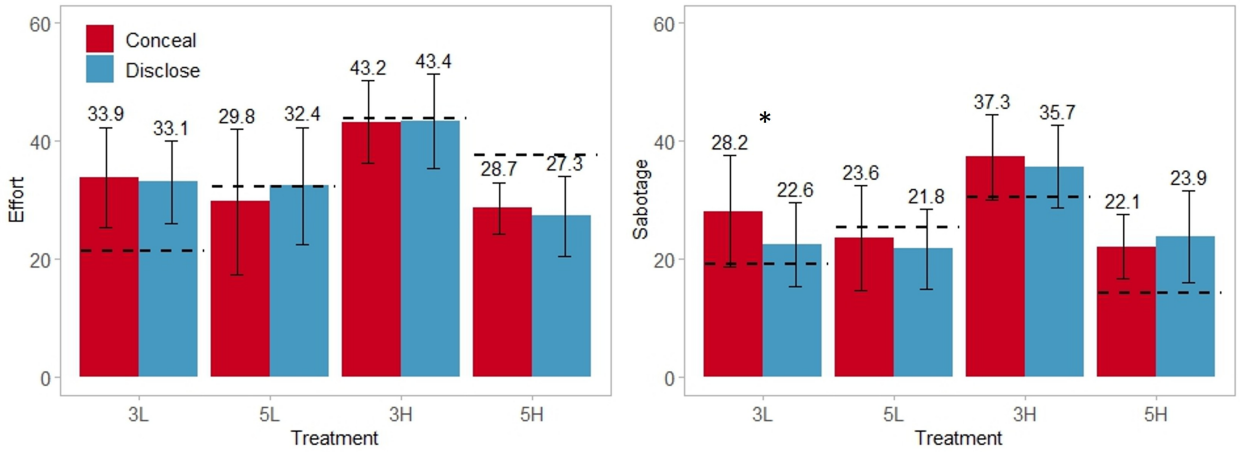


Figure 30: The bar charts show the average effort (left panel) and sabotage (right panel) conditional on the disclosure policy and the treatments for rounds 11-20 and 21-30 (around policy changes). The error bars show 95% confidence intervals. Significance levels: * $p < 0.05$

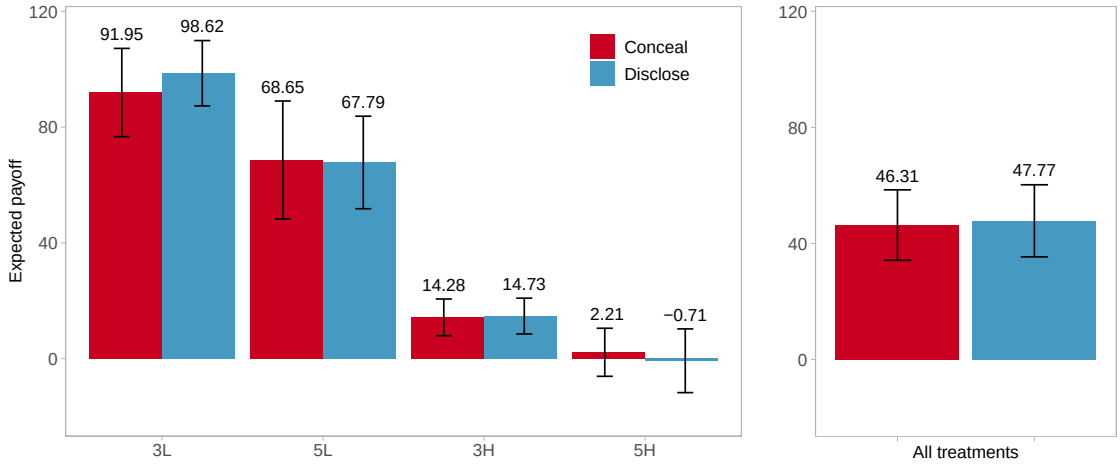


Figure 31: The figure depicts the average expected individual payoffs based on the subjects' choices conditional on the treatments (left panel) or pooled over all treatments (right panel) for the rounds 11-20 and 26-30. Error bars show 95% confidence intervals.

I cluster standard errors at the group level. I include the pre-registered controls[53] and additionally include the treatments as controls. Table 2 shows no significant difference in effort levels and expected payoffs between the disclosure policies. It further shows that concealment significantly ($p < 0.05$) increases sabotage levels.

Next, I run the same regressions, but conditional on the specific treatments. Table 3 shows

[53]The controls are: being active in the round before, having won in the round before, average sabotage and effort levels of other participants in the round before, round, determined group size in the round before, how often won in the rounds before, SVO, spite, risk, loss and ambiguity aversion, age, gender, highest degree, the field of study, the degree of concentration and understanding

Table 2: Linear regression expected effort, sabotage, and payoff on concealment and controls

	Dependent variable:		
	effort	sabotage	expected payoff
	(1)	(2)	(3)
Concealment	0.43	2.20*	−0.18
	(1.09)	(1.11)	(1.50)
Round	−0.59***	−0.40***	0.65***
	(0.10)	(0.08)	(0.11)
Risk Aversion	−1.72	−1.64	1.18
	(1.51)	(1.34)	(1.82)
Loss Aversion	−0.59	−1.03	−1.54
	(0.86)	(0.88)	(1.02)
Ambiguity Aversion	−3.09*	−2.04+	3.34**
	(1.35)	(1.08)	(1.26)
SVO	0.19	0.11	−0.19
	(0.18)	(0.17)	(0.17)
Spite	10.43	18.35*	0.63
	(8.75)	(8.89)	(9.63)
Female	6.99+	5.87+	−6.03+
	(3.94)	(3.39)	(3.53)
Age	0.18	−0.58	−0.42
	(0.63)	(0.48)	(0.63)
Constant	46.39*	64.84***	39.78*
	(20.75)	(18.75)	(20.10)
Treatment Dummies	✓	✓	✓
Other Controls	✓	✓	✓
Observations	6,630	6,630	6,630
# Clusters	52	52	52
R^2	0.16	0.15	0.69
Note: SE clustered at group level	+ $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$		

linear regressions of the expected average effort and sabotage levels on concealment but for each treatment separately. It confirms the significant ($p < 0.05$) increase in sabotage levels under concealment for the treatment $3L$ and additionally shows a significant ($p < 0.05$) increase in effort under concealment in treatment *5H*. Finally, I run the same regressions but for expected payoffs. Again, it does not show any significant differences between the disclosure policies for any of the treatments (Table 4).

Table 3: Linear regression expected effort and sabotage on disclosure and controls for different treatments

	Dependent variable:							
	effort	sabotage	effort	sabotage	effort	sabotage	effort	sabotage
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Concealment	0.58	4.98*	−0.89	3.18	0.62	1.91	3.99*	1.03
	(2.61)	(2.08)	(2.82)	(3.34)	(1.76)	(1.44)	(1.84)	(2.10)
Round	−0.28	−0.01	−0.82**	−0.64*	−0.34*	−0.34**	−1.01***	−0.76***
	(0.19)	(0.09)	(0.28)	(0.28)	(0.16)	(0.12)	(0.22)	(0.18)
Risk Aversion	−8.53**	−8.90***	−2.71*	−1.78	−5.58*	−6.75***	0.93	1.52
	(2.85)	(1.85)	(1.16)	(1.29)	(2.52)	(1.94)	(3.31)	(2.57)
Loss Aversion	2.83	2.47	−0.12	0.09	−4.30**	−3.11*	−2.06	−3.51**
	(2.31)	(2.02)	(1.47)	(1.49)	(1.58)	(1.56)	(1.41)	(1.27)
Ambiguity Aversion	−2.90	−2.99**	−2.68	−2.33	−3.00	−1.02	−0.24	0.50
	(1.89)	(0.99)	(2.19)	(2.01)	(3.16)	(3.35)	(1.57)	(1.13)
SVO	−0.05	0.09	−0.30	−0.52+	0.50	−0.04	−0.14	0.11
	(0.38)	(0.20)	(0.35)	(0.30)	(0.33)	(0.29)	(0.45)	(0.44)
Spite	11.10	9.64	47.84+	23.49	35.91+	35.65**	−3.31	31.31
	(12.18)	(10.22)	(25.69)	(25.58)	(18.58)	(12.65)	(43.03)	(34.87)
Female	−9.91	−0.79	−5.10	0.80	4.25	−1.24	17.83***	17.81***
	(6.64)	(2.99)	(11.66)	(11.13)	(7.13)	(6.60)	(3.96)	(4.05)
Age	3.57**	1.94**	−1.20	−1.08	−1.21	−1.34	−0.46	−2.44*
	(1.21)	(0.69)	(2.23)	(2.01)	(1.54)	(1.23)	(1.58)	(1.01)
Constant	−4.72	48.41**	80.16	83.76	124.48*	177.78***	72.55	86.67*
	(41.64)	(17.68)	(73.26)	(65.27)	(53.93)	(43.48)	(45.57)	(37.52)
Treatments	3L	3L	5L	5L	3H	3H	5H	5H
Other Controls	✓	✓	✓	✓	✓	✓	✓	✓
Observations	1,632	1,632	1,700	1,700	1,632	1,632	1,666	1,666
# Clusters	16	16	10	10	16	16	10	10
R^2	0.48	0.41	0.23	0.24	0.30	0.30	0.26	0.24

Note: SE clustered at group level + $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Table 4: Linear regression expected payoff on concealment and controls for different treatments

	Dependent variable:			
	expected payoff			
	(1)	(2)	(3)	(4)
Concealment	-3.82	-0.68	-0.17	0.55
	(3.59)	(4.78)	(1.54)	(1.97)
Round	0.29^{+}	1.19^{**}	0.20^{*}	0.77^{***}
	(0.16)	(0.40)	(0.08)	(0.19)
Risk Aversion	12.13^{***}	0.34	2.76^{**}	0.87
	(3.18)	(1.05)	(0.86)	(2.17)
Loss Aversion	-4.30	-2.63^{+}	0.68	2.28^{+}
	(3.07)	(1.36)	(0.71)	(1.28)
Ambiguity Aversion	3.93^{*}	4.18^{+}	2.21	1.27
	(1.54)	(2.28)	(1.48)	(1.09)
SVO	0.04	0.13	-0.05	-0.15
	(0.40)	(0.35)	(0.10)	(0.27)
Spite	-15.20	11.14	14.88	-37.64
	(18.09)	(25.08)	(9.95)	(26.72)
Female	-3.02	14.94	-0.75	-13.09^{***}
	(5.10)	(9.77)	(2.17)	(2.96)
Age	-5.58^{***}	2.46^{*}	-0.34	2.49^{**}
	(1.54)	(1.11)	(0.55)	(0.76)
Constant	145.54^{***}	16.50	-2.48	-40.58^{*}
	(39.04)	(39.86)	(14.93)	(20.64)
Treatments	3L	5L	3H	5H
Other Controls	✓	✓	✓	✓
Observations	1,632	1,700	1,632	1,666
# Clusters	16	10	16	10
R^2	0.52	0.36	0.36	0.32

Note: SE clustered at group level $+\ p < 0.1$; $*\ p < 0.05$; $**\ p < 0.01$; $***\ p < 0.001$

C.1.4 Robustness Check Group Performance

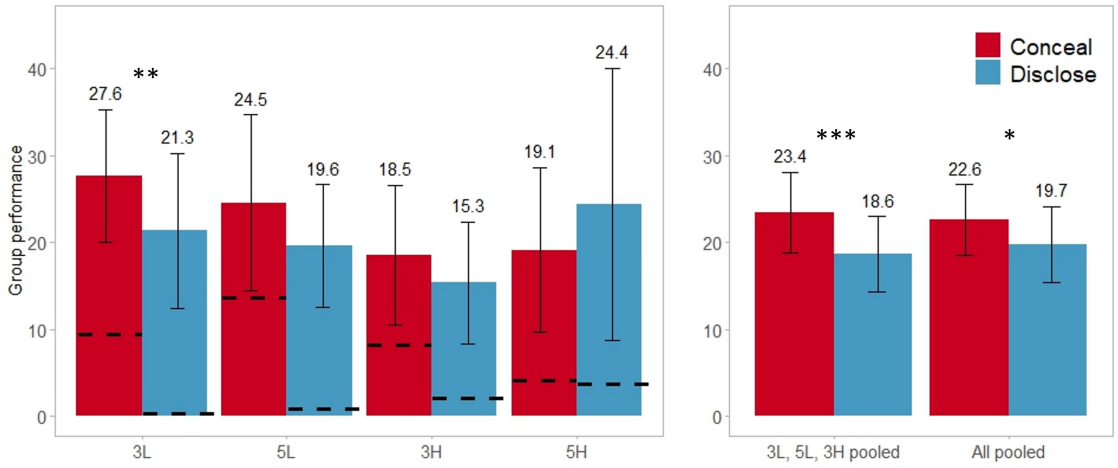


Figure 32: The bar charts show the average group performance conditional on the disclosure policy and on the treatments. The error bars show 95% confidence intervals. Significance levels: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Figure 32 depicts the expected group performance conditional on the disclosure policy for each treatment separately (left panel) and over all treatments pooled, excluding and including treatment 5H (right panel). It shows the averages of the subsets around the policy changes, i.e., rounds 11-20 and 21-30. It confirms the results from the main section in all cases, except in the case of treatment *3H*, where it shows no statistically significant increase under concealment. Linear regressions reveal the same significance levels and effects of concealment on group performance (see Table 5).

Table 5: Linear regression group performance on concealment and controls

	Dependent variable:					
	group performance					
	(1)	(2)	(3)	(4)	(5)	(6)
Concealment	7.78^{**}	6.86	4.98	-5.39	3.41^{+}	6.07^{**}
	(2.43)	(4.73)	(3.48)	(3.58)	(1.89)	(2.05)
Round	-0.34^{+}	-0.73^{**}	0.25	0.73^{*}	-0.005	-0.21
	(0.18)	(0.26)	(0.21)	(0.31)	(0.15)	(0.14)
Risk Aversion	-3.92	0.43	-0.74	-2.71^{**}	-0.68	-0.24
	(2.54)	(0.58)	(0.99)	(0.92)	(0.86)	(1.04)
Loss Aversion	2.76	0.95	1.88^{*}	0.28	1.50^{***}	1.62^{**}
	(2.30)	(0.64)	(0.93)	(0.54)	(0.43)	(0.58)
Ambiguity Aversion	-1.39	-0.78	-2.45^{**}	-0.70	-1.79^{*}	-1.73^{+}
	(1.61)	(0.71)	(0.92)	(0.62)	(0.82)	(1.03)
SVO	-0.10	0.01	-0.06	-0.16	0.04	0.004
	(0.31)	(0.11)	(0.13)	(0.29)	(0.12)	(0.14)
Spite	10.36	2.76	-16.09^{***}	19.36^{*}	-0.49	-0.19
	(9.49)	(8.49)	(4.71)	(8.88)	(4.98)	(5.45)
Female	-10.68^{+}	-7.09^{+}	-4.44^{+}	1.09	-1.25	-2.18
	(6.46)	(3.74)	(2.51)	(3.42)	(2.65)	(3.06)
Age	3.33^{**}	-0.54	0.89^{+}	0.33	0.81^{*}	1.12^{*}
	(1.17)	(0.39)	(0.53)	(0.23)	(0.36)	(0.47)
Constant	-38.52	54.06^{***}	-1.53	-5.61	-14.97	-14.04
	(33.11)	(10.90)	(23.84)	(10.36)	(12.37)	(15.08)
Treatments	3L	5L	3H	5H	All	No 5H
Other Controls	✓	✓	✓	✓	✓	✓
Observations	1,632	1,700	1,632	1,666	6,630	4,964
# Clusters	16	10	16	10	52	42
R^2	0.41	0.30	0.18	0.28	0.13	0.19

Note: SE clustered at group level $+ p < 0.1$; $* p < 0.05$; $** p < 0.01$; $*** p < 0.001$

C.1.5 Robustness Check Implemented Choices

In this section, I replicate the main results by focusing on the choices that were implemented in each round of the experiment. Specifically, this includes only the choices for the realized group size in each round and only the participants that were chosen to become active in each round. I run several regression analyses with the pre-registered controls[54] and additionally add the implemented realized group size of the current round as a control. Table 6 shows the regression of the implemented effort, sabotage, and resulting payoffs conditional on concealment, revealing no significant differences between the disclosure policies. Table 7 and Table 8 show the same regressions but for each treatment separately. It shows significantly higher sabotage levels under concealment for treatment *3L* and significantly higher effort levels for *5L*. Apart from these significant differences, the regressions do not show any other significant differences in sabotage, effort, or payoffs between the disclosure policies. Finally, Table 9 shows the regression of the implemented group performance on concealment and replicates the results from the main section, but other than in the main section also provides support for a significant increase in group performance for treatment *5L*.

[54]The controls are: being active in the round before, having won in the round before, average sabotage and effort levels of other participants in the rounds before, round, determined group size in the round before, how often won in the rounds before, SVO, spite, risk, loss and ambiguity aversion, age, gender, highest degree, the field of study, the degree of concentration and understanding.

Table 6: Linear regression effort, sabotage, and payoffs on concealment and controls for implemented choices

	Dependent variable:		
	effort	sabotage	expected payoff
	(1)	(2)	(3)
Concealment	1.42	2.35^{+}	-1.49
	(1.39)	(1.23)	(1.89)
Round	-0.61^{***}	-0.47^{***}	0.82^{***}
	(0.11)	(0.08)	(0.12)
Risk Aversion	-1.59	-1.17	1.68
	(1.66)	(1.45)	(1.57)
Loss Aversion	-1.31	-1.76^{+}	-1.30
	(0.86)	(0.90)	(1.27)
Ambiguity Aversion	-2.40^{+}	-1.17	1.89
	(1.35)	(1.19)	(1.49)
SVO	0.18	0.11	0.07
	(0.20)	(0.19)	(0.26)
Spite	8.27	18.19^{+}	15.11
	(9.67)	(9.62)	(14.14)
Female	7.66^{*}	6.11^{+}	-10.97^{**}
	(3.48)	(3.20)	(4.22)
Age	0.16	-0.61	-0.51
	(0.64)	(0.53)	(0.83)
Constant	43.48^{+}	61.39^{**}	345.35^{***}
	(22.32)	(21.35)	(28.16)
Treatment Dummies	✓	✓	✓
Group Size Realization Dummies	✓	✓	✓
Other Controls	✓	✓	✓
Observations	3,751	3,751	3,751
# Clusters	52	52	52
R^2	0.18	0.16	0.41

Note: SE clustered at group level $+ p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Table 7: Linear regression effort, sabotage, and payoffs on concealment and controls for implemented choices for different treatments

	Dependent variable:							
	effort	sabotage	effort	sabotage	effort	sabotage	effort	sabotage
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Concealment	1.73	6.81*	0.68	2.73	0.62	2.63	5.51**	2.22
	(2.83)	(2.74)	(4.05)	(3.82)	(2.62)	(1.84)	(2.02)	(2.37)
Round	−0.62**	−0.48*	−0.36	−0.53	−0.32*	−0.29+	−1.11***	−0.81***
	(0.22)	(0.19)	(0.53)	(0.37)	(0.16)	(0.15)	(0.22)	(0.17)
Risk Aversion	−8.55***	−8.07**	−2.19	−1.94	−6.03*	−7.72***	1.20	1.86
	(2.14)	(2.48)	(1.36)	(1.39)	(2.48)	(1.97)	(3.24)	(2.50)
Loss Aversion	3.77+	2.09	1.66	2.04	−4.73**	−3.46+	−1.93	−3.35**
	(2.23)	(2.84)	(1.36)	(1.44)	(1.55)	(1.78)	(1.34)	(1.14)
Ambiguity Aversion	−2.06	−1.69	−5.31**	−3.95*	−2.93	−1.16	0.17	0.91
	(1.62)	(1.35)	(1.85)	(1.66)	(3.10)	(3.32)	(1.54)	(1.03)
SVO	−0.22	−0.03	−0.36	−0.53+	0.56	−0.02	−0.27	0.04
	(0.37)	(0.25)	(0.34)	(0.31)	(0.34)	(0.31)	(0.45)	(0.44)
Spite	7.56	4.78	13.58	0.27	37.35+	36.68**	−11.52	23.63
	(12.83)	(10.67)	(22.80)	(23.10)	(19.59)	(13.11)	(40.85)	(32.63)
Female	−14.24*	−1.27	0.62	3.75	4.57	−0.65	14.67***	16.01***
	(6.55)	(3.58)	(9.87)	(8.59)	(7.54)	(6.98)	(3.27)	(3.46)
Age	4.33**	1.99*	−1.61	−1.89	−0.84	−0.82	−0.15	−2.31*
	(1.34)	(0.78)	(2.08)	(1.75)	(1.55)	(1.25)	(1.58)	(0.96)
Constant	11.55	69.00*	95.21	108.82+	118.75*	179.86***	69.56	89.37*
	(41.95)	(29.38)	(64.42)	(55.40)	(57.06)	(48.30)	(47.70)	(39.94)
Treatments	3L	3L	5L	5L	3H	3H	5H	5H
Other Controls	✓	✓	✓	✓	✓	✓	✓	✓
Observations	713	713	544	544	1,237	1,237	1,257	1,257
# Clusters	16	16	10	10	16	16	10	10
R^2	0.39	0.22	0.20	0.22	0.27	0.27	0.25	0.22

Note: SE clustered at group level + $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Table 8: Linear regression implemented payoff on concealment and controls

	Dependent variable:			
	payoff			
	(1)	(2)	(3)	(4)
Concealment	−1.02	1.28	−2.67	−3.78
	(4.26)	(8.58)	(3.16)	(2.45)
Round	1.03**	0.14	0.52*	0.91***
	(0.39)	(0.79)	(0.22)	(0.18)
Risk Aversion	9.35***	0.51	7.03*	2.47
	(2.33)	(2.36)	(2.73)	(2.74)
Loss Aversion	−5.81+	−5.91**	5.22**	−0.08
	(3.23)	(2.25)	(1.76)	(1.25)
Ambiguity Aversion	2.99+	3.90	5.24	−0.19
	(1.66)	(2.89)	(3.31)	(1.57)
SVO	1.23*	0.53	−0.14	−0.03
	(0.50)	(0.49)	(0.40)	(0.33)
Spite	12.61	46.86	14.28	−10.11
	(23.28)	(39.48)	(18.32)	(24.51)
Female	−4.93	8.41*	−3.34	−18.17***
	(9.84)	(3.95)	(7.49)	(4.62)
Age	−7.62***	2.35+	−2.55+	2.29**
	(1.92)	(1.30)	(1.49)	(0.79)
Constant	348.62***	325.98***	313.77***	324.74***
	(69.24)	(44.88)	(49.81)	(28.59)
Treatments	3L	5L	3H	5H
Other Controls	✓	✓	✓	✓
Observations	713	544	1,237	1,257
# Clusters	16	10	16	10
R^2	0.59	0.50	0.24	0.10

Note: SE clustered at group level + $p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

Table 9: Linear regression implemented group performance on concealment and controls

	Dependent variable:					
	group performance					
	(1)	(2)	(3)	(4)	(5)	(6)
Concealment	6.34^{+}	14.84^{*}	7.88^{**}	-7.39^{+}	5.60^{*}	9.51^{***}
	(3.28)	(6.34)	(3.03)	(3.78)	(2.55)	(2.73)
Round	-0.18	-0.34	-0.03	0.59^{+}	0.05	-0.13
	(0.11)	(0.39)	(0.20)	(0.33)	(0.14)	(0.11)
Risk Aversion	-0.31	-0.33	-1.05	-3.26^{***}	-0.73	-0.41
	(1.06)	(0.47)	(0.75)	(0.98)	(0.52)	(0.55)
Loss Aversion	1.40	0.31	2.41^{***}	0.05	0.89^{**}	0.99^{**}
	(1.08)	(0.32)	(0.63)	(0.48)	(0.32)	(0.36)
Ambiguity Aversion	-0.07	-0.22	-1.59^{**}	-1.87^{**}	-0.82^{**}	-0.60^{+}
	(0.42)	(0.52)	(0.55)	(0.64)	(0.31)	(0.34)
SVO	-0.09	0.06	-0.04	-0.15	0.04	0.04
	(0.15)	(0.09)	(0.11)	(0.33)	(0.10)	(0.09)
Spite	8.46	15.03	-8.61^{*}	24.61^{**}	3.53	6.17
	(7.19)	(11.38)	(3.91)	(8.85)	(3.58)	(3.78)
Female	-5.39^{*}	-4.21	-2.01	-1.53	-0.81	-1.62
	(2.59)	(2.61)	(2.04)	(3.64)	(1.53)	(1.33)
Age	0.81	-0.35	0.45	0.11	0.21	0.34
	(0.66)	(0.31)	(0.43)	(0.26)	(0.19)	(0.23)
Constant	1.04	22.97^{*}	13.93	-5.80	-6.13	1.63
	(16.89)	(10.79)	(20.48)	(20.53)	(11.04)	(10.29)
Treatments	3L	5L	3H	5H	All	No 5H
Other Controls	✓	✓	✓	✓	✓	✓
Realized Group Size	✓	✓	✓	✓	✓	✓
Observations	1,632	1,700	1,632	1,666	6,630	4,964
# Clusters	16	10	16	10	52	42
R^2	0.09	0.16	0.13	0.22	0.07	0.09

Note: SE clustered at group level $+ p < 0.1$; $* p < 0.05$; $** p < 0.01$; $*** p < 0.001$

C.1.6 Heterogeneities

Figure 33 shows effort (line 1), sabotage (line 2), received sabotage (line 3), and individual performance (line 4) depending on the rank within a group conditional on the treatments. The rank is based on the exerted effort by group for each round separately. The figure shows that there are heterogeneities between the group members. The group member that exerts the highest effort also exerts the highest sabotage. Therefore, the group member with the highest effort also receives the least sabotage of the others. This results in a high individual performance for this group member. Therefore, different to theory, less effort is destroyed, as the highest effort group member receives the least sabotage.

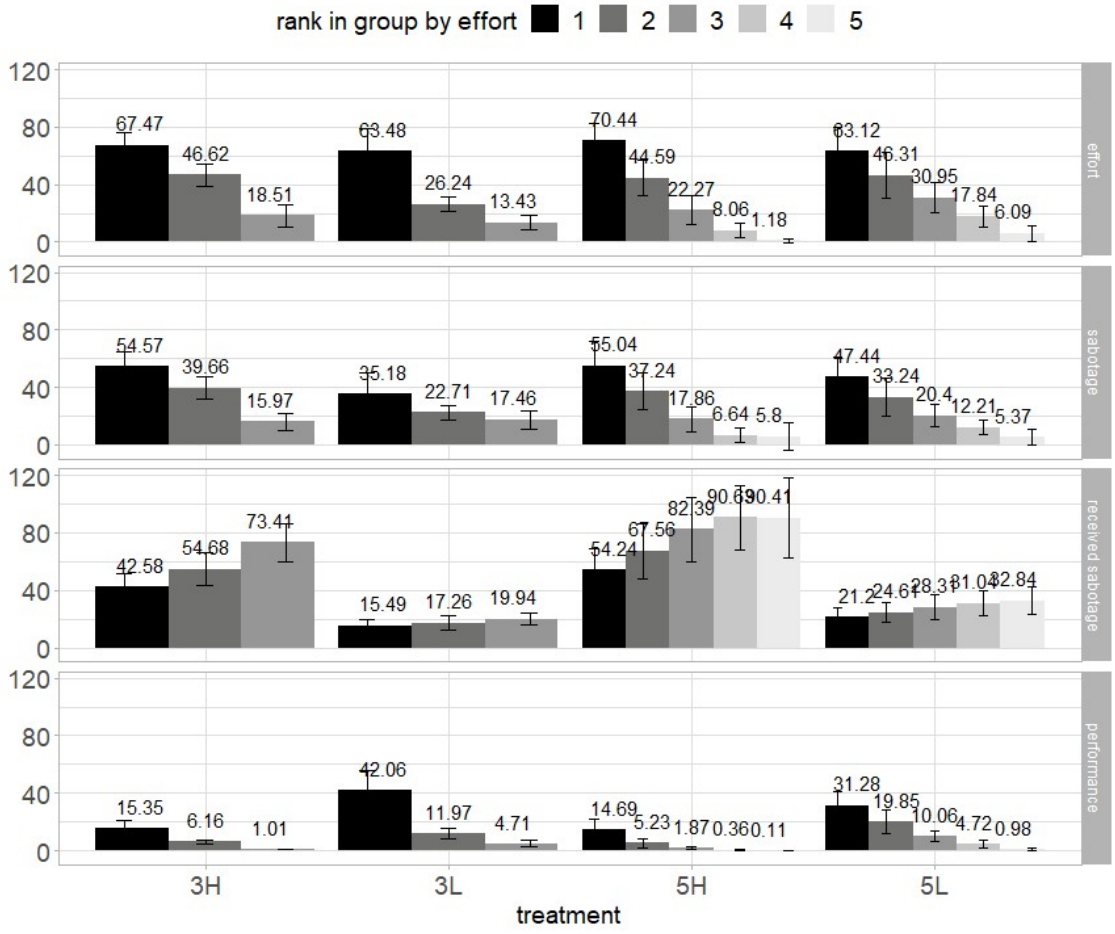


Figure 33: The bar charts show the individual averages based on the rank by effort within a group. The x-axis shows the four treatments, whereas the y-axis in the four panels shows either effort, sabotage, received sabotage, or individual performance. The error bars show 95% confidence intervals.

C.2 Known Group Sizes (Group Size Disclosure)

C.2.1 Subsets of Rounds

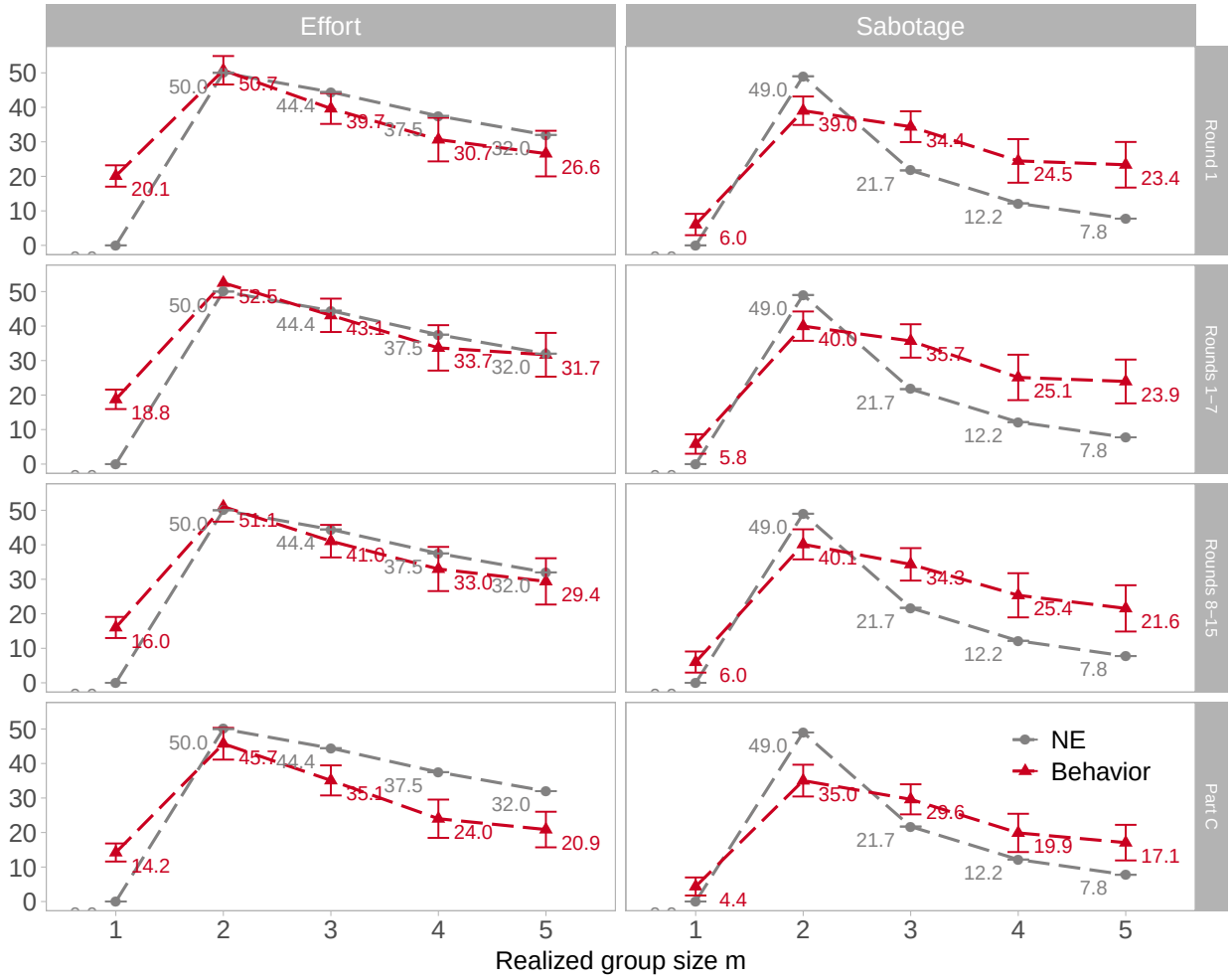


Figure 34: Effort and sabotage levels and Nash equilibria under group size disclosure for the realized group sizes m conditional on a specific subset. Red lines show the elicited behavior, averaged over the specified subset of rounds. Blue lines show the Nash equilibrium predictions. The error bars show 95% confidence intervals.

Figure 34 shows the effort and sabotage levels under disclosure for the realized group sizes conditional on a specific subset of rounds. The first line shows decisions only from the first round, the second the average from the first part of Part A, the third line from the second part of Part A, and the fourth from Part C.

The effort and sabotage levels are very similar between these subsets of rounds. Importantly, the significant and substantive decrease in effort and sabotage levels for an increase in the group

size (except $m = 1$) is very prevalent in all of the panels. Additionally, non-parametric tests show a significant difference between the sabotage (effort) decisions for $m = 2$ and $m = 5$ at a significance level of $p < 0.001$ ($p < 0.001$) in all panels. Moreover, most of the piece-wise comparisons are statistically significant at at least $p < 0.05$.

C.2.2 Time Trends

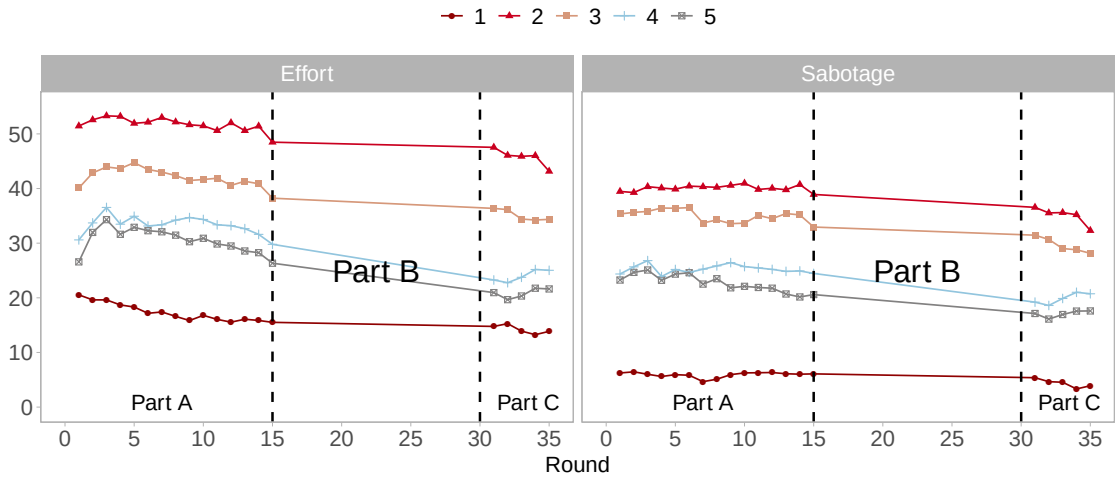


Figure 35: The panels show the time trends for effort and sabotage levels in Part A and Part C. The vertical line at round 15 indicates the end of Part A and the beginning of Part C. The colors indicate the average choices for a specific group size.

In this section, I analyze changes in effort and sabotage levels over time. Figure 35 depicts these time trends for parts A and C conditional on the specific realized group size m. Overall, there is a slight decrease in effort and sabotage levels over the rounds. Importantly, the differences between the realized group sizes are not affected by the slight decrease over time – they remain relatively stable over all rounds.

C.2.3 Regression Results

Table 10 shows the results of linear regressions with clustered standard errors at the matching group level, where I regress effort and sabotage on the realized group size under group size disclosure (parts A and C). I only include realized group size of $m > 1$, as being alone in the contest ($m = 1$) is a special case. I include the pre-registered controls[55] and a dummy for Part C.

[55]The controls are: being active in the round before, having won in the round before, average sabotage and effort levels of other participants in the round before, round, determined group size in the round before, how often won in

Models (1) and (2) confirm the results of the main section and show a significant negative effect of the realized group size on effort and sabotage levels. It further confirms the slight time trend, as round and Part C have significant negative effects on effort and sabotage. In all models, the spite score has a significant and substantive effect on the elicited choices, showing that subjects with spiteful preferences are more competitive.

Table 10: Linear regression effort and sabotage on realized group size based on Part A and Part C

	Dependent variable:	
	effort	sabotage
	(1)	(2)
Realized Group Size	-6.36^{***}	-4.77^{***}
	(0.74)	(0.64)
Round	-0.47^{**}	-0.28^{*}
	(0.16)	(0.13)
Part C	-7.83^{***}	-6.58^{***}
	(1.84)	(1.67)
Risk Aversion	-1.47	-0.15
	(1.51)	(1.29)
Ambiguity Aversion	-2.48^{+}	-1.60
	(1.45)	(1.27)
Loss Aversion	-0.91	-1.59^{+}
	(1.12)	(0.87)
SVO	0.06	0.04
	(0.18)	(0.18)
Spite	18.67^{*}	29.76^{***}
	(9.07)	(8.69)
Female	9.65^{*}	5.43
	(4.47)	(4.22)
Age	-0.92	-1.78^{***}
	(0.71)	(0.51)
Constant	99.99^{***}	100.46^{***}
	(23.93)	(20.68)
Treatment Dummies	✓	✓
Other Controls	✓	✓
Observations	11,172	11,172
# Clusters	52	52
R^2	0.18	0.17
Note: Se clustered at group level	$+ p < 0.1$; $* p < 0.05$; $** p < 0.01$; $*** p < 0.001$	

the rounds before, SVO, spite, risk, loss and ambiguity aversion, age, gender, highest degree, the field of study, the degree of concentration and understanding

C.3 Group Size Uncertainty

C.3.1 Subsets of Rounds

	effort	effort levels		
treatment	NE	round 1	rounds 1-7	rounds 8-15
3L	21.53	39.19	36.85	30.17
		(5.01)	(4.36)	(3.81)
5L	32.00	31.78	31.89	27.82
		(4.68)	(5.91)	(4.53)
3H	43.75	44.21	43.58	43.22
		(4.87)	(3.23)	(3.53)
5H	37.66	33.71	32.13	26.43
		(4.97)	(2.65)	(2.21)

Table 11: Average elicited effort in Part B by treatment based on different subsets, as well as the Nash equilibrium (NE). Standard errors by group and in round 1 by individual.

	sabotage	sabotage levels		
treatment	NE	round 1	rounds 1-7	rounds 8-15
3L	19.17	27.73	29.89	26.29
		(4.09)	(4.81)	(4.03)
5L	25.50	24.74	25.25	22.73
		(4.04)	(4.09)	(3.67)
3H	30.45	36.48	37.45	36.97
		(4.47)	(3.17)	(3.62)
5H	14.36	23.55	25.04	20.67
		(4.05)	(2.87)	(2.10)

Table 12: Average elicited sabotage levels in Part B by treatment based on different subsets, as well as the Nash equilibrium (NE). Standard errors by group and in round 1 by individual.

Tables 11 and 12 show average effort and sabotage levels for different subsets of rounds. The tables show that both effort and sabotage decrease when the group size increases from 3H and 5H for high enter probabilities, for all shown subsets of rounds. Additionally, non-parametric tests confirm these decreases as significant (at least $p < 0.05$) for all subsets of rounds apart from effort in the single round 1 ($p = 0.1994$). Sabotage and effort decisions are not significantly different between *3L* and *5L* in all shown subsets of rounds.

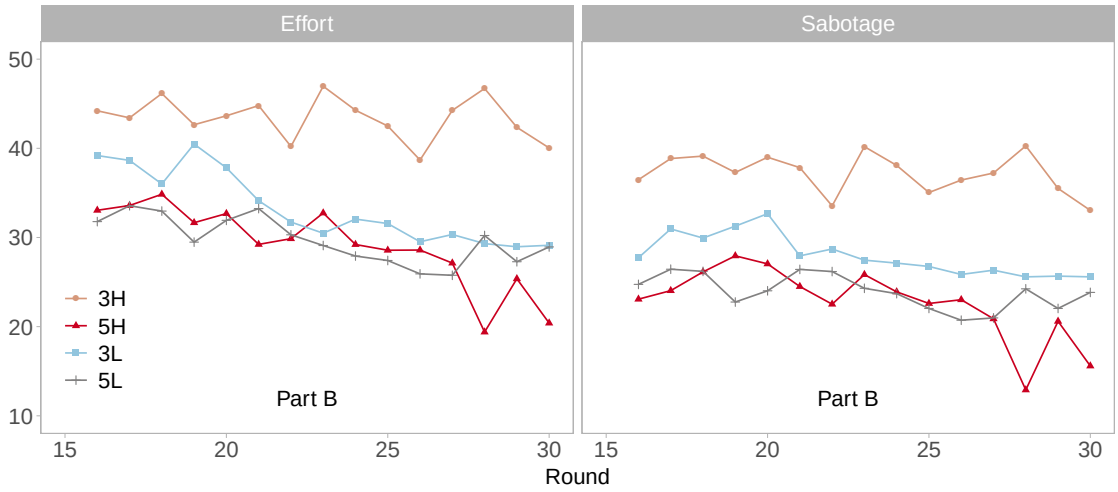


Figure 36: The panels show the time trends for average effort and sabotage levels in Part B. The colors indicate the average choices for the specific treatment.

C.3.2 Time Trends

Figure 36 shows the time trends for Part B conditional on the treatments. It shows a slight decrease over time and the treatment differences remain relatively stable across the rounds.

C.3.3 Regression Results

Table 13 shows the results of a linear regression of effort and sabotage on the treatments and controls under group size uncertainty (Part B) with clustered standard errors at the matching group level. I split the sample into the treatments with a high enter probability (*3H* and *5H*) and into the treatments with a low enter probability (*3L* and *5L*) as from theory, I expect differential effects depending on the enter probabilities. Models (1) and (2) are based on a sample of treatments *3H* and *5H*, and models (3) and (4) of *3L* and *5L*. Furthermore, I include the pre-registered controls.[56]

The models confirm the results from the main section. In models (1) and (2), I find a significant decrease in effort and sabotage for Treatment *5H* compared to *3H*. In models (3) and (4), I do not find any significant effect of Treatment *5L* compared to *3L*.

Furthermore, I confirm the negative time trend, as the round variable has a significant negative effect on effort and sabotage in all models. Under high enter probabilities, I find a significant neg-

[56]The controls are: being active in the round before, having won in the round before, average sabotage and effort levels of other participants in the round before, round, determined group size in the round before, how often won in the rounds before, SVO, spite, risk, loss and ambiguity aversion, age, gender, highest degree, the field of study, the degree of concentration and understanding

ative correlation between loss aversion and effort and sabotage, and under low enter probabilities, I find a significant negative correlation between ambiguity aversion and effort and sabotage levels.

Table 13: Linear regression effort and sabotage on treatments under group size uncertainty

	Dependent variable:			
	effort	sabotage	effort	sabotage
	(1)	(2)	(3)	(4)
Round	-0.94^{***}	-0.72^{**}	-0.83^{**}	-0.40^{*}
	(0.28)	(0.24)	(0.29)	(0.19)
Treatment 5H	-13.49^{***}	-10.76^{*}		
	(3.94)	(5.33)		
Treatment 5L			−3.18	3.55
			(5.90)	(3.35)
Risk Aversion	−1.12	−0.27	−1.15	-3.14^{*}
	(2.02)	(2.19)	(1.71)	(1.36)
Ambiguity Aversion	−0.72	−0.20	-3.88^{*}	-3.03^{**}
	(1.89)	(1.91)	(1.86)	(1.08)
Loss Aversion	-3.82^{**}	-4.11^{***}	1.64	1.16
	(1.22)	(1.18)	(1.55)	(1.49)
SVO	0.36	0.29	0.09	0.13
	(0.31)	(0.32)	(0.27)	(0.24)
Spite	24.33	30.94^{+}	10.99	3.53
	(18.54)	(16.31)	(13.88)	(11.62)
Female	12.12^{**}	8.63^{+}	2.72	5.21
	(4.63)	(5.07)	(6.76)	(6.30)
Age	−0.41	−0.71	0.18	−0.29
	(0.94)	(0.80)	(1.30)	(1.07)
Constant	66.61^{*}	74.67^{**}	3.55	35.55
	(31.44)	(25.13)	(37.69)	(33.99)
Treatments	3H, 5H	3H, 5H	3L, 5L	3L, 5L
Other Controls	✓	✓	✓	✓
Observations	1,455	1,455	1,470	1,470
# Clusters	52	52	52	52
R^2	0.24	0.21	0.17	0.23

Note: SE clustered at group level $+ p < 0.1$; * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$

D Instructions for Treatment $5H$

D.1 Tutorial

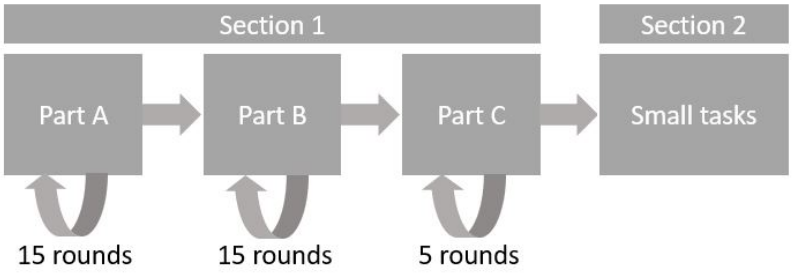
Welcome!

Welcome to this experiment, and thank you for your participation. In this experiment, you have the opportunity to earn points. The number of points depends on your decisions, the decisions of the other participants of this experiment, and luck. After the experiment is finished, we will translate the number of points into Euros at an exchange rate of 100 points = 9 Euros (1 point = 9 Euro cents). On top of that, you will receive 1 Euro.

One note before we begin: It is very important to us, that all participants stay concentrated only on the experiment. If you have a question during the experiment, please write to us in the Zoom chat.

Procedure

The experiment consists of two sections. Section 1 is the main part of this experiment, where you will interact with other participants of this experiment. In section 2, you will complete several small tasks. Section 1 consists of part A, part B, and part C. In total, there will be 35 rounds, as shown in the picture.



At the end of the experiment, 3 of the 35 rounds will be randomly chosen for your payment. The average earnings of these 3 selected rounds determine your payoff of section 1. **Hence, any of the rounds can be payoff-relevant for you. Therefore, it is advisable to think about each decision carefully.**

Additionally, we will donate to one of the following five charities. We will explain in the following pages, how the amount will be determined. One charity will be randomly determined by

the computer at the end of the experiment. The average donation of the 3 out of 35 randomly selected rounds is taken.

- Amnesty International
- Doctors Without Borders
- German Red Cross
- Greenpeace
- UNICEF

We start with a tutorial for part A. Please go through the tutorial attentively and contact us in the Zoom chat in case you have any question. At the end of the tutorial, the computer will ask you several comprehension questions about the instructions. After that, part A will begin.

Part A - Tutorial 1/5

Welcome to the tutorial of part A. The tutorial will prepare you step-by-step for the actual experiment. We start with the most simple version and successively add layers.

Opportunity to win 200 points

You are grouped with one other participant of this experiment. One of you will win a prize of 200 points.

Option A

You will begin each round with a start balance of 200 points. You can choose to keep these points or invest some of them in Option A. The maximum you can invest is 100 points. Any number invested in Option A increases your 'performance'. The other group member can also invest points in Option A to increase his/her 'performance'. The higher your performance is in comparison to the other group member's performance, the higher is your probability to win the 200 point prize. Your and the other group member's performance and probability to win are calculated as follows:

$$\text{Performance} = \text{Points invested in Option A ('Option-A points')}$$

$$\text{Your probability to win} = \frac{\text{Your performance}}{\text{Your performance} + \text{Other group member's performance}}$$

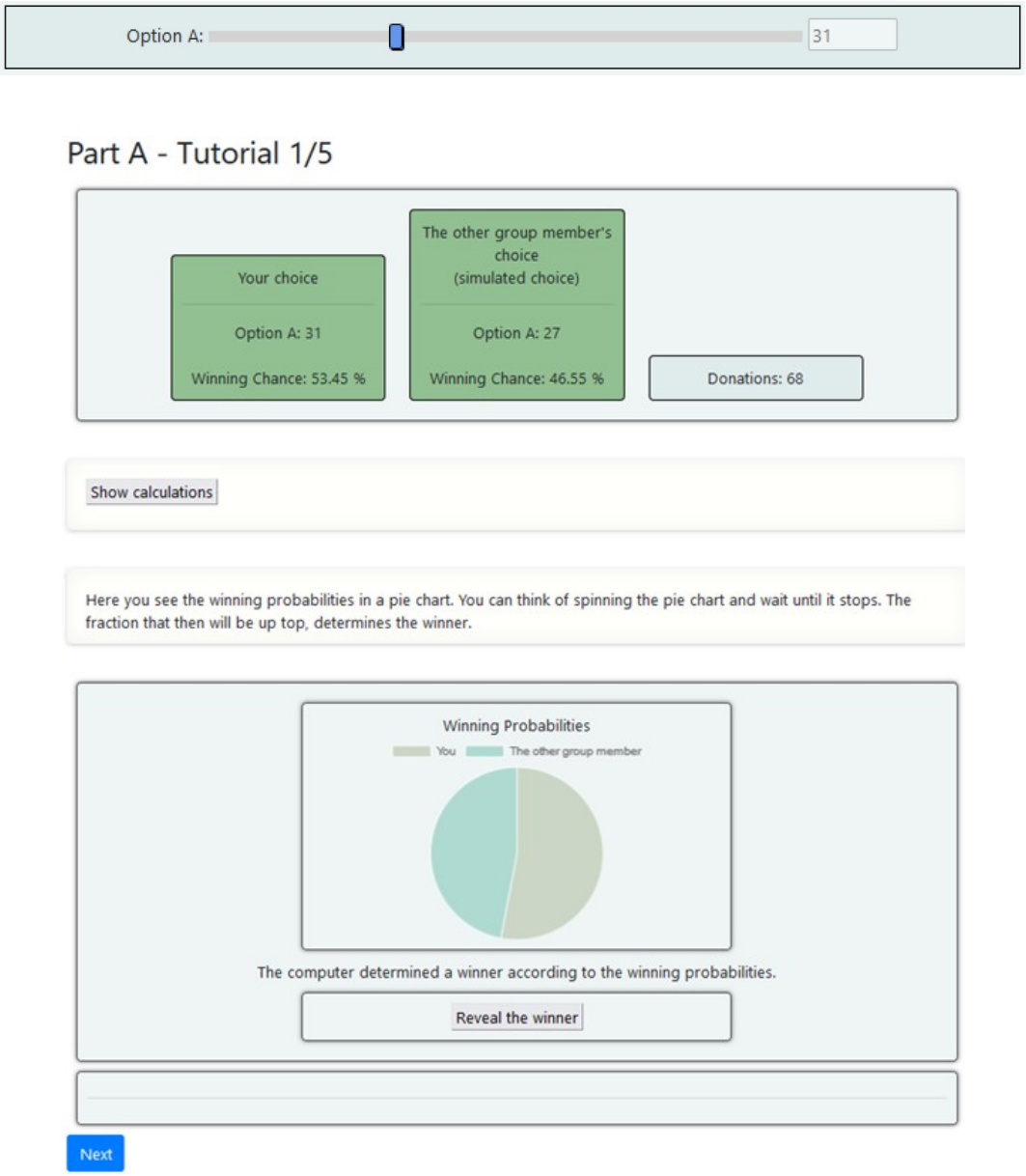
If both performances are 0, the winning probability is 50% for each group member. Your and

the other group member's performance not only influence the winning probabilities. We will also donate money to a charity depending on the total performance. The donations are calculated as follows:

$$\text{Donations} = \text{Your performance} + \text{Other group member's performance} + 10$$

TRY IT OUT!

Please choose how much to invest in Option A. Any points that you don't invest (out of the 200) are yours to keep. The computer will simulate a random choice for the other group member.



Part A - Tutorial 2/5

Option B

In the actual experiment, you will have a second option. Additionally to Option A, you can invest up to 100 points into Option B. With Option B you decrease the other group member's performance. Likewise, the other group member can decrease your performance by investing into Option B. Your and the other group member's performance and probability to win are then calculated as follows. It is not important to remember this formula or to fully understand it. We show it for full transparency. For the actual experiment, you will have access to a calculator which helps you get a sense of how the choices affect your performance and probability to win.

$$\text{Your performance} = \frac{\text{Your Option-A points}}{1 + \text{The other group member's Option-B points}}$$

$$\text{Your probability to win} = \frac{\text{Your performance}}{\text{Your performance} + \text{The other group member's performance}}$$

If both performances are 0, the winning probability is 50% for each group member. The donations are calculated as before:

$$\text{Donations} = \text{Your performance} + \text{Other group member's performance} + 10$$

Hence, Option A increases the donations and Option B decreases the donations.

TRY IT OUT!

Please choose how much to invest in Option A and Option B. Any points that you don't invest (out of the 200) are yours to keep. The computer will simulate random choices for the other group member.



[[*Feedback shown similar to before*]]

Part A Tutorial 3/5

Your Group

In the actual experiment, instead of one other participant, you will be paired with 4 other participants of this experiment. You stay in this group until the end of the first section of this experiment.

Now, one of the group members will win 200 points. Every group member can invest into Option A and Option B. As before, Option A increases the own performance. Option B decreases the performance of **all other group members simultaneously**. Your and the other group members' performances and probabilities to win are calculated as follows:

$$\text{Your performance} = \frac{\text{Your Option-A points}}{1 + \text{All other group members' Option-B points}}$$

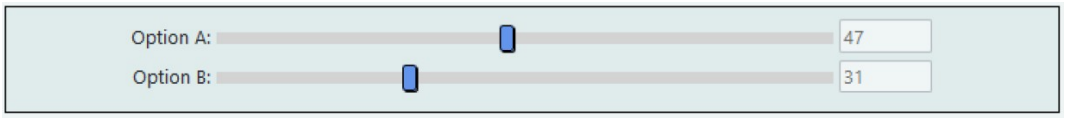
$$\text{Your probability to win} = \frac{\text{Your performance}}{\text{Your performance} + \text{All other group members' performances}}$$

If all performances are 0, the winning probabilities are the same for all group members (20% for every group member). The donations are calculated as before:

$$\text{Donations per group} = \text{Sum of all performances} + 10$$

TRY IT OUT!

Please choose how much to invest in Option A and Option B and then press on next. The computer will simulate random choices for all other group members.



[[*Feedback shown similar to before*]]

Part A - Tutorial 4/5 *Active and inactive group members*

In the actual experiment, not every group member will be active in each round. In every round, the computer will randomly choose a number of active group members between 1 and 5 and determines randomly who they are.

If you are inactive...

... you do not interact with anyone in this round.

If you are active...

... you will interact with all other active group members. One of the active group members will win the 200-point prize. Depending on the random choice of the computer, you interact with either 0, 1, 2, 3, or 4 other active group members.

Procedure

In every round, you make your decisions before the computer determines whether you are active or not. We will ask you how many points you would want to invest if

- you are the only active group member
- there is 1 other active group member
- there are 2 other active group members
- there are 3 other active group members
- there are 4 other active group members

After that, the computer will randomly determine the number of active group members. The chances of the different numbers of active group members are not even. As you make your decisions depending on the specific number of other active group members, the chances are not further important for part A. The chance of being active is NOT influenced by the choices.

If you are active, the computer will implement your decisions for the specific number of active players.

If you are the only active group member, you will win the prize of 200 points for sure, independent of how much you invest into Option A and Option B. However, you still have to pay for your investment choices.

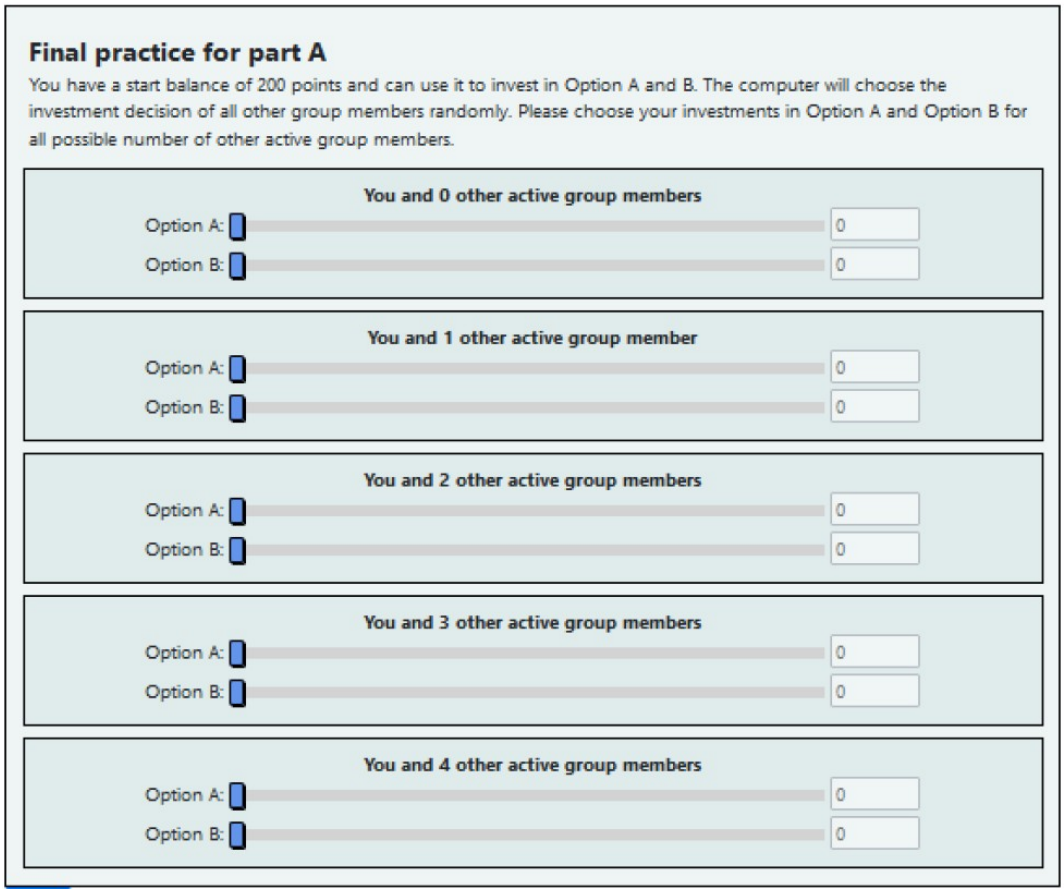
If you are inactive, your choices do not matter neither for the performances and winning probabilities, nor for the donations. You will not have to pay for your decisions.

Payoff

Your payoff in this round is determined as follows:

If you are inactive:	**If you are active and win:**	**If you are active and lose:**
+200 (start balance)	+200 (start balance)	+200 (start balance)
	+200 (winning prize)	+0 (no prize)
	- points you invested	- points you invested
200	400 - points you invested	200 - points you invested

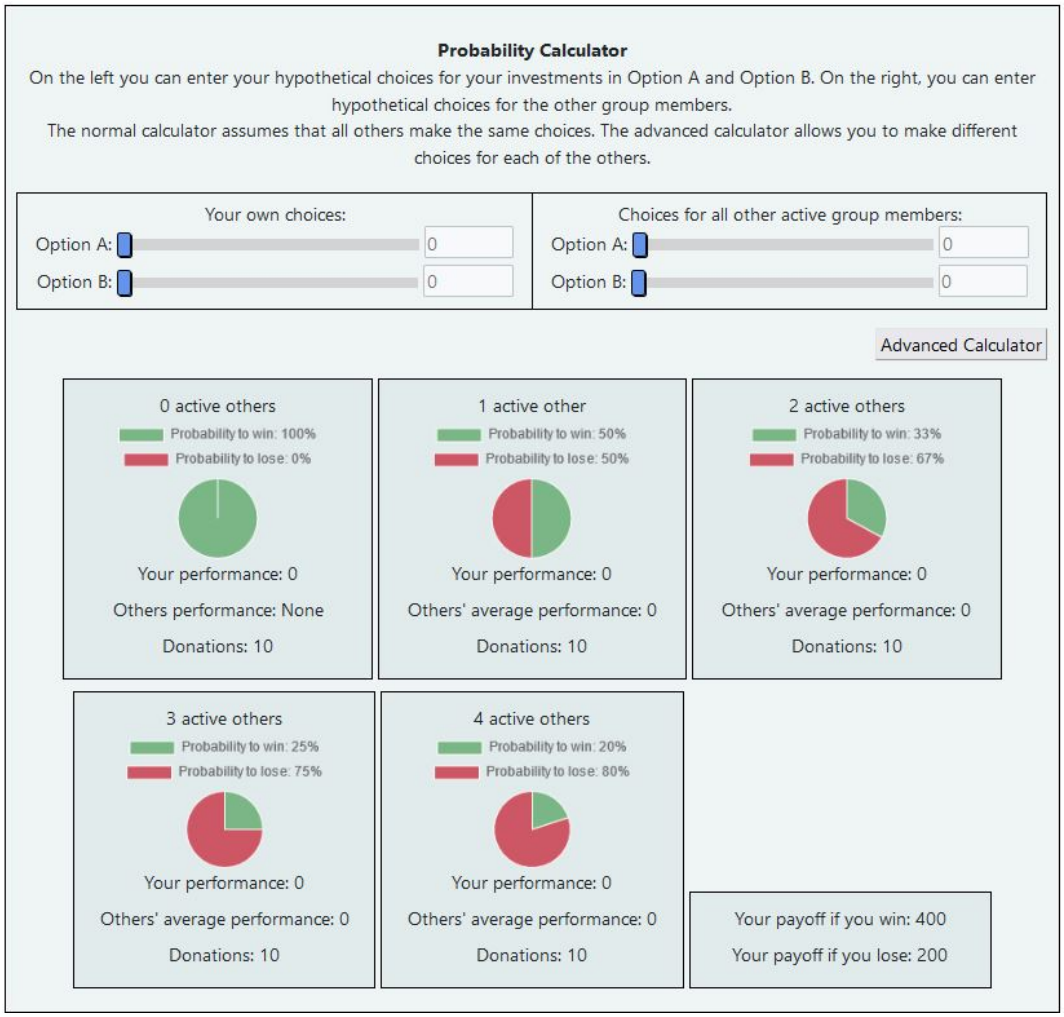
There will be a time limit for your decisions. In the first rounds you will have more time than in later rounds.



[[*Feedback shown similar to before*]]

Part A - Tutorial 5/5

You will have access to a calculator throughout the entire experiment. The probability calculator shows you how your choices and the other active group members' choices affect your probability to win and the donations. Please take your time to familiarize yourself with the underlying mechanisms.



Just for your information, you will always be able to access a summary of the instructions by scrolling down.

Summary of Instructions

1. Everyone decides how many points of the start balance to invest in Option A and Option B.

$$\textbf{Your performance} = \frac{\textbf{Your Option-A points}}{\textbf{1} + \textbf{All other group members' Option-B points}}$$

- Option A increases your own performance
- Option B decreases all other active group members' performances simultaneously

$$\textbf{Your probability to win} = \frac{\textbf{Your performance}}{\textbf{Your performance} + \textbf{All other group members' performances}}$$

- The higher your performance in comparison to the other active group members' performances, the higher your probability to win

The performances of all active group members influence the donations:

$$\textbf{Donations per group} = \textbf{Sum of all performances} + \textbf{10}$$

- The higher the performances, the higher the donations
- Option A increases the donations
- Option B decreases the donations

2. The computer determines a number of active group members between 1 and 5 and determines the active group members. Everyone has the same probability to become active.

3. The computer determines the winning probabilities with the choices of the active group members

- The choices of all active group members for the specific number of active group members are picked.
- The performances and winning probabilities are calculated based on the choices of the active group member.
- The donations are calculated with the performance of every active group member.

4. The computer will determine the winner according to the winning probabilities. The winner receives the 200-point prize. If you are inactive, your choices do not count.

Your individual payoff is determined as follows:

If you are inactive:	**If you are active and win:**	**If you are active and lose:**
+200 (start balance)	+200 (start balance)	+200 (start balance)
	+200 (winning prize)	+0 (no prize)
	- points you invested	- points you invested
200	400 - points you invested	200 - points you invested

Quiz

Here is a little quiz. After you have answered all quiz questions correctly (you have several tries), we can begin with part A. Remember, you can always scroll down to see an overview of the instructions.

Q1: How many participants will be in your group, including you?

- 1
- 2
- 3
- 4
- 5

Q2: Provided that you are active, how can you increase your probability to win?

- Increase my performance and reduce the other active group members' performances
- I can't
- Increase the other active group members' performances

Q3: What can you do with Option A?

- Increase my performance
- Increase the other active group members' performances
- Decrease the other active group members' performances

Q4: What can you do with Option B?

- Decrease my performance
- Increase the other active group members' performances
- Decrease all other active group members' performances simultaneously
- Decrease the performance of another active group member of my choice

Q5: How can you increase the donations?

- Increase my performance (by investing in Option A)
- Decrease my performance (by investing less in Option A)

Q6: How can you decrease the donations?

- Increase my performance (by investing in Option A)

- Decrease the other active group members' performances (by investing in Option B)

Q7: Suppose that you are active and that a round was selected for payment. Who is affected by your decisions?

- Me and the charity
- Me, the other active group members of my group, and the charity
- Everyone of my group and the charity

Q8: How many group members are active in one round?

- 3
- 5
- This is determined randomly in every round
- This is determined randomly in the first round and stays the same until the end of the experiment

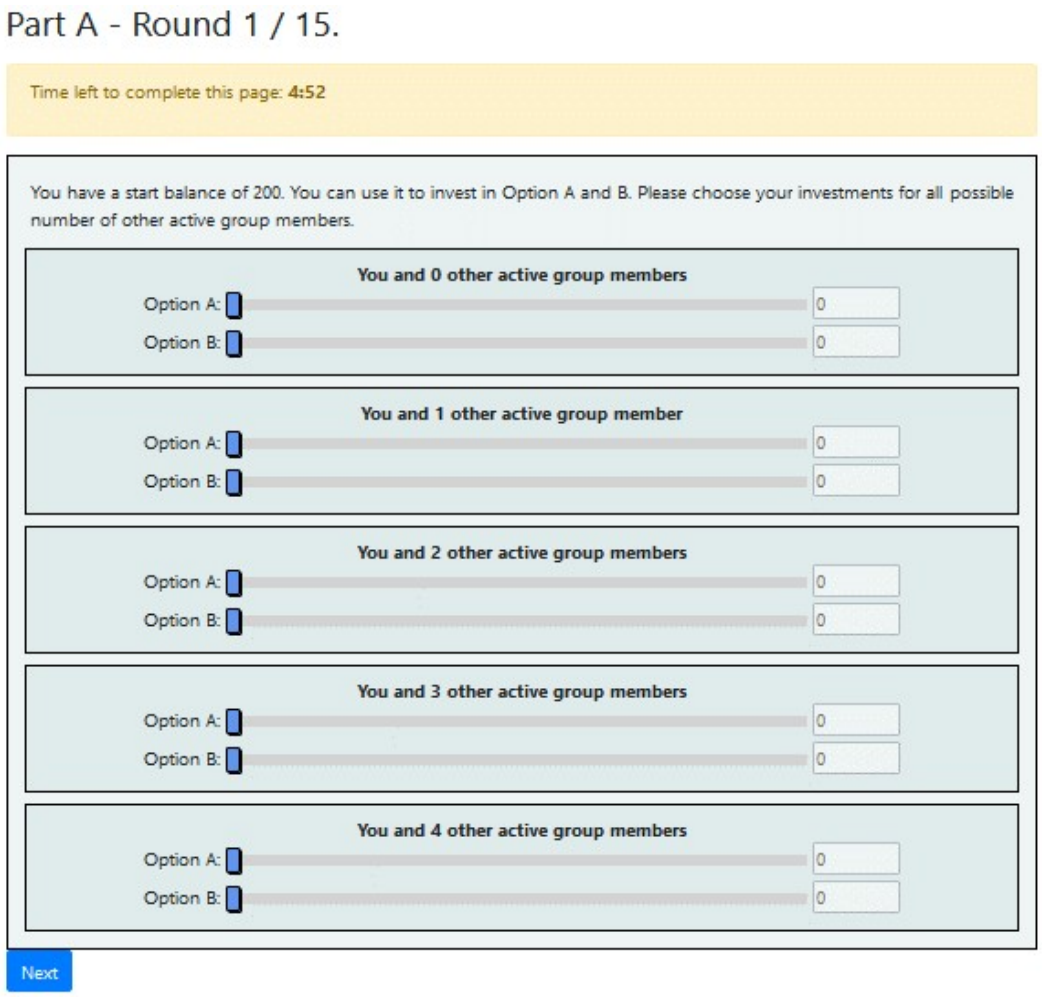
Q9: What happens if only one group member becomes active?

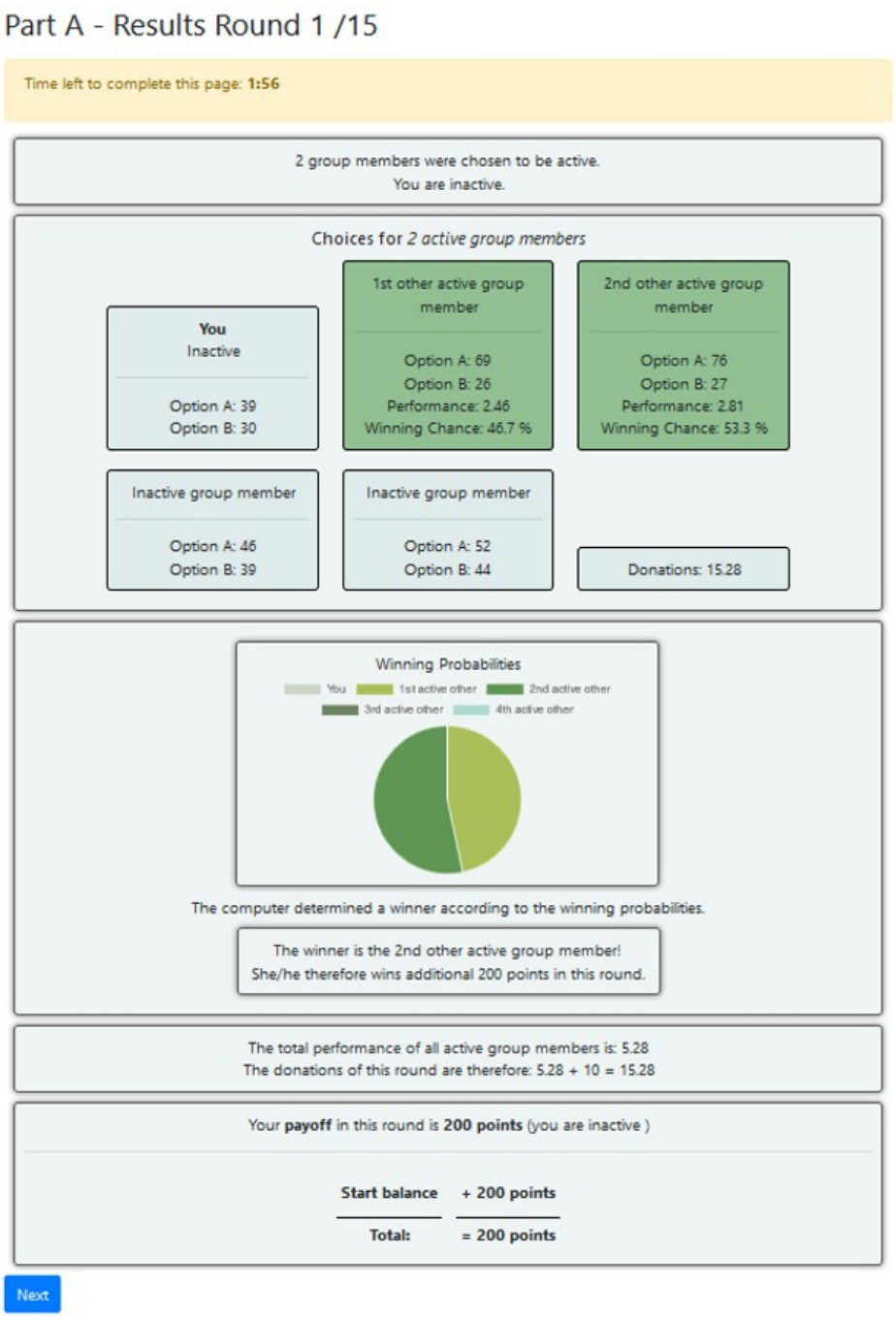
- This group member wins the 200-point prize independently of his/her choices
- There will never be just one active person
- There is a 50% chance that this persons wins the 200-point prize

Q10: In each round, you receive a start balance of 200 points. What happens with the points that you do not invest in Option A or Option B?

- These points are destroyed and will not be added to my payoff of this round
- I can keep the points and they will be added to my payoff of this round

D.2 Section 1 - Part A





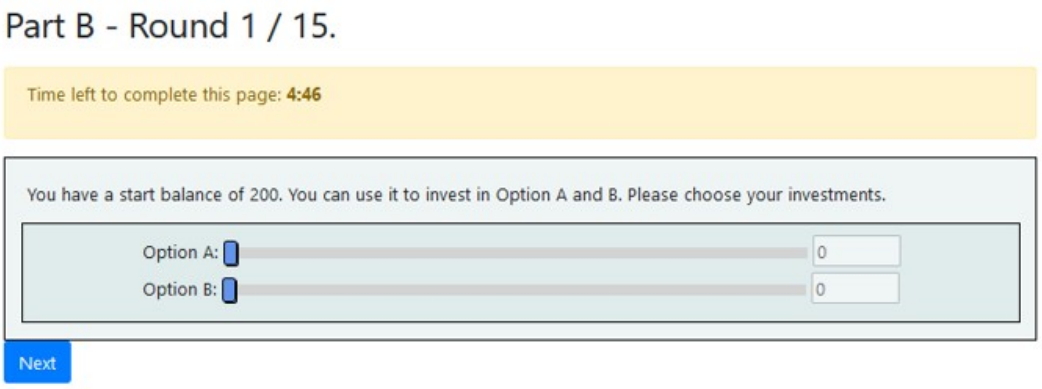
D.3 Section 1 - Part B

Part B is very similar to Part A. You stay in the same group as in part A. The only difference to part A is that we do not ask you for your decisions for every possible number of other active group members. Instead, we ask you for **one** decision for Option A and for **one** decision for Option B. In other words, you only decide once for Option A and once for Option B, and this one decision

each has to fit all possible scenarios (0 others, 1 other, 2 others, 3 others, 4 others). Therefore, it is advisable that you think about how likely these scenarios are and adjust your decisions accordingly.

The following table and pie chart show the probabilities for the number of other active group members in each round, given that you are active.

Number of other active group members	0	1	2	3	4
Probability of Occurrence	<1%	5%	21%	42%	32%



Part B - Results Round 1 /15

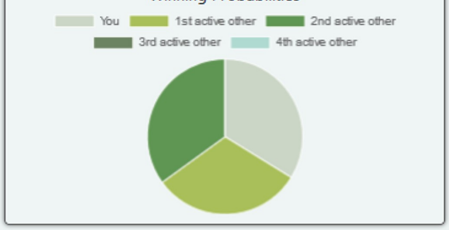
Time left to complete this page: **1:50**

3 group members were chosen to be active.
You are active.

Choices

You Active	1st other active group member	2nd other active group member
Option A: 48 Option B: 32 Performance: 1.07 Winning Chance: 33.8 %	Option A: 62 Option B: 14 Performance: 0.98 Winning Chance: 31.2 %	Option A: 52 Option B: 30 Performance: 1.11 Winning Chance: 35.0 %
Inactive group member Option A: 56 Option B: 38	Inactive group member Option A: 54 Option B: 37	Donations: 13.16

Winning Probabilities



The computer determined a winner according to the winning probabilities.

The winner is the 1st other active group member!
She/He therefore wins additional 200 points in this round.

The total performance of all active group members is: 3.16
The donations of this round are therefore: 3.16 + 10 = 13.16

Your **payoff** in this round is **120 points** (you are active)

Start balance	**+ 200 points**
Costs Option A	**- 48 points**
Costs Option B	**- 32 points**
Total:	**= 120 points**

Next

D.4 Section 1 - Part C

The rules for part C are identical to the rules of part A. Part Consists of 5 rounds. You remain in the same group as before.

[[Elicitation and Feedback the same as in Part A.]]

D.5 Section 2

In this final section 2, we continue with several small tasks. Unlike in section 1, we do not use points anymore. Instead, you will be deciding about Euro cents.

In the following task, you will be randomly paired with another participant. You will be making a series of decisions about allocating cents between you and this other person. You will make 9 choices. The other person also makes the same 9 choices. It will be randomly determined whether your choices or the other person's choices constitute the payoff for the two of you. 1 of the 9 choices will be randomly picked at the end of the experiment.

Table 14: Section 2 - Choices

You receive	85	85	85	85	85	85	85	85	85
	○	○	○	○	○	○	○	○	○
Other receives	85	76	68	59	50	41	32	24	15
You receive	85	87	89	91	92	94	96	98	100
	○	○	○	○	○	○	○	○	○
Other receives	15	19	24	28	32	37	41	46	50
You receive	50	54	59	63	68	72	76	81	85
	○	○	○	○	○	○	○	○	○
Other receives	100	98	96	94	92	91	89	87	85
You receive	50	54	59	63	68	72	76	81	85
	○	○	○	○	○	○	○	○	○
Other receives	100	89	79	68	58	47	36	26	15
You receive	100	94	88	81	75	69	62	56	50
	○	○	○	○	○	○	○	○	○
Other receives	50	56	62	69	75	81	88	94	100
You receive	100	98	96	94	92	91	89	87	85
	○	○	○	○	○	○	○	○	○
Other receives	50	54	59	63	68	72	76	81	85
You receive	70	70	70	70	70	70	70	70	70
	○	○	○	○	○	○	○	○	○
Other receives	100	98	96	94	92	91	89	87	85
You receive	70	68	65	62	60	58	55	52	50
	○	○	○	○	○	○	○	○	○
Other receives	100	96	92	89	85	81	78	74	70
You receive	100	100	100	100	100	100	100	100	100
	○	○	○	○	○	○	○	○	○
Other receives	100	98	96	94	92	91	89	87	85

The following three pages each present 10 scenarios for which you should make a decision. In each case, you decide between a lottery and a fixed payment.

After you have made all choices, one of the pages and one of the scenarios will be randomly chosen for your payment. If you have chosen the fixed payment, you will get the corresponding payoff for sure. If you have chosen the lottery, it will be randomly determined (according to the corresponding probabilities) whether you receive the low or high outcome.

Table 15: Section 2 - Page [[1 or 2]]

Please choose for every row, whether you prefer the lottery or the fixed payment

Lottery			Fixed payment
50% prob. of winning 0 cents, 50% prob. of winning 50 cents	○	○	winning 5 cents for sure
50% prob. of winning 0 cents, 50% prob. of winning 50 cents	○	○	winning 10 cents for sure
50% prob. of winning 0 cents, 50% prob. of winning 50 cents	○	○	winning 15 cents for sure
50% prob. of winning 0 cents, 50% prob. of winning 50 cents	○	○	winning 20 cents for sure
50% prob. of winning 0 cents, 50% prob. of winning 50 cents	○	○	winning 25 cents for sure
50% prob. of winning 0 cents, 50% prob. of winning 50 cents	○	○	winning 30 cents for sure
50% prob. of winning 0 cents, 50% prob. of winning 50 cents	○	○	winning 35 cents for sure
50% prob. of winning 0 cents, 50% prob. of winning 50 cents	○	○	winning 40 cents for sure
50% prob. of winning 0 cents, 50% prob. of winning 50 cents	○	○	winning 45 cents for sure
50% prob. of winning 0 cents, 50% prob. of winning 50 cents	○	○	winning 50 cents for sure

Table 16: Section 2 - Page [[1 or 2]]

Please choose for every row, whether you prefer the lottery or the fixed payment

Lottery			Fixed payment
50% prob. of winning 0 cents, 50% prob. of winning 50 cents	○	○	losing 5 cents for sure
50% prob. of winning 0 cents, 50% prob. of winning 50 cents	○	○	losing 10 cents for sure
50% prob. of winning 0 cents, 50% prob. of winning 50 cents	○	○	losing 15 cents for sure
50% prob. of winning 0 cents, 50% prob. of winning 50 cents	○	○	losing 20 cents for sure
50% prob. of winning 0 cents, 50% prob. of winning 50 cents	○	○	losing 25 cents for sure
50% prob. of winning 0 cents, 50% prob. of winning 50 cents	○	○	losing 30 cents for sure
50% prob. of winning 0 cents, 50% prob. of winning 50 cents	○	○	losing 35 cents for sure
50% prob. of winning 0 cents, 50% prob. of winning 50 cents	○	○	losing 40 cents for sure
50% prob. of winning 0 cents, 50% prob. of winning 50 cents	○	○	losing 45 cents for sure
50% prob. of winning 0 cents, 50% prob. of winning 50 cents	○	○	losing 50 cents for sure

Table 17: Section 2 - Page 3

For every row, please choose whether you prefer the lottery or the fixed payment. In the lottery, **p denotes the probability in percent with which you lose.** The computer will randomly determine this probability after your decisions. p can be between 0 and 100.

Lottery			Fixed payment
p% prob. of winning 0 cents, (100-p)% prob. of winning 50 cents	○	○	winning 5 cents for sure
p% prob. of winning 0 cents, (100-p)% prob. of winning 50 cents	○	○	winning 10 cents for sure
p% prob. of winning 0 cents, (100-p)% prob. of winning 50 cents	○	○	winning 15 cents for sure
p% prob. of winning 0 cents, (100-p)% prob. of winning 50 cents	○	○	winning 20 cents for sure
p% prob. of winning 0 cents, (100-p)% prob. of winning 50 cents	○	○	winning 25 cents for sure
p% prob. of winning 0 cents, (100-p)% prob. of winning 50 cents	○	○	winning 30 cents for sure
p% prob. of winning 0 cents, (100-p)% prob. of winning 50 cents	○	○	winning 35 cents for sure
p% prob. of winning 0 cents, (100-p)% prob. of winning 50 cents	○	○	winning 40 cents for sure
p% prob. of winning 0 cents, (100-p)% prob. of winning 50 cents	○	○	winning 45 cents for sure
p% prob. of winning 0 cents, (100-p)% prob. of winning 50 cents	○	○	winning 50 cents for sure

Final questionnaire

Lastly, please enter the following information.

Your age: _____

Please indicate the gender you most identify with:

- female
- male
- other

Please indicate your field of study: _____

In which semester are you? _____

Please indicate your highest degree:

- HighSchoolDegree
- Bachelor
- Master
- PhD
- Other

How concentrated were you during the experiment?

- 1 - not at all
- 2 - little concentrated
- 3 - medium
- 4 - mostly concentrated
- 5 - very concentrated

How well did you understand the experiment?

- 1 - not at all
- 2 - not well
- 3 - medium
- 4 - mostly understood it
- 5 - understood it well

Your Payoff

Thank you for participating in this experiment. You earned a total of **XY Euros. IMPORTANT!** Please write the following payment code on the formula that you received by E-Mail. Without this code, the payment can not be made: XYZ123456

[[*Calculations of payoff were shown.*]]